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(54) **CUTTERS WITH PULSATING VACUUM CONTROL**

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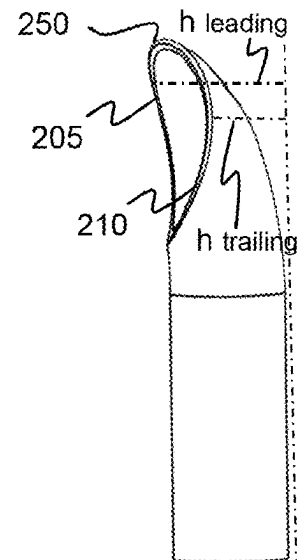
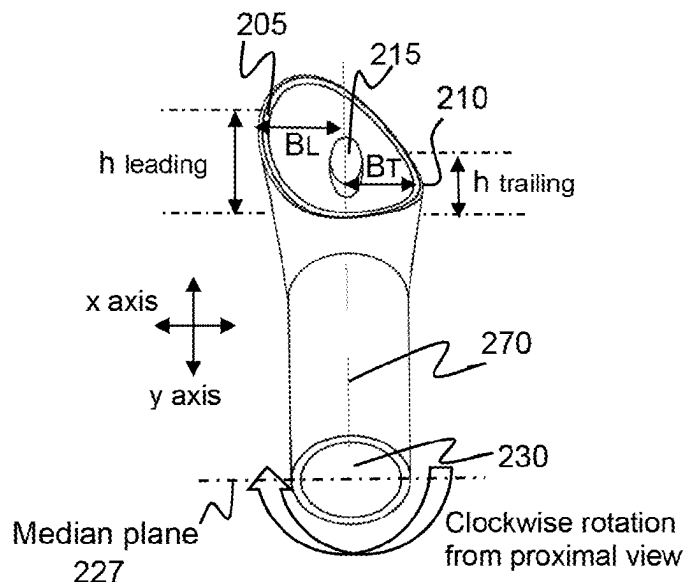
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(57) **ABSTRACT**

Vascular cutter devices, systems, and methods are provided to effectively cut and remove all types of thrombus and atheroma tissue, whether soft, tough, or hard, and can do so with a reduced risk of suffering vessel injuries while avoiding an undesirable volume of blood loss during procedures. Thrombectomy systems having curved tubular cutting devices, and methods of using them, are provided. These systems, devices, and methods can (i) effectively cut and remove a variety of thrombus tissue from blood vessels, including soft, tough, and hard tissue; (ii) safely self-collect and remove tissue particles to avoid release of emboli; (iii) effectively treat a blood vessel with a reduced risk of suffering vessel injuries that can lead to increased stenosis; and (iv) effectively reduce blood loss during a procedure.

20 Claims, 28 Drawing Sheets



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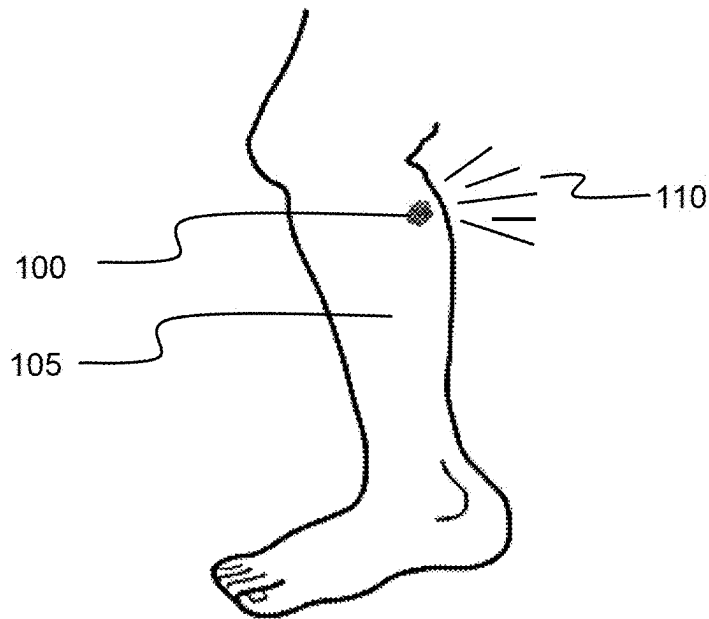


FIG. 1A

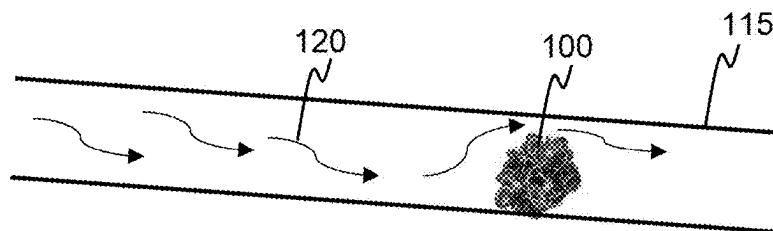


FIG. 1B

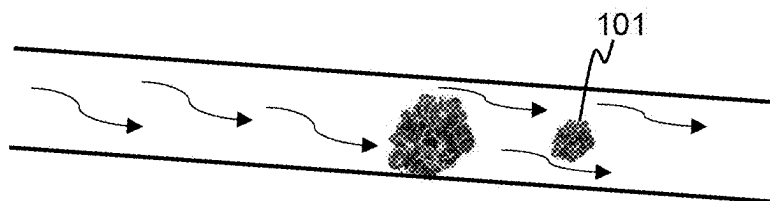


FIG. 1C

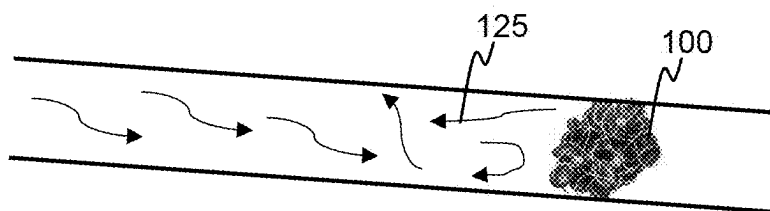
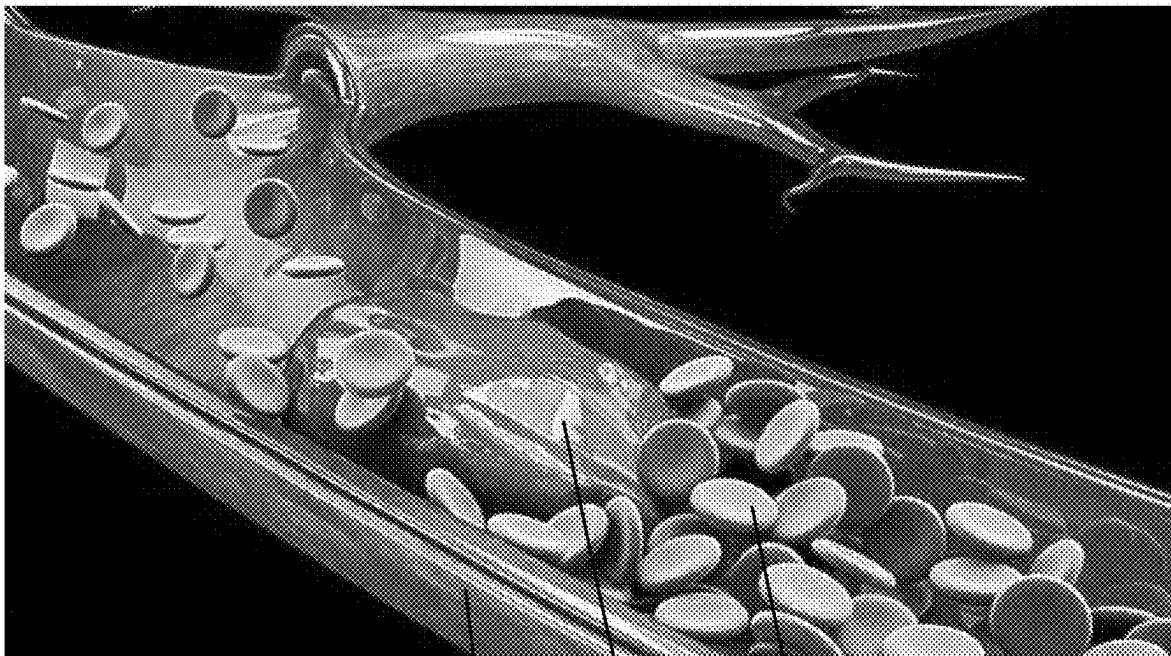


FIG. 1D

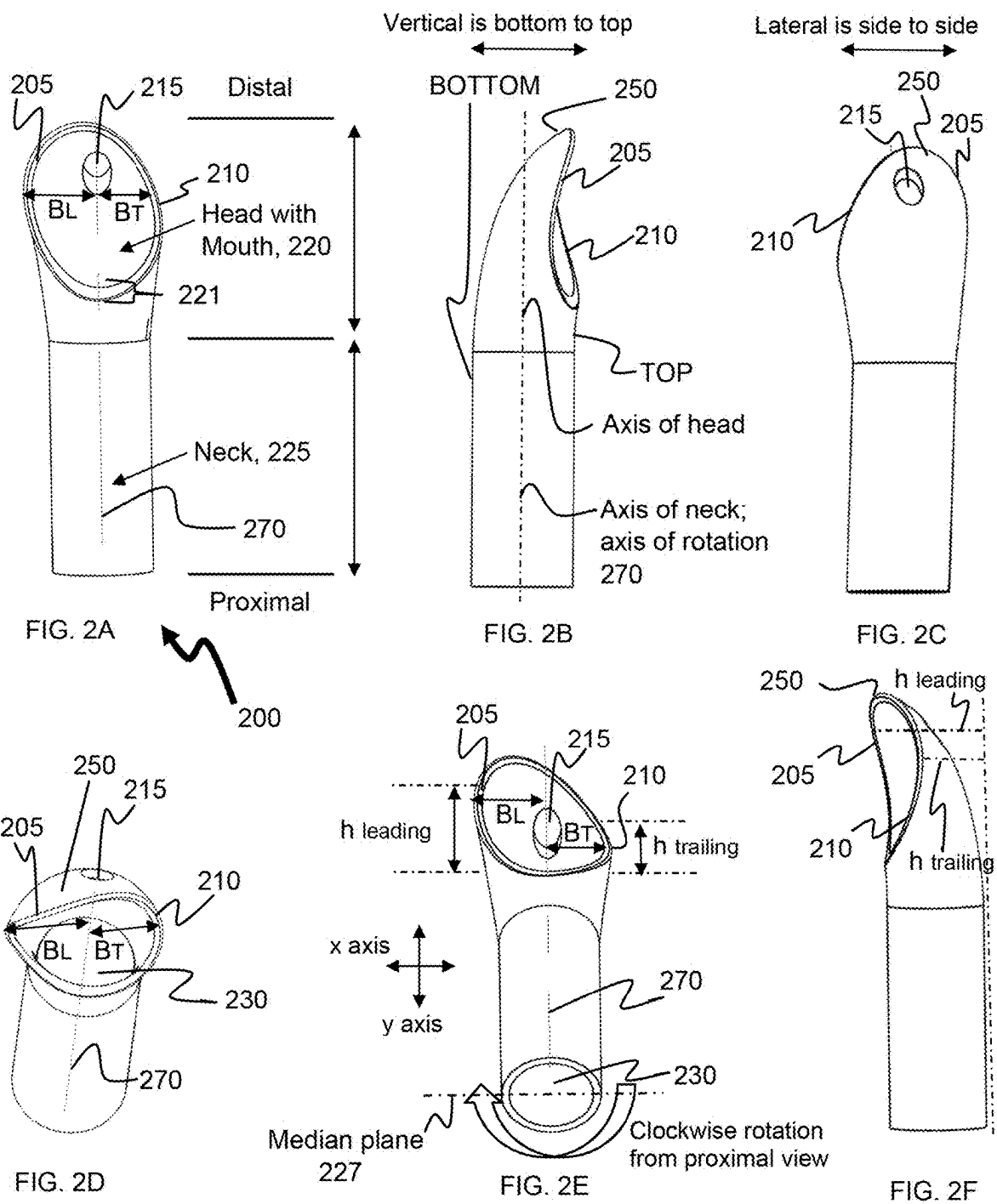


115

100

FIG. 1E

130



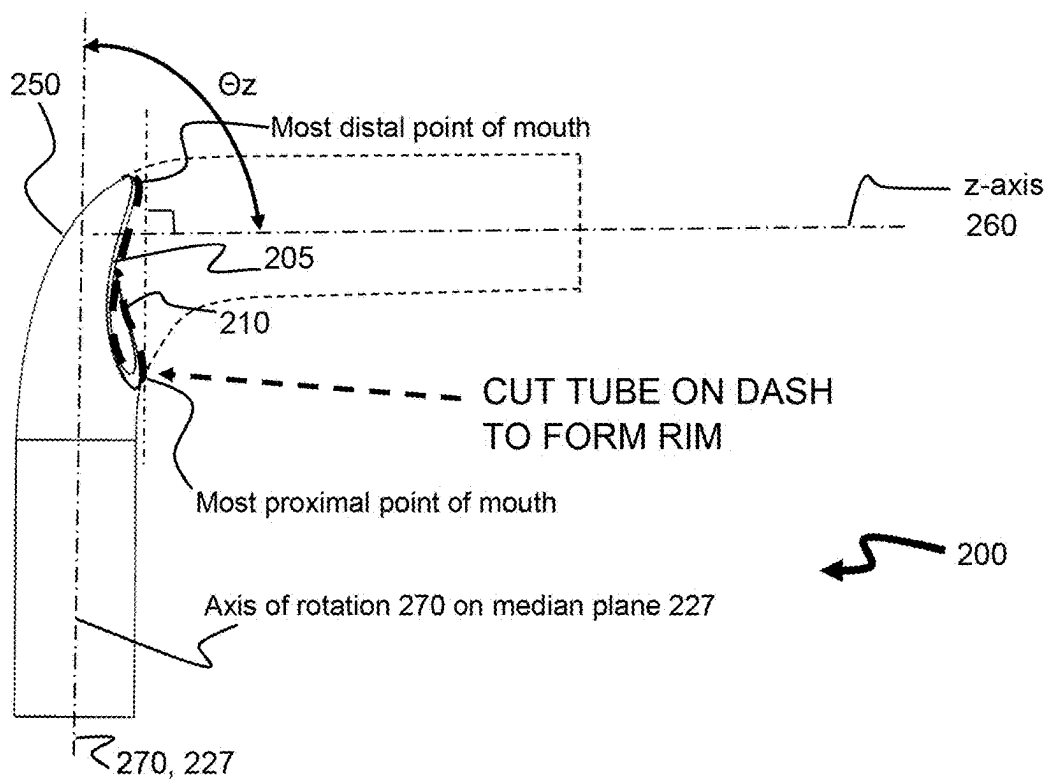


FIG. 2G

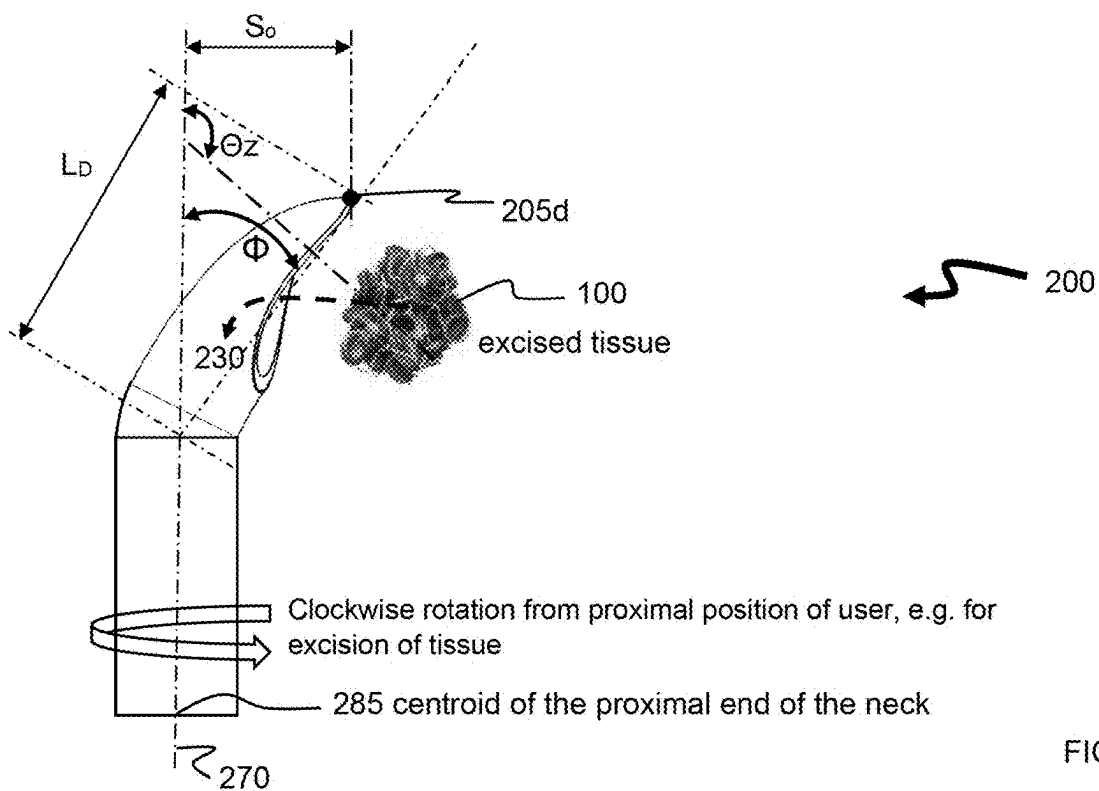
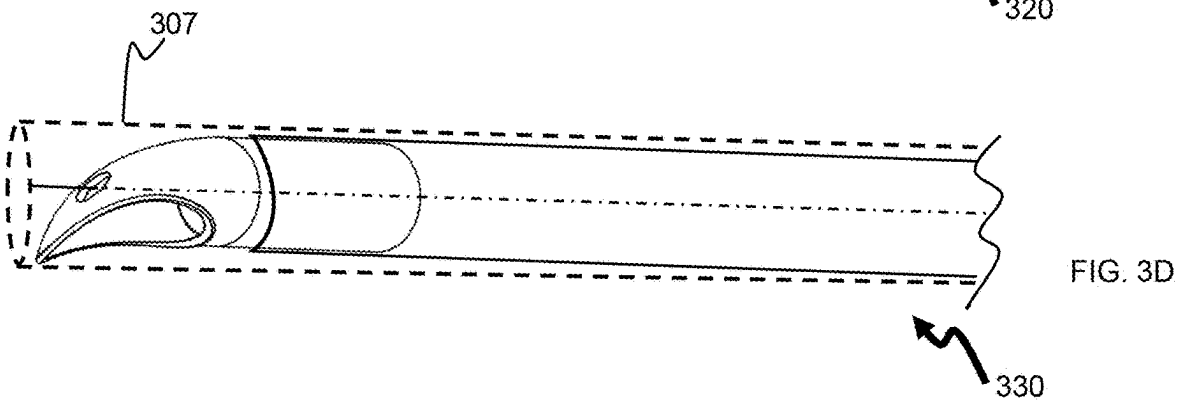
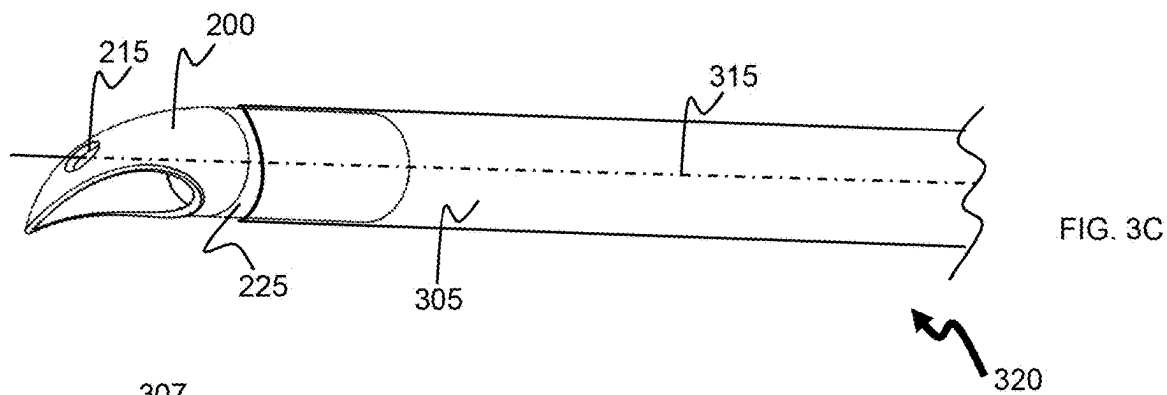
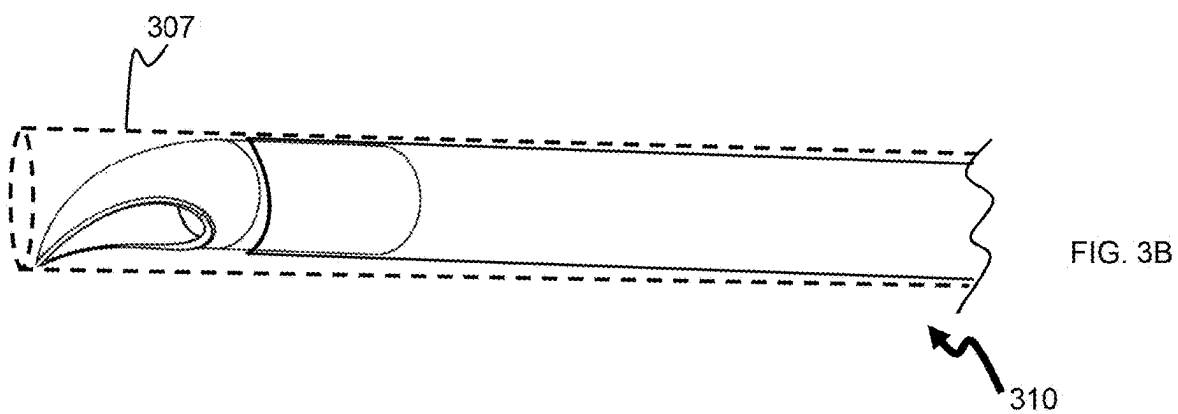
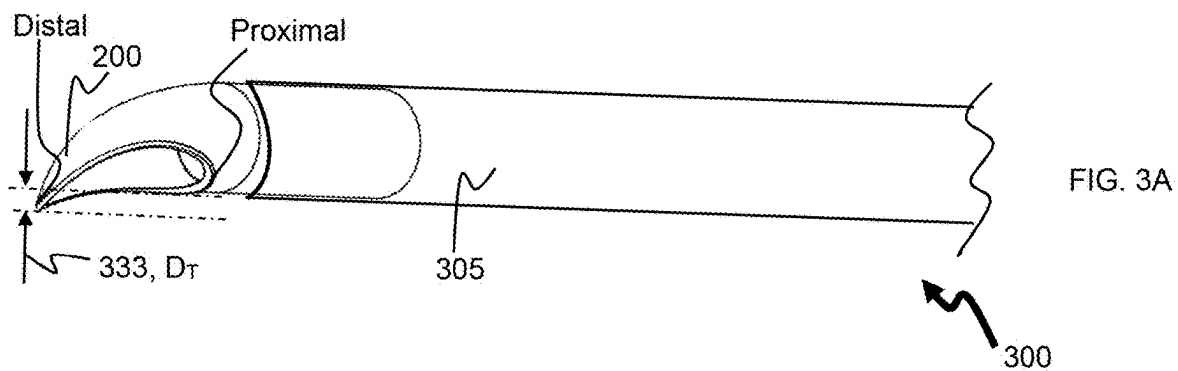
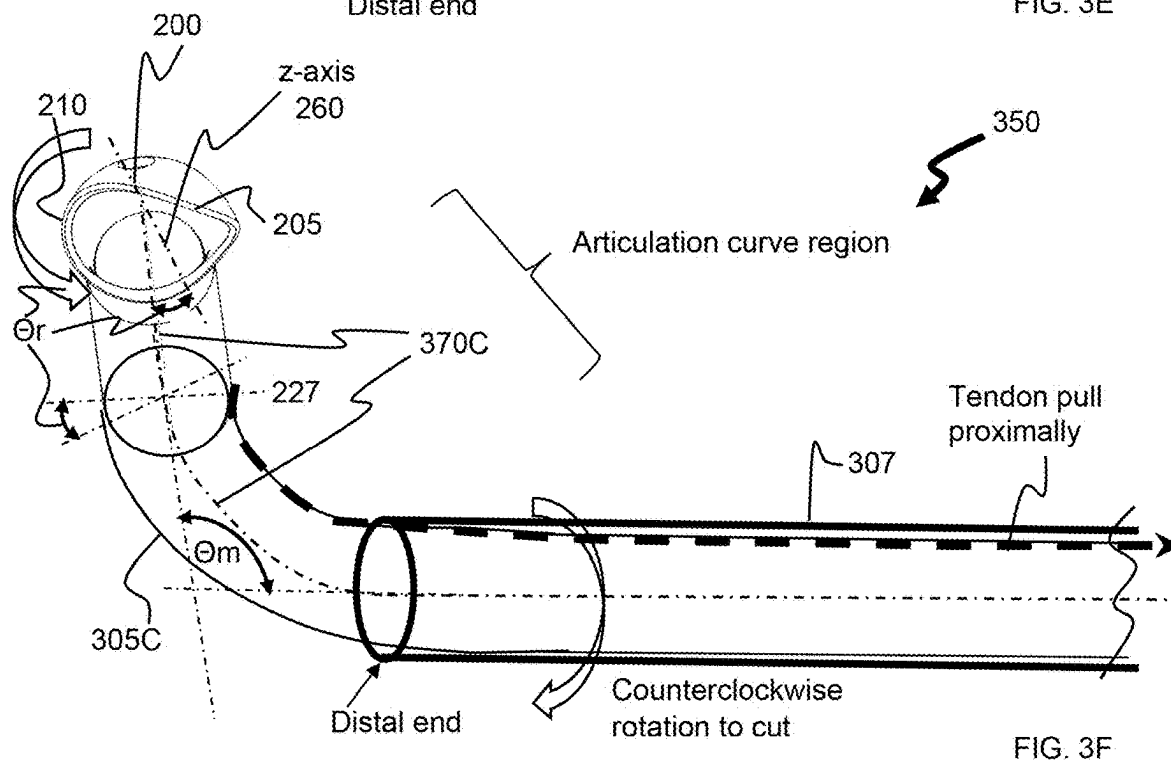
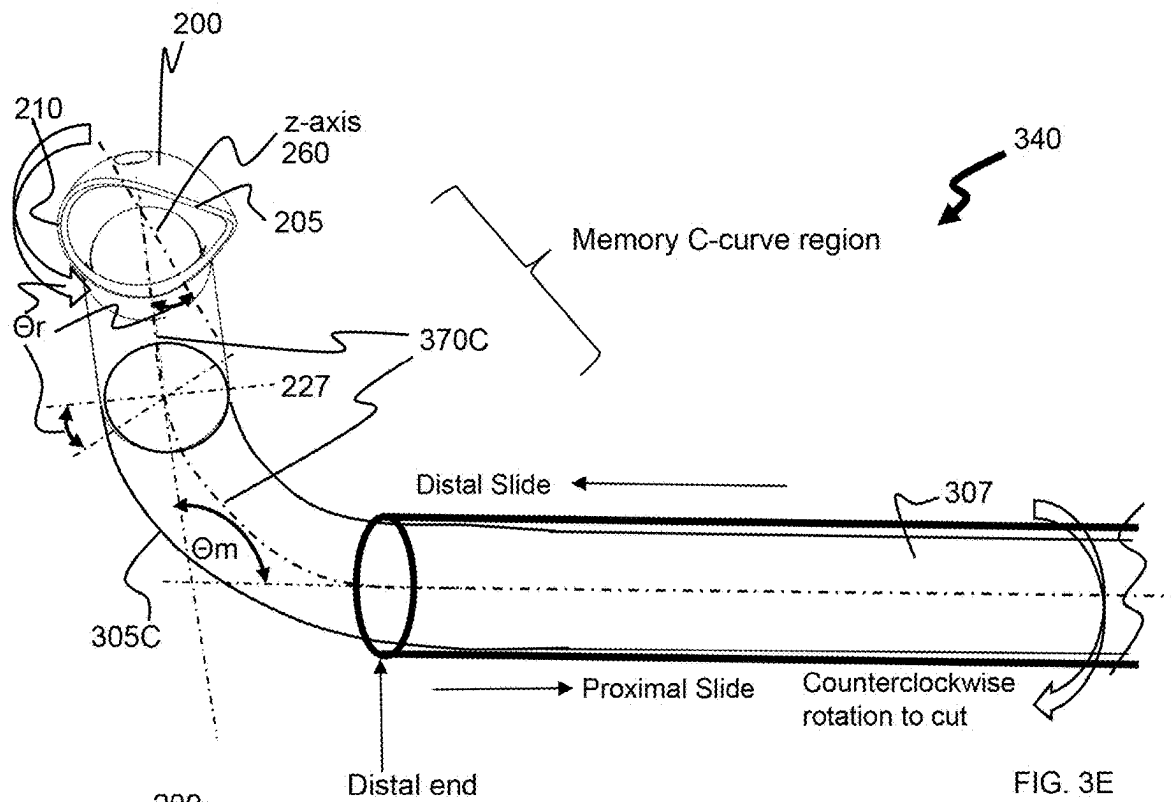
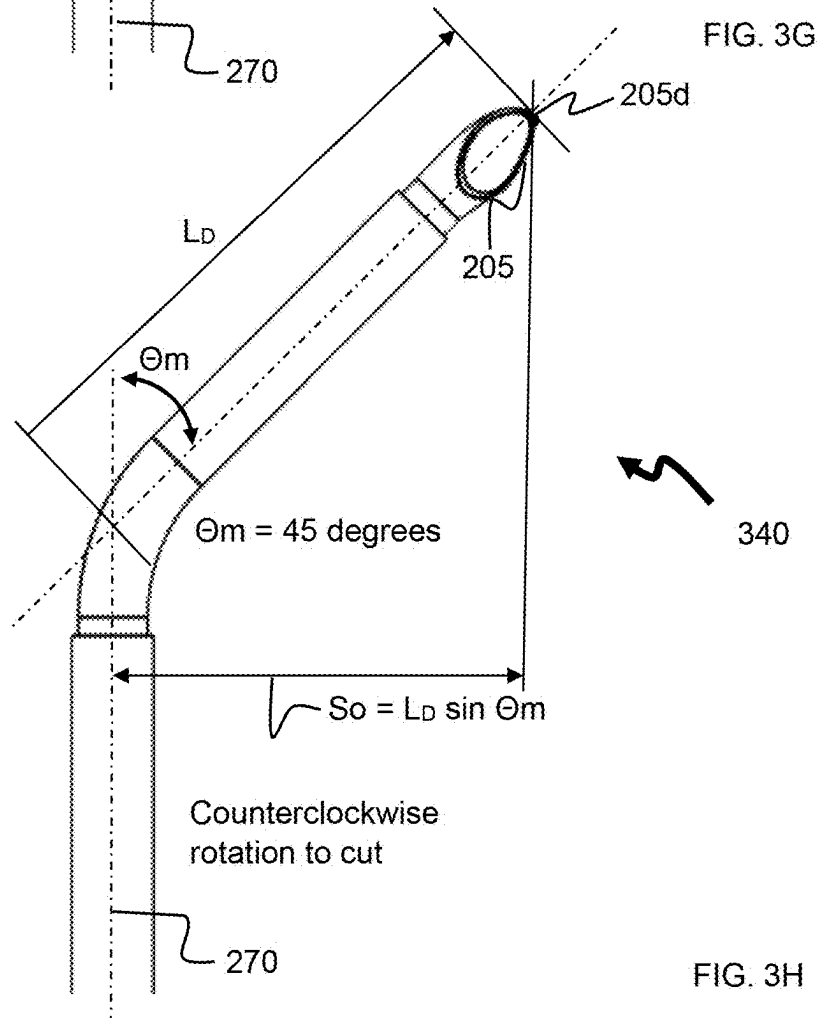
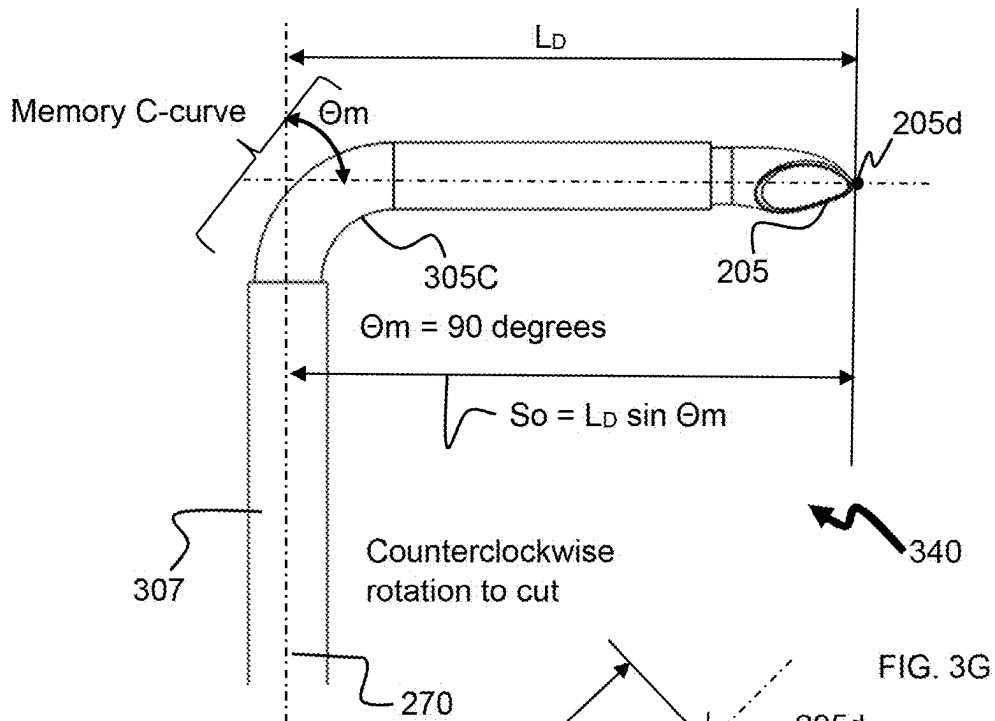
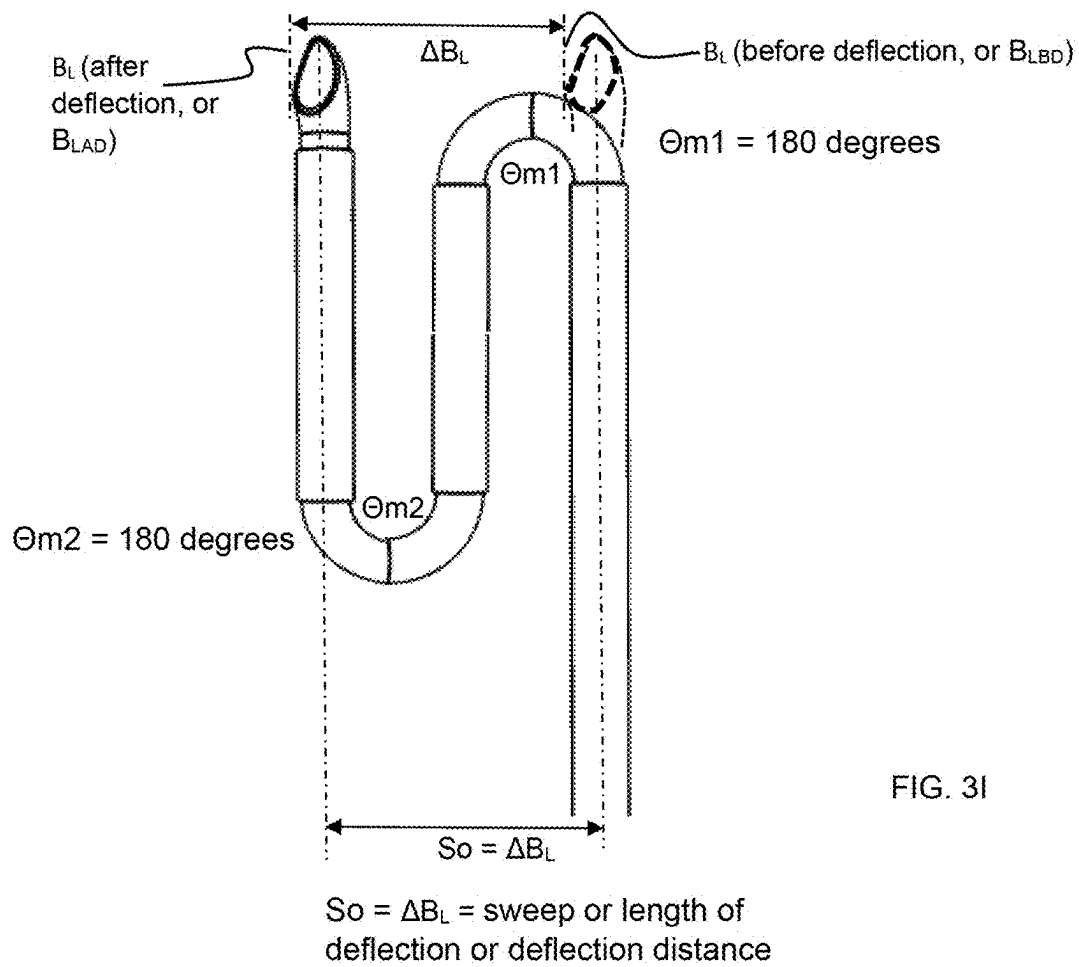


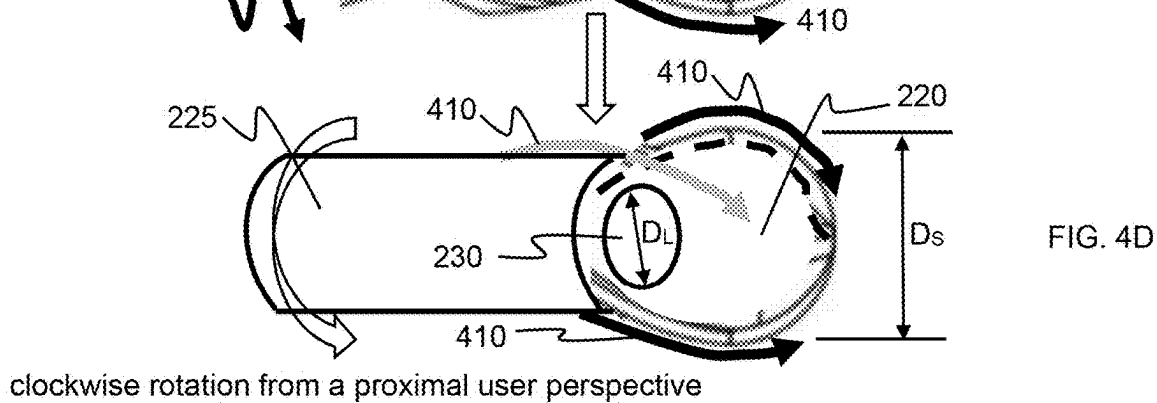
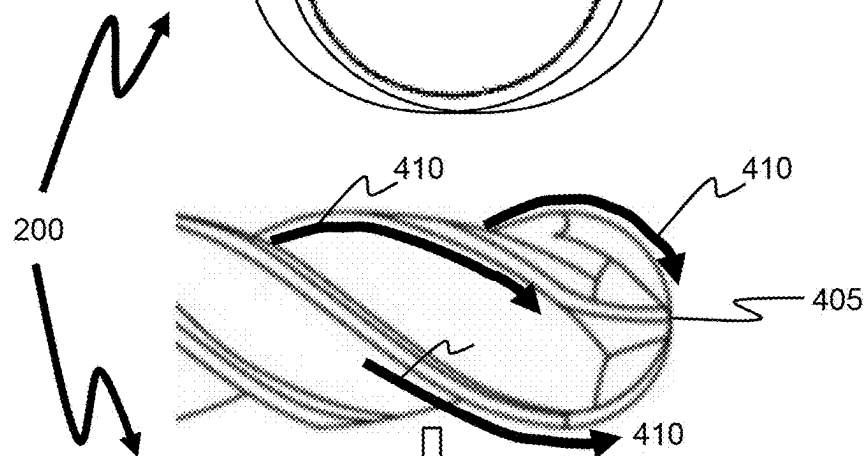
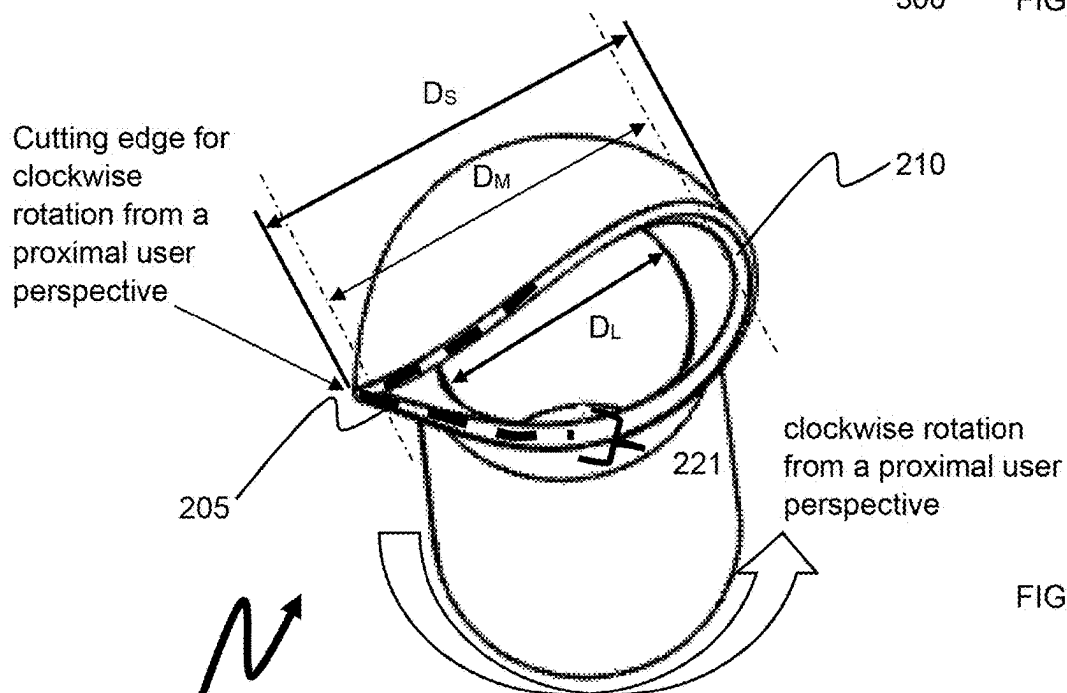
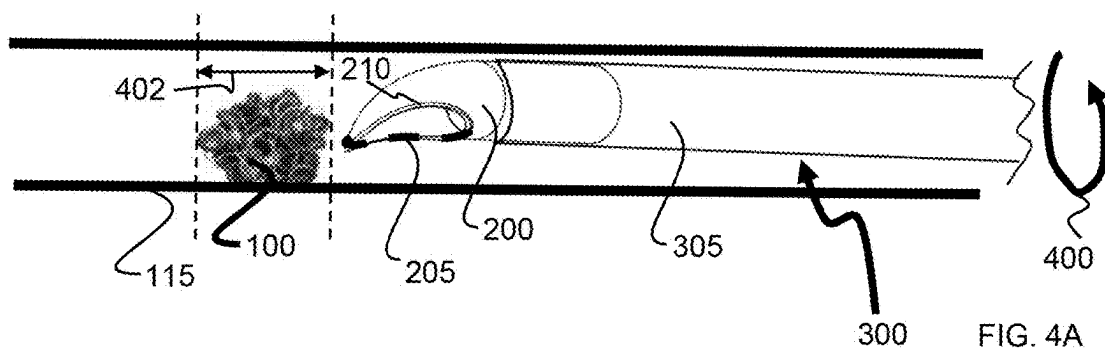
FIG. 2H

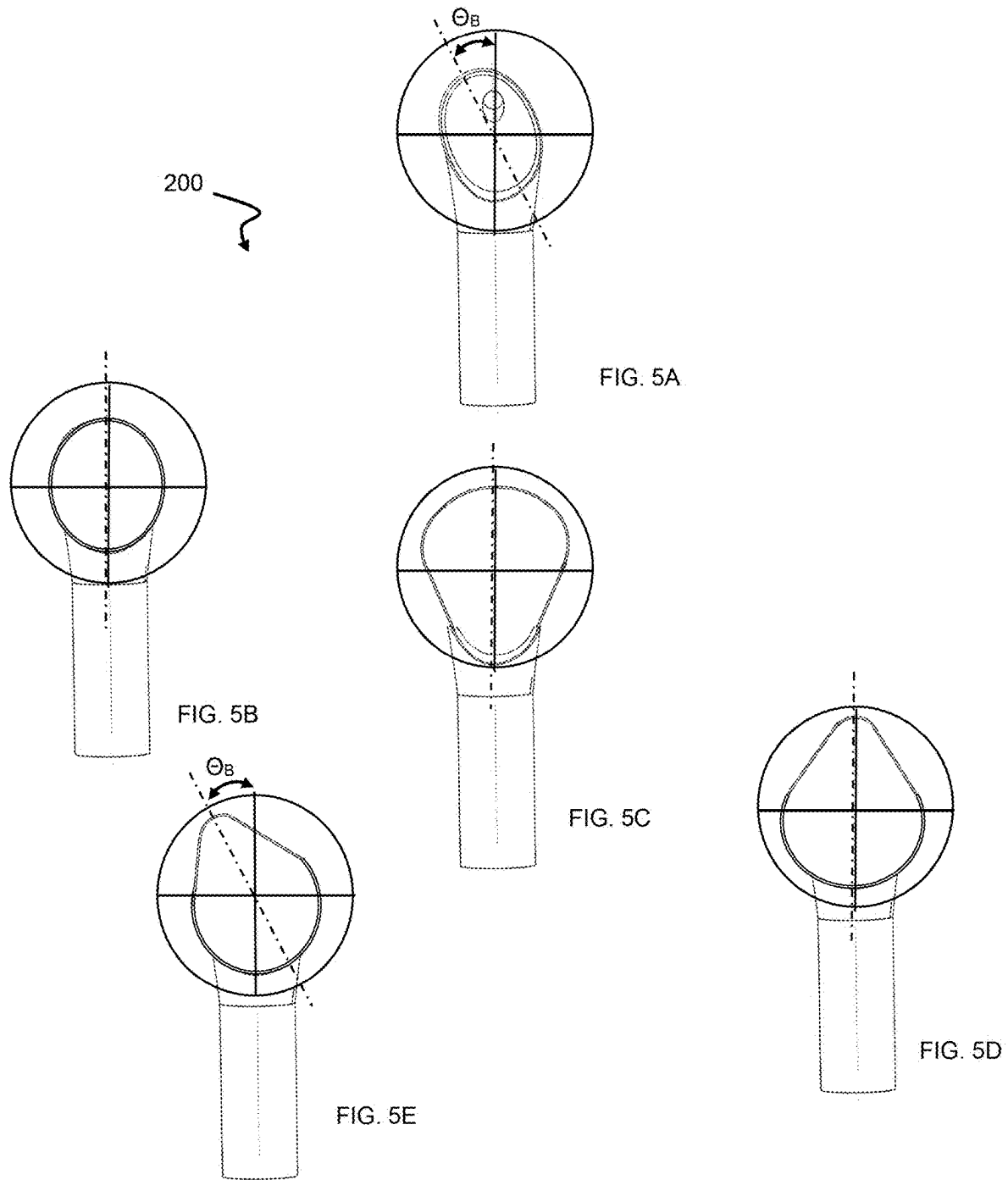


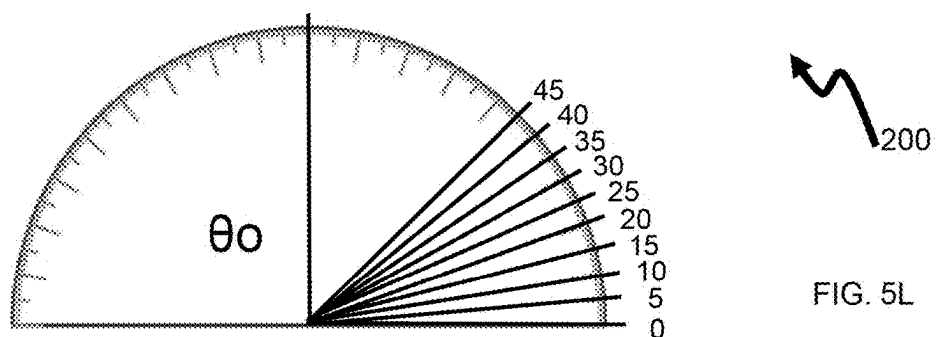
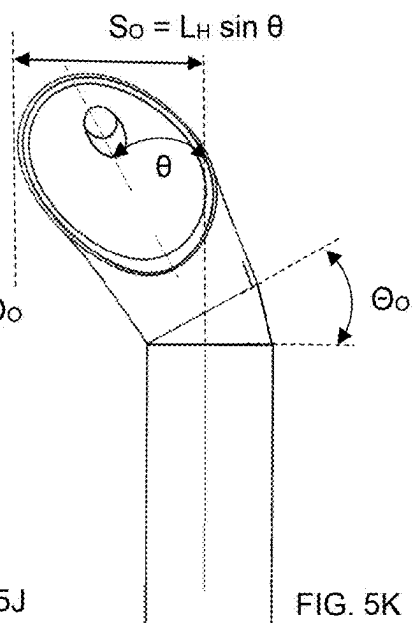
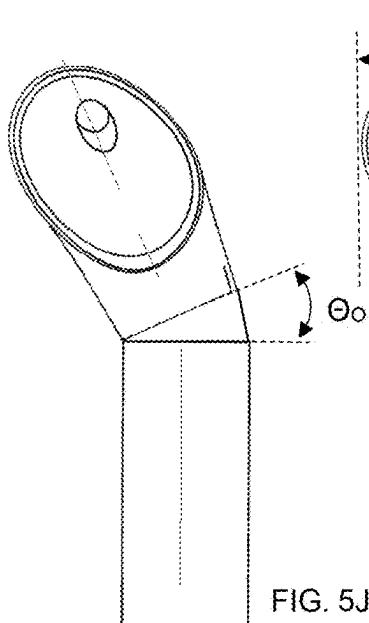
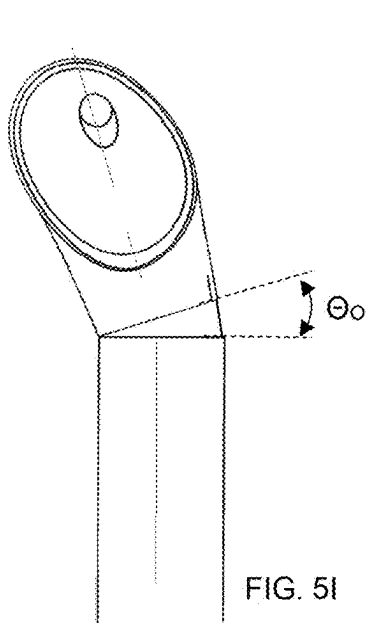
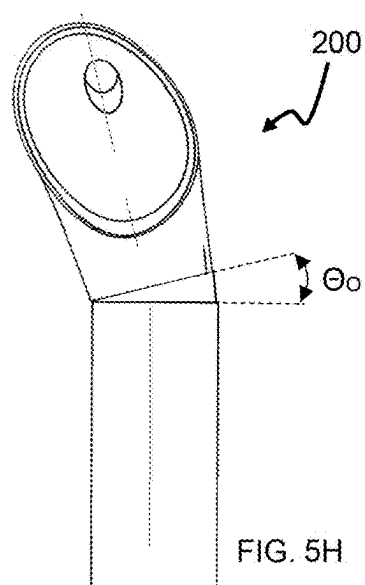
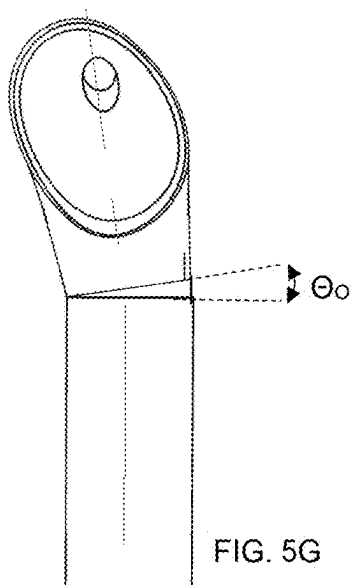
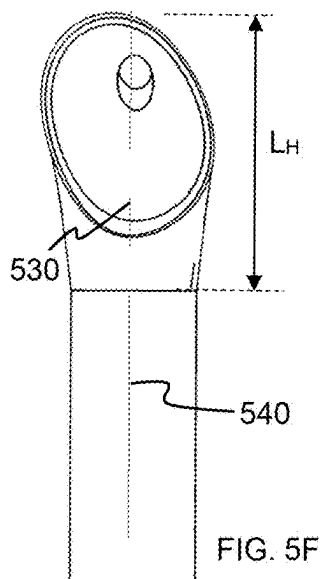












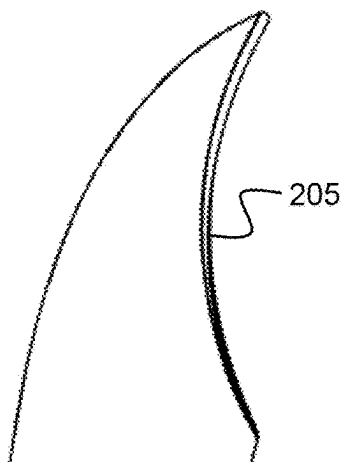


FIG. 6A

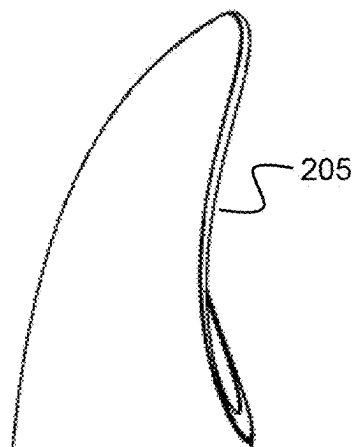


FIG. 6B



FIG. 6C

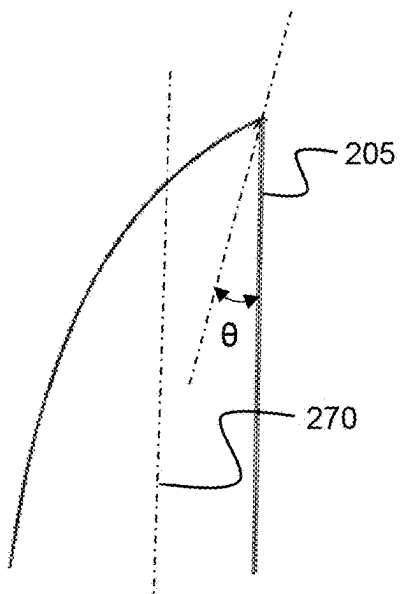


FIG. 6D

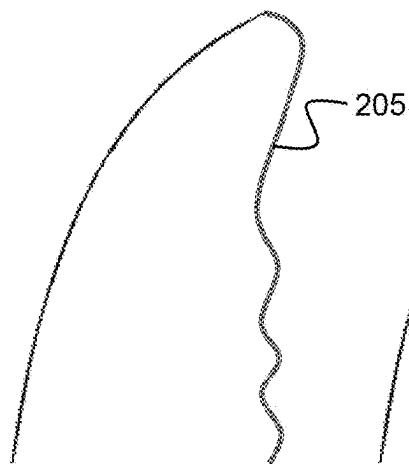


FIG. 6E

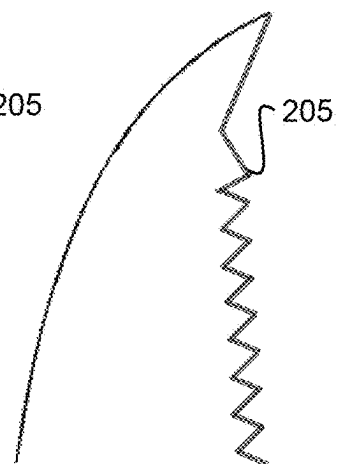


FIG. 6F

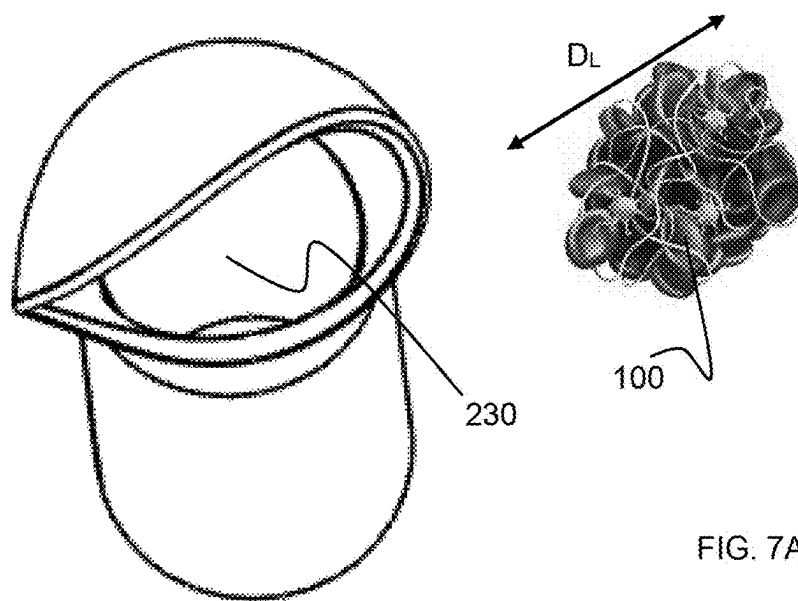


FIG. 7A

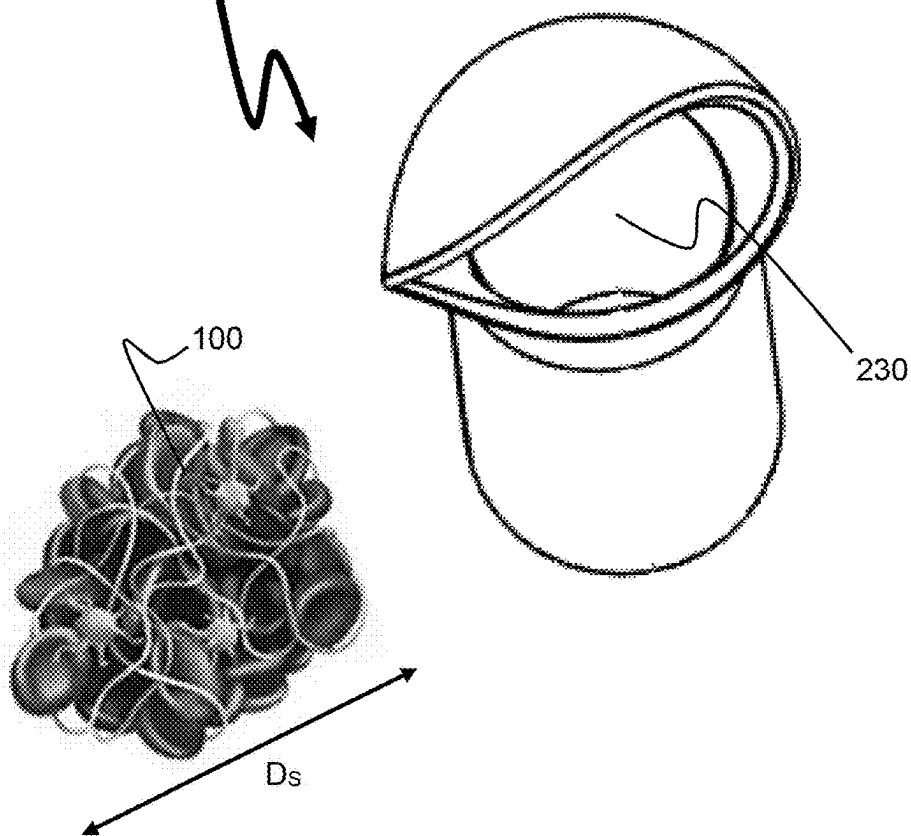


FIG. 7B

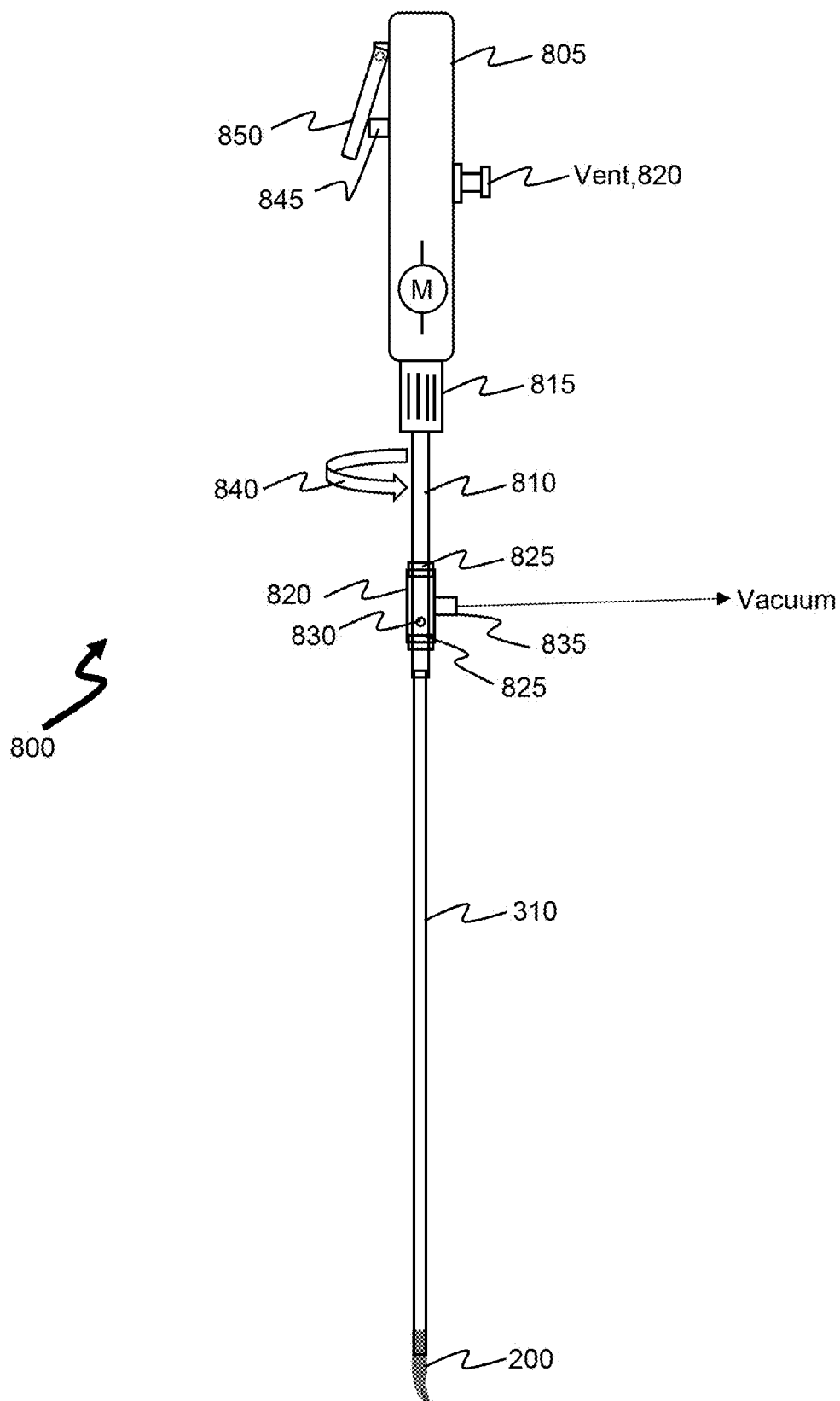


FIG. 8

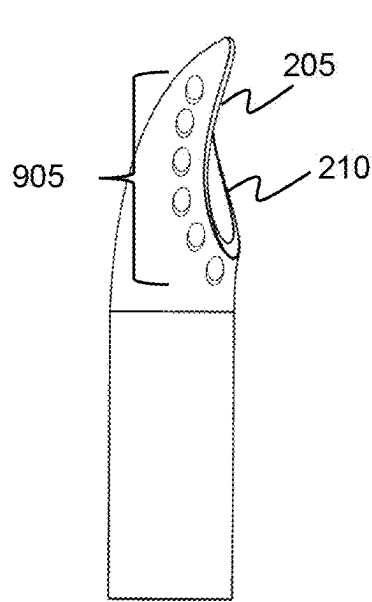


FIG. 9A

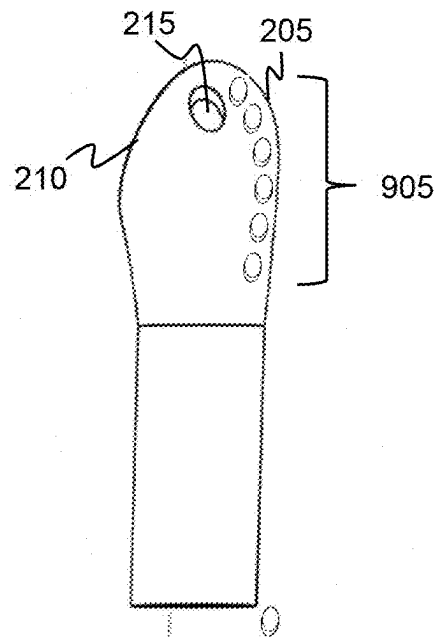


FIG. 9B

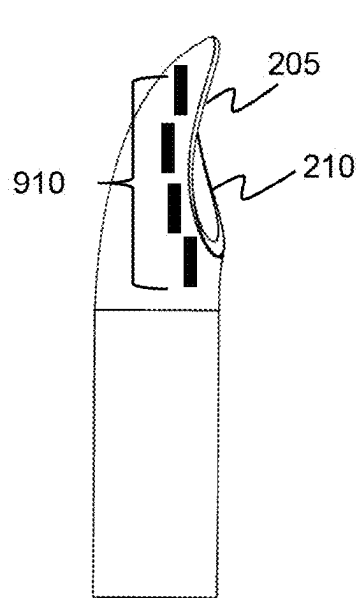


FIG. 9D

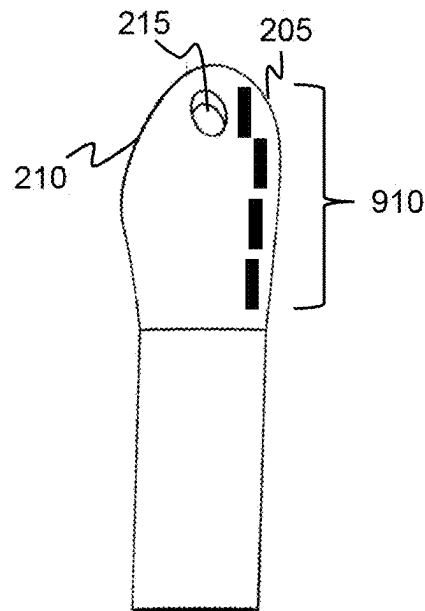


FIG. 9E



FIG. 9F

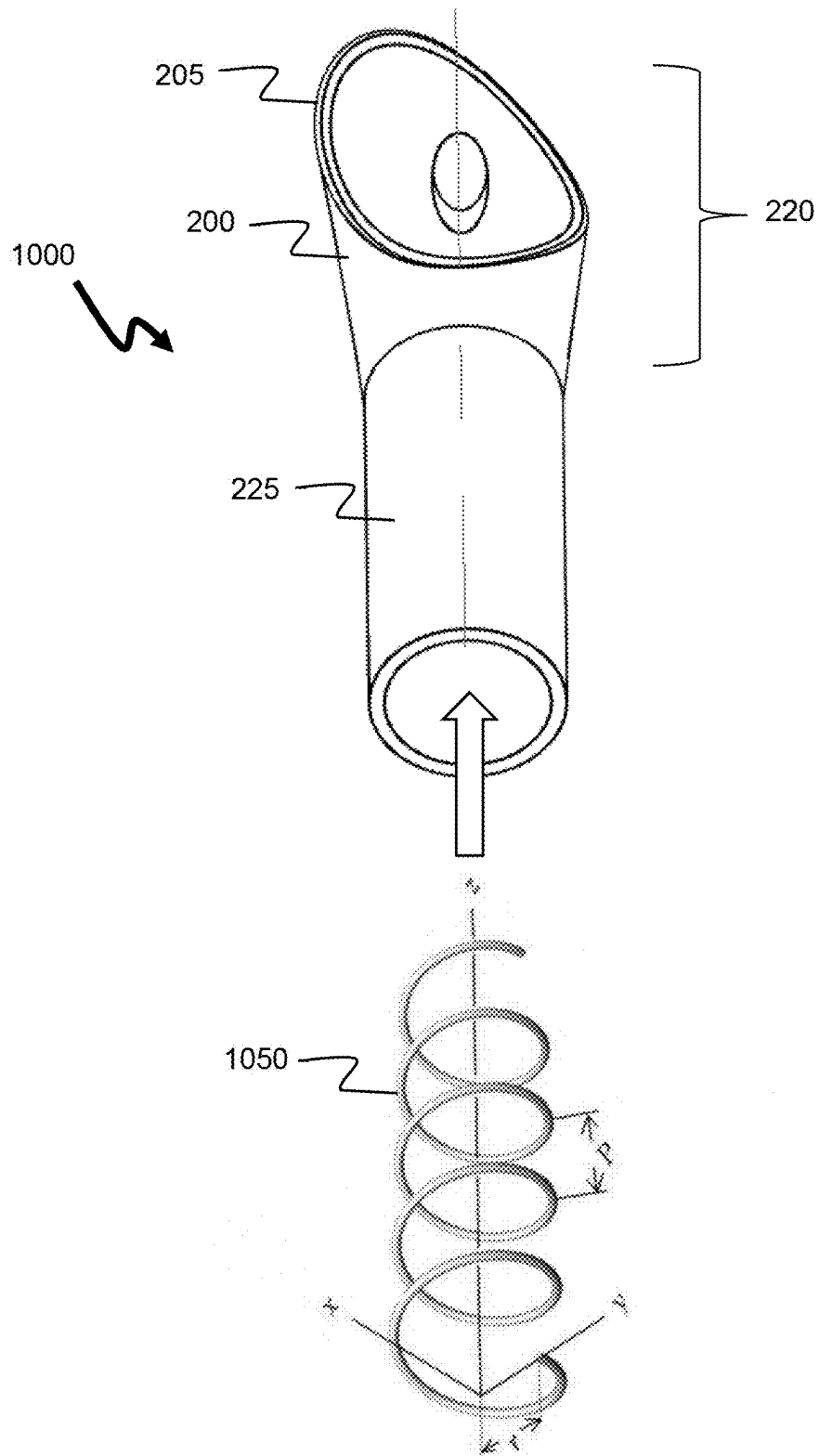


FIG. 10A

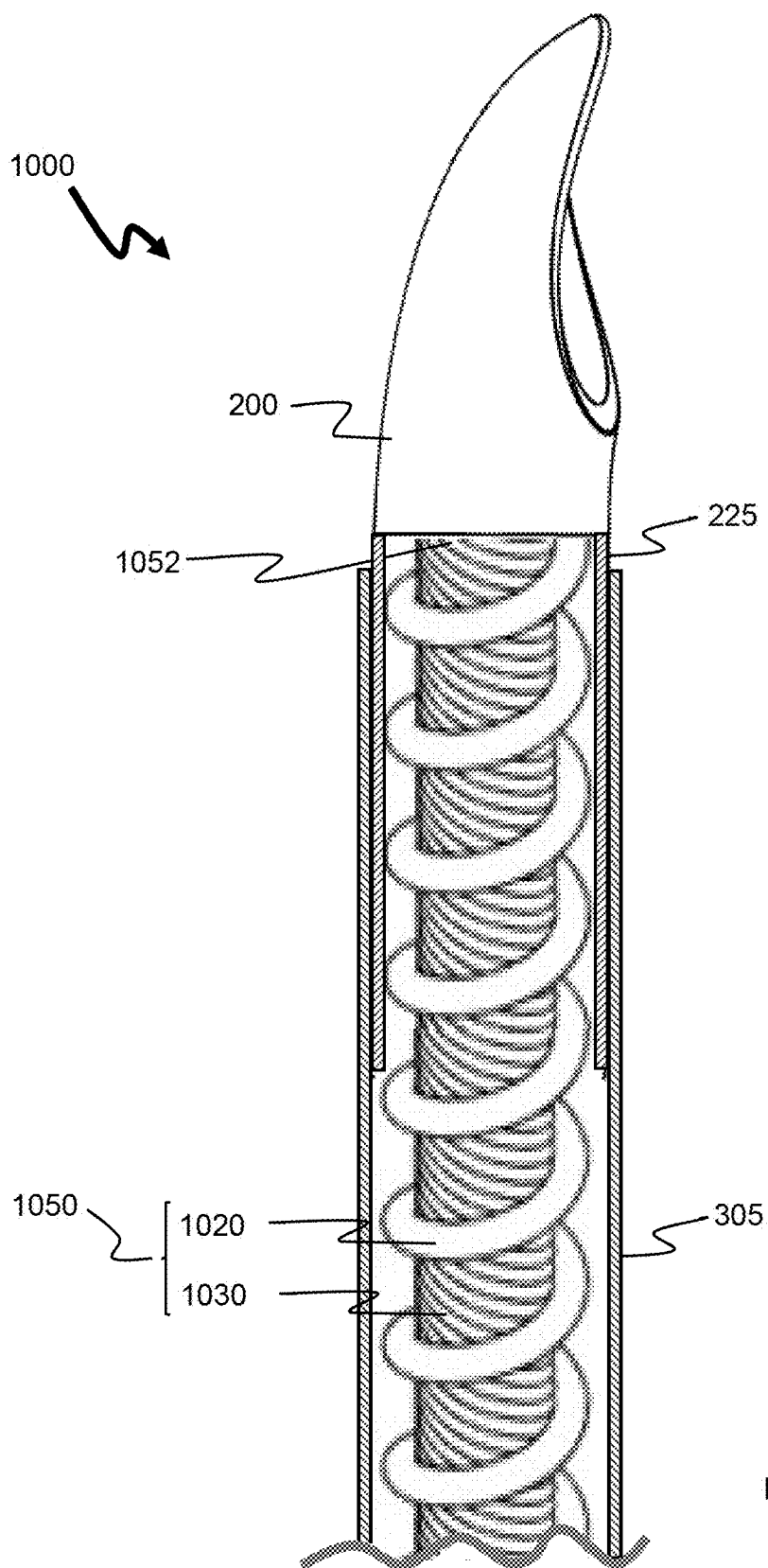


FIG. 10B

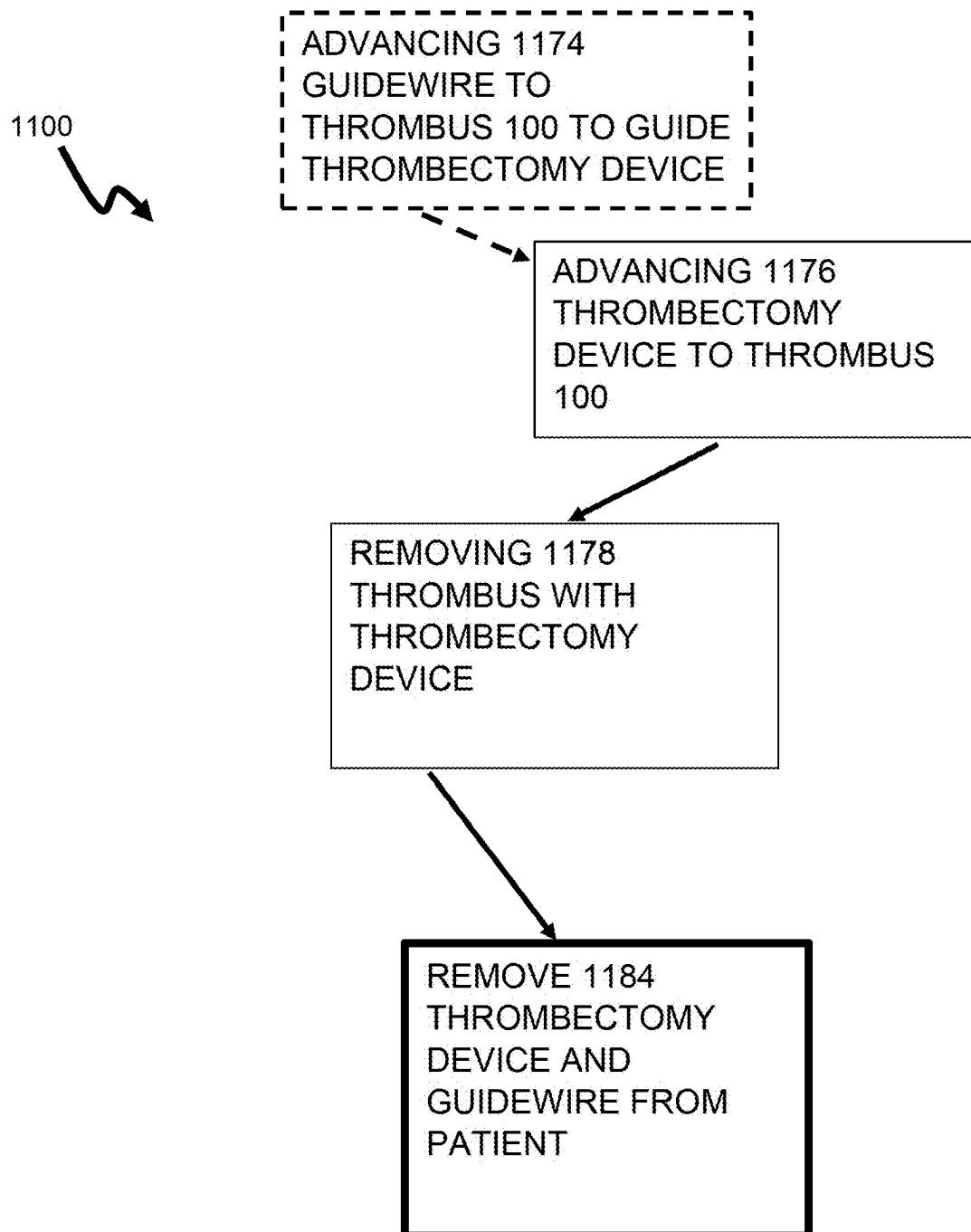
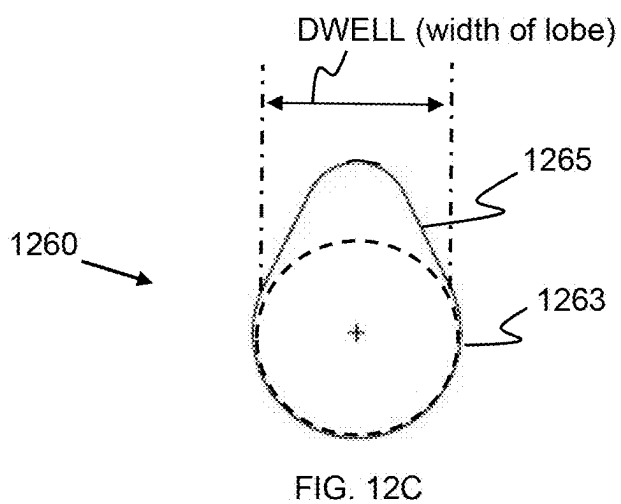
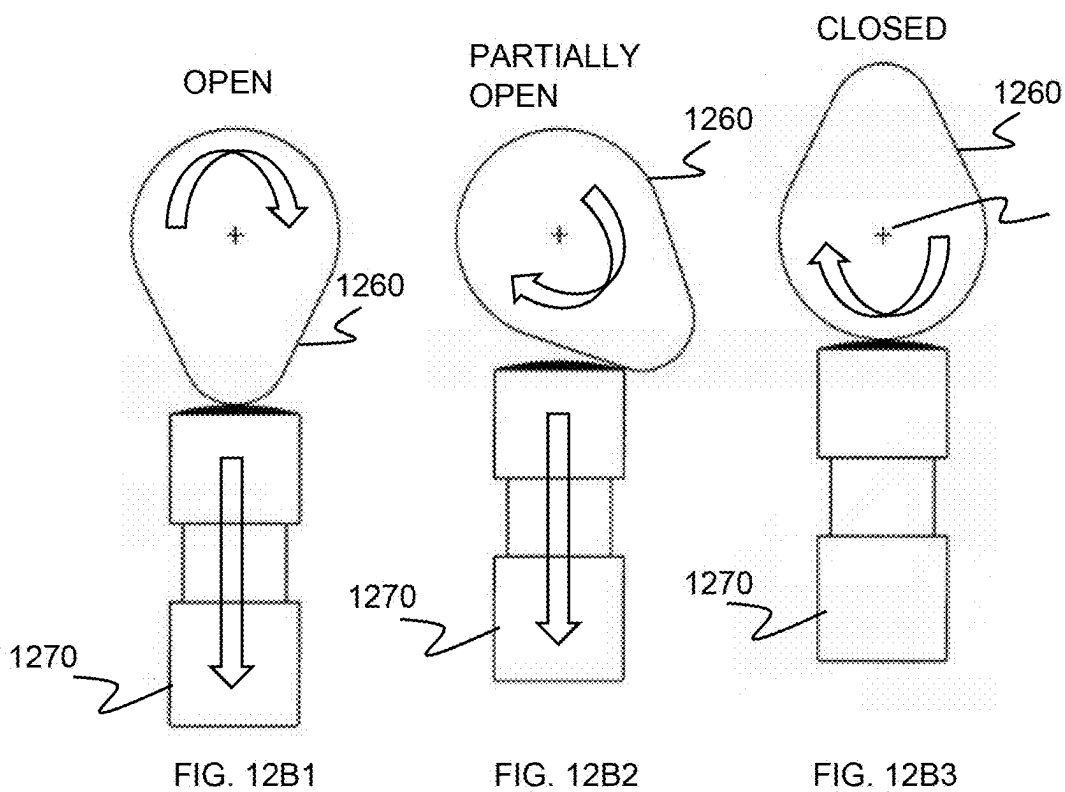
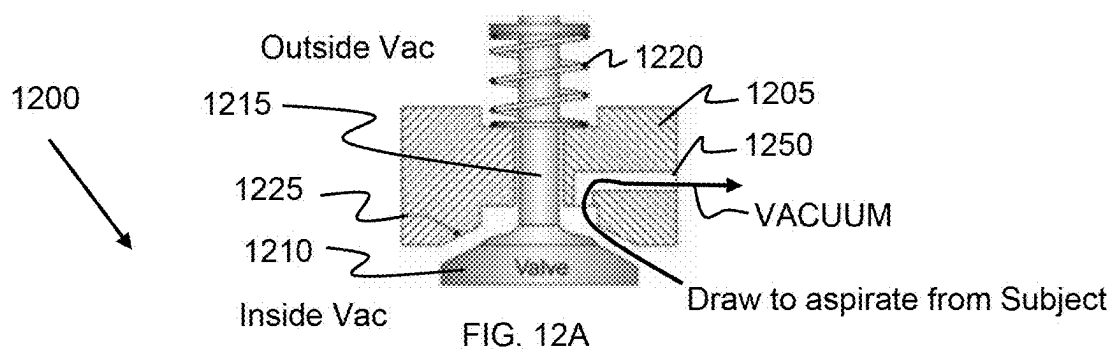


FIG. 11



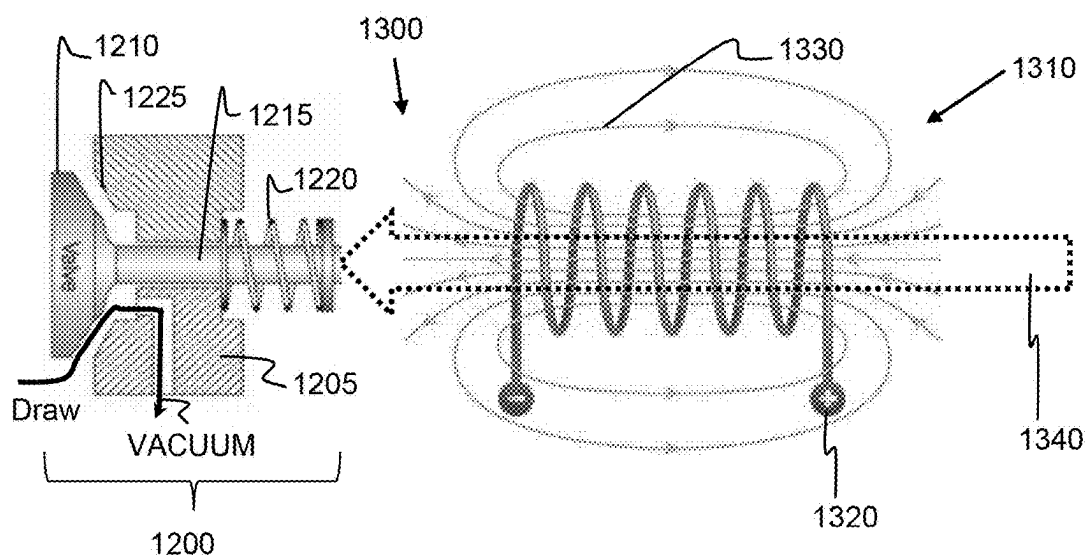


FIG. 13

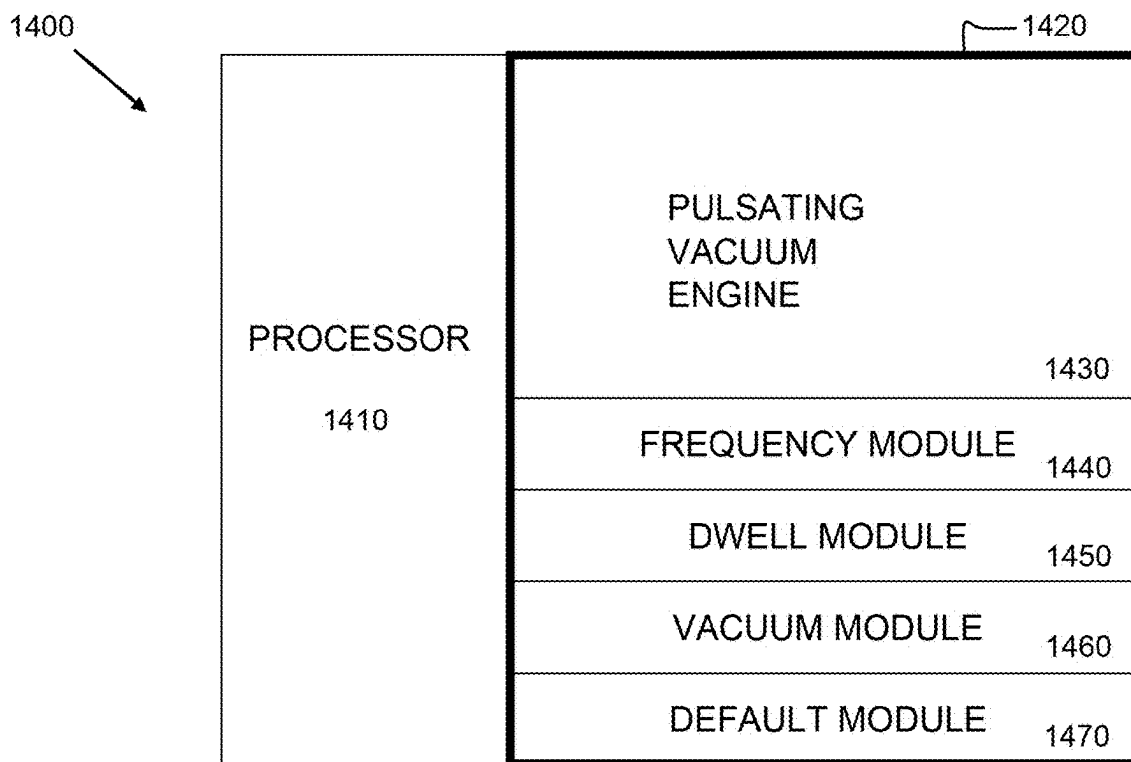


FIG. 14

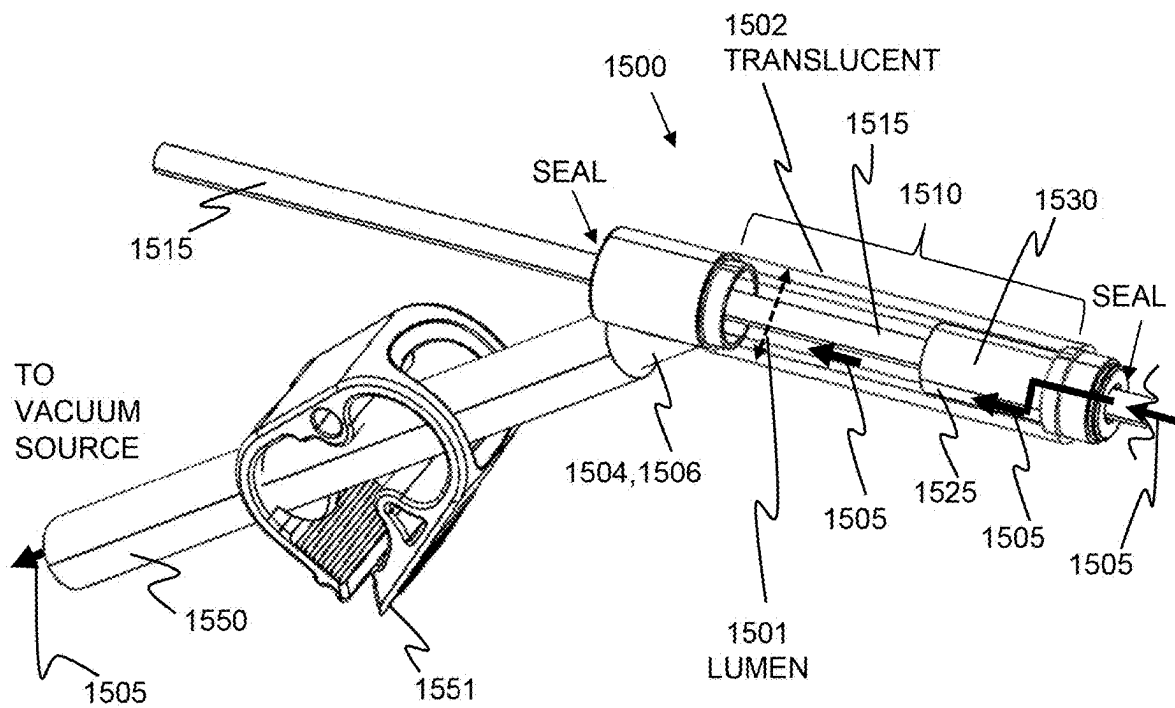
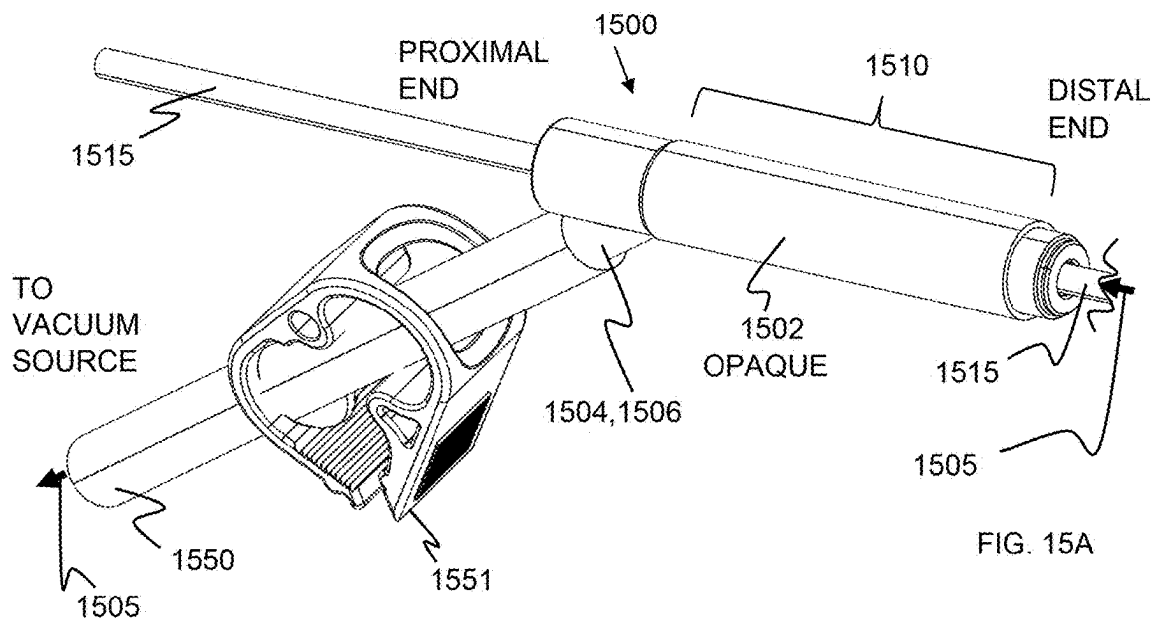


FIG. 15B

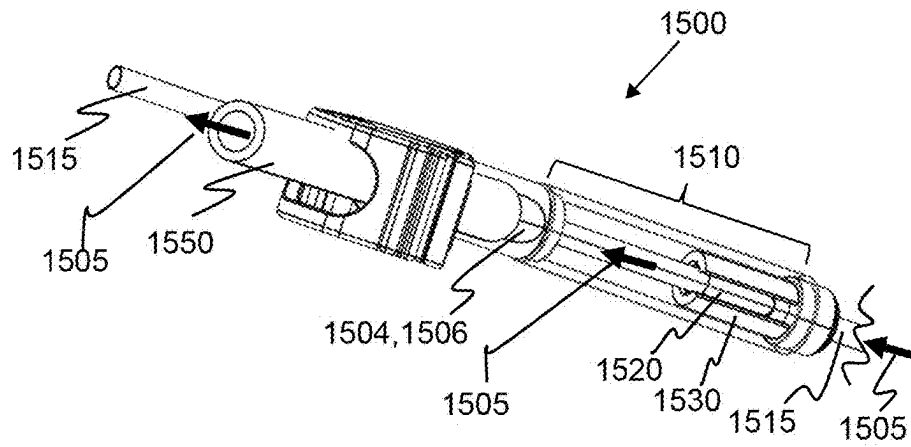


FIG. 15C

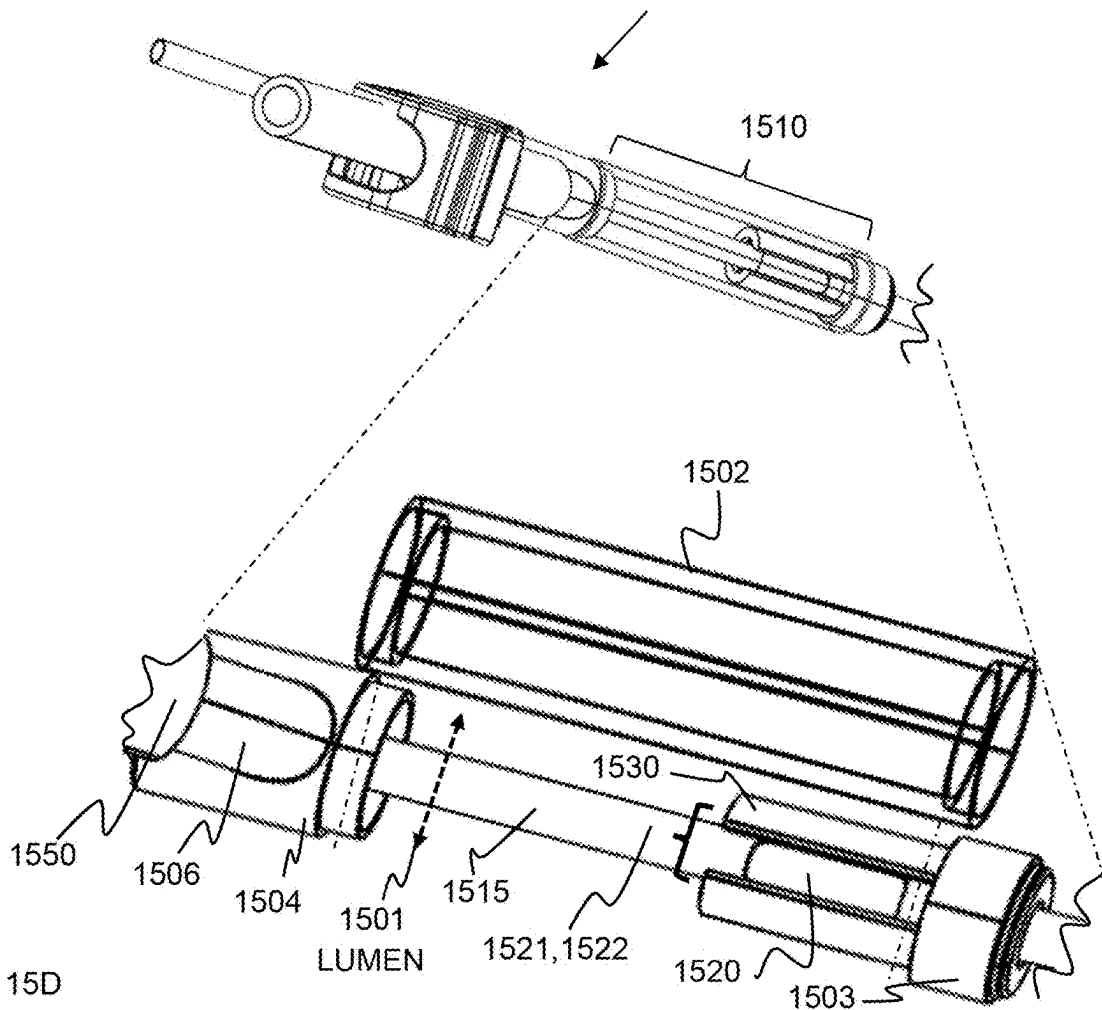
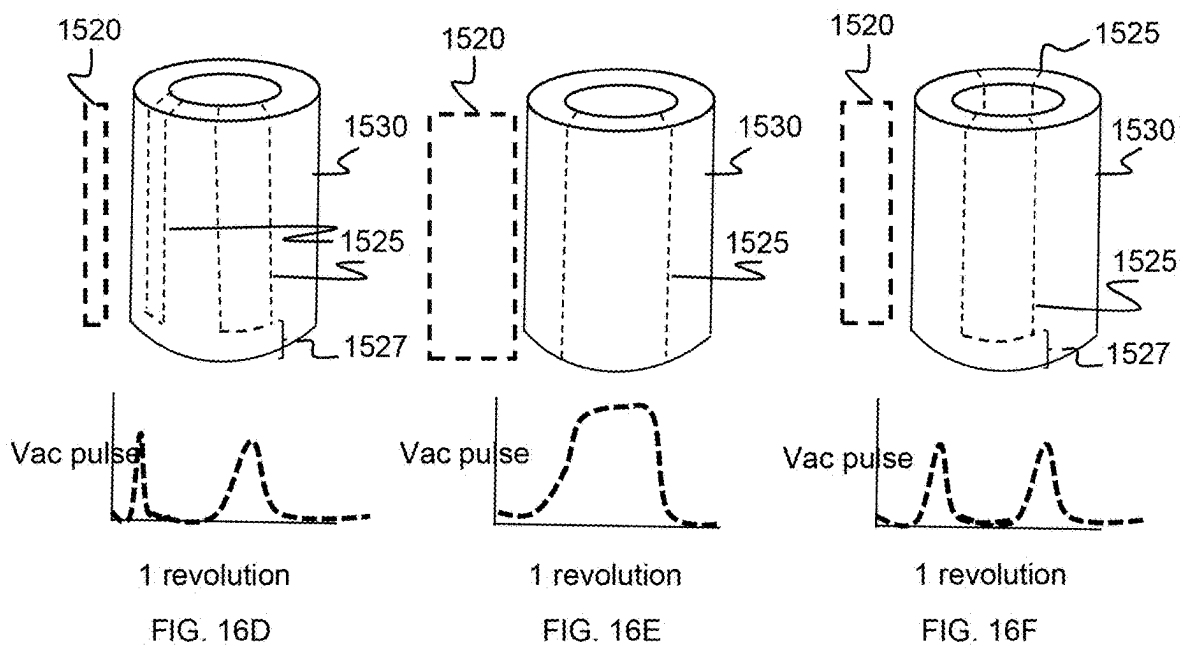
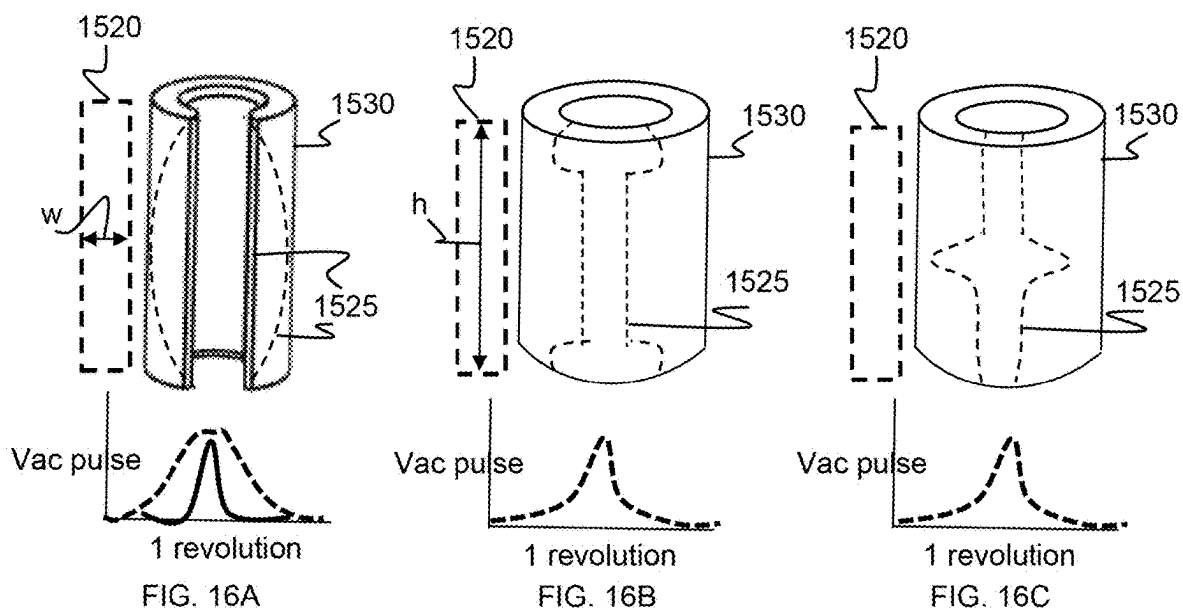


FIG. 15D



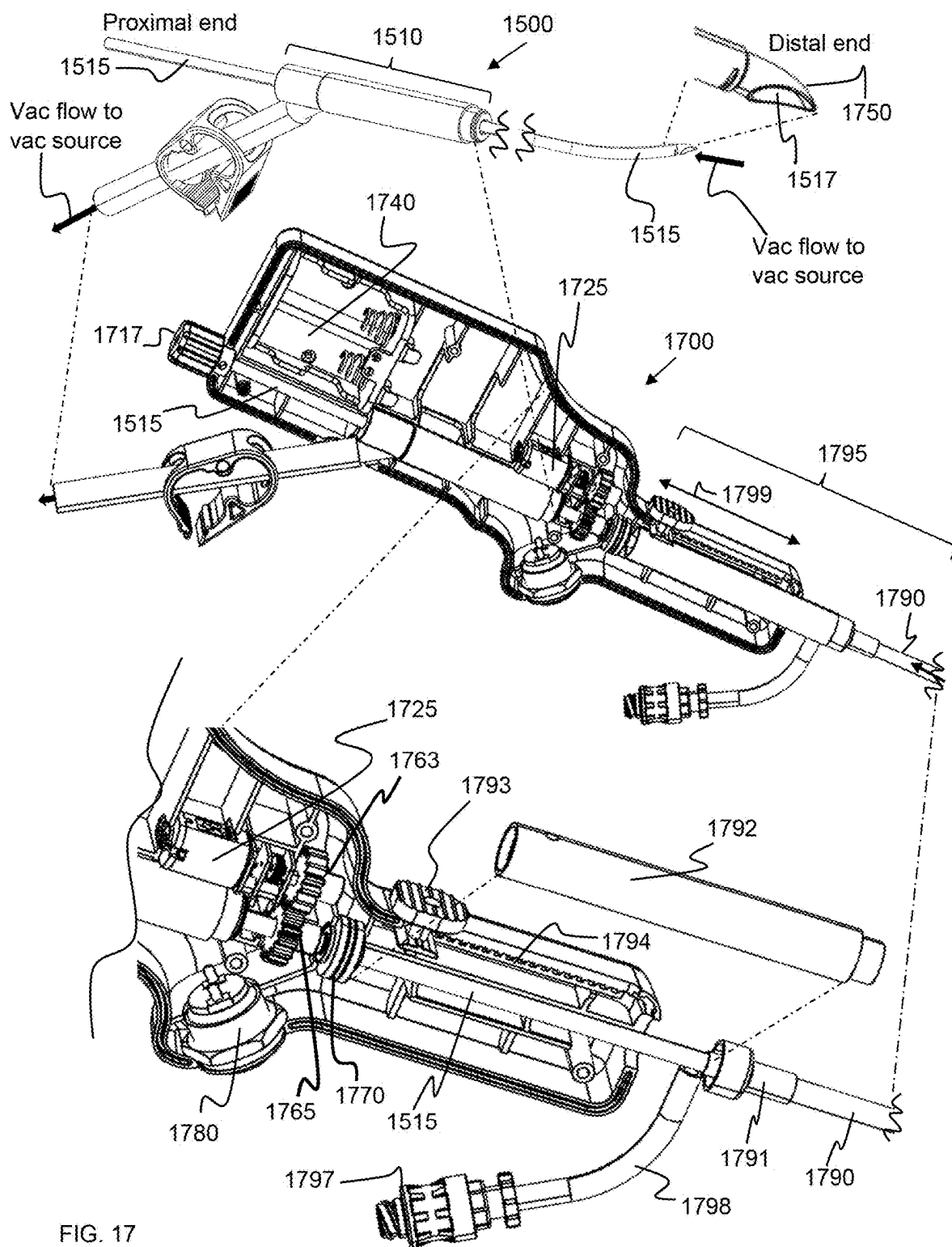


FIG. 17

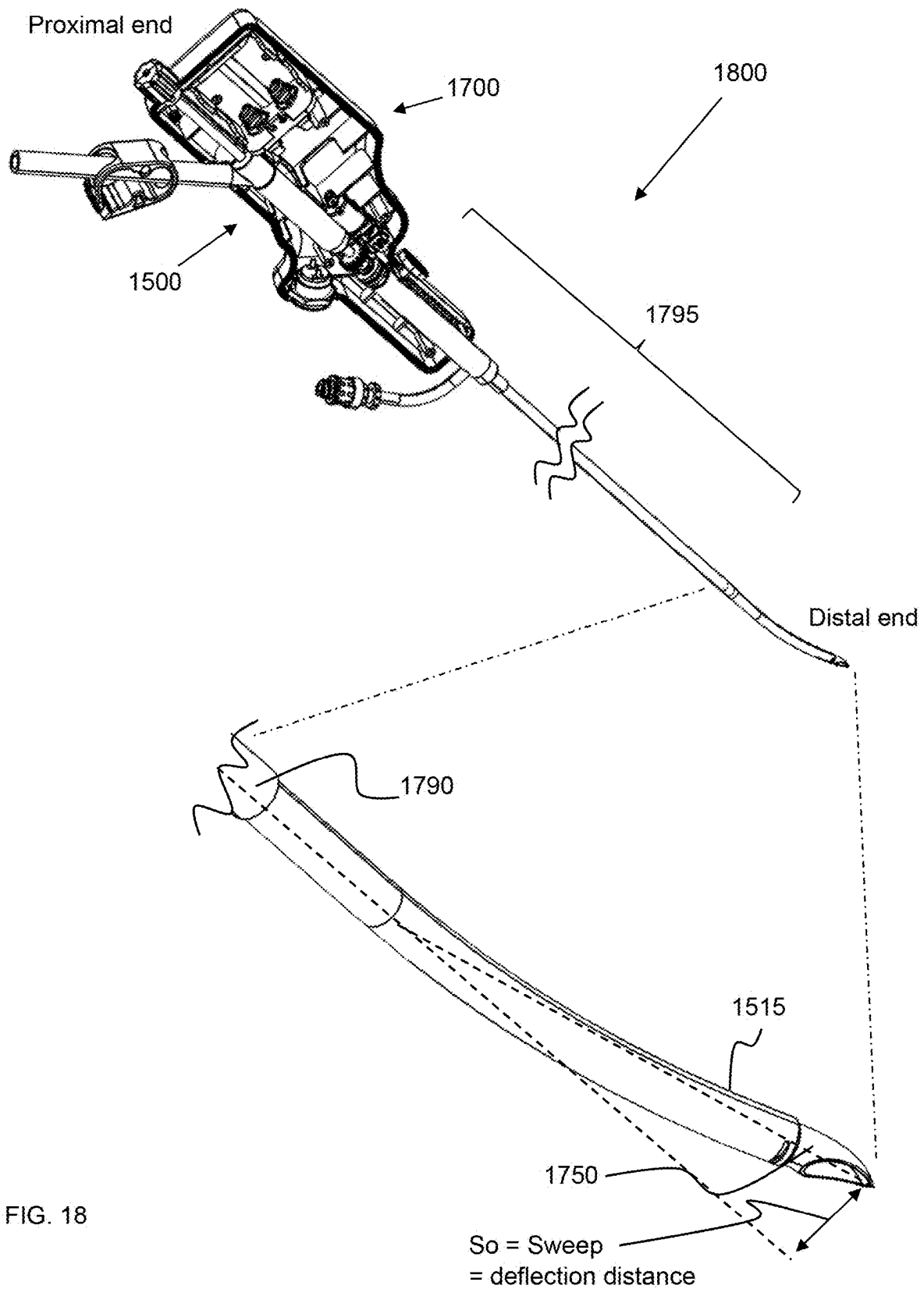


FIG. 18

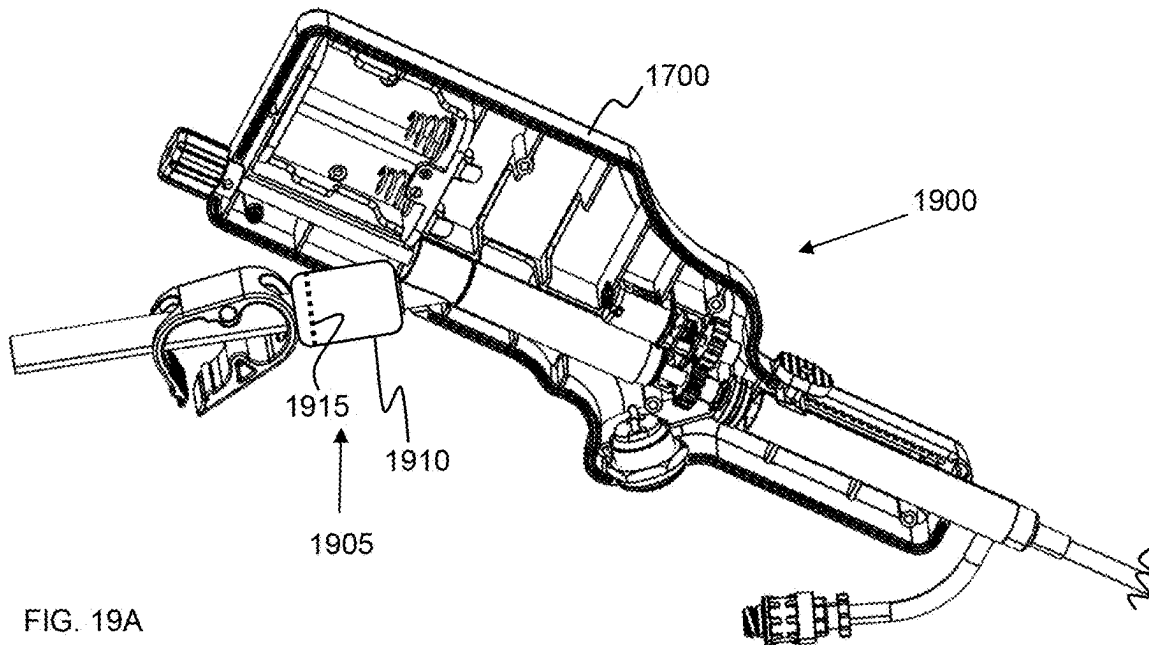


FIG. 19A

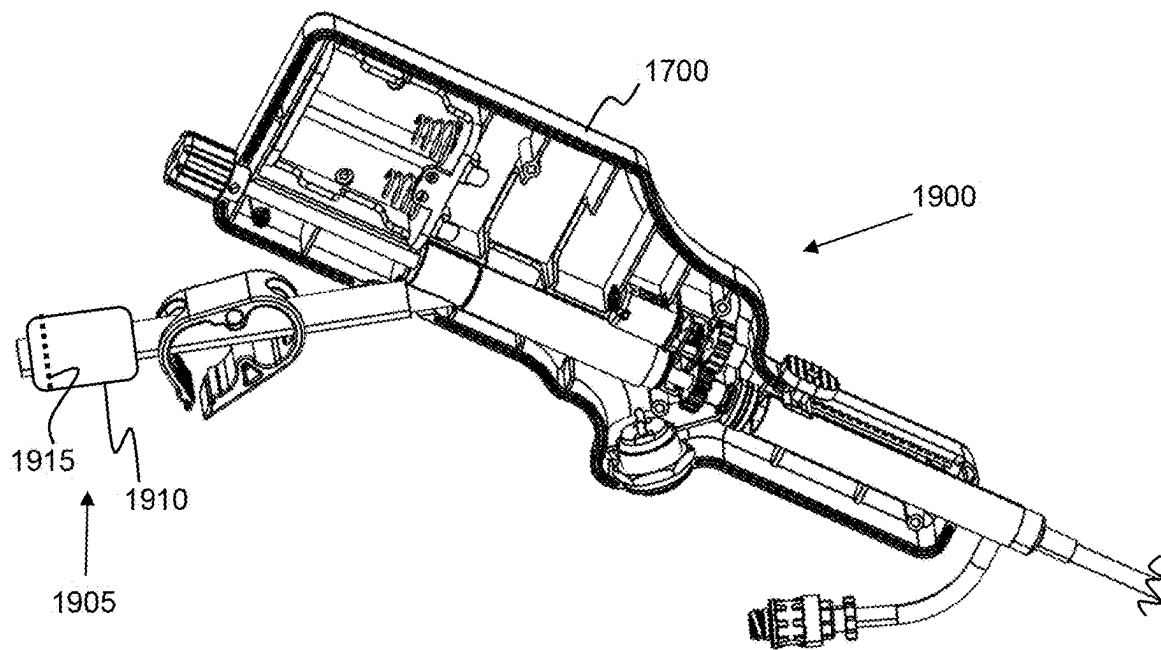


FIG. 19B

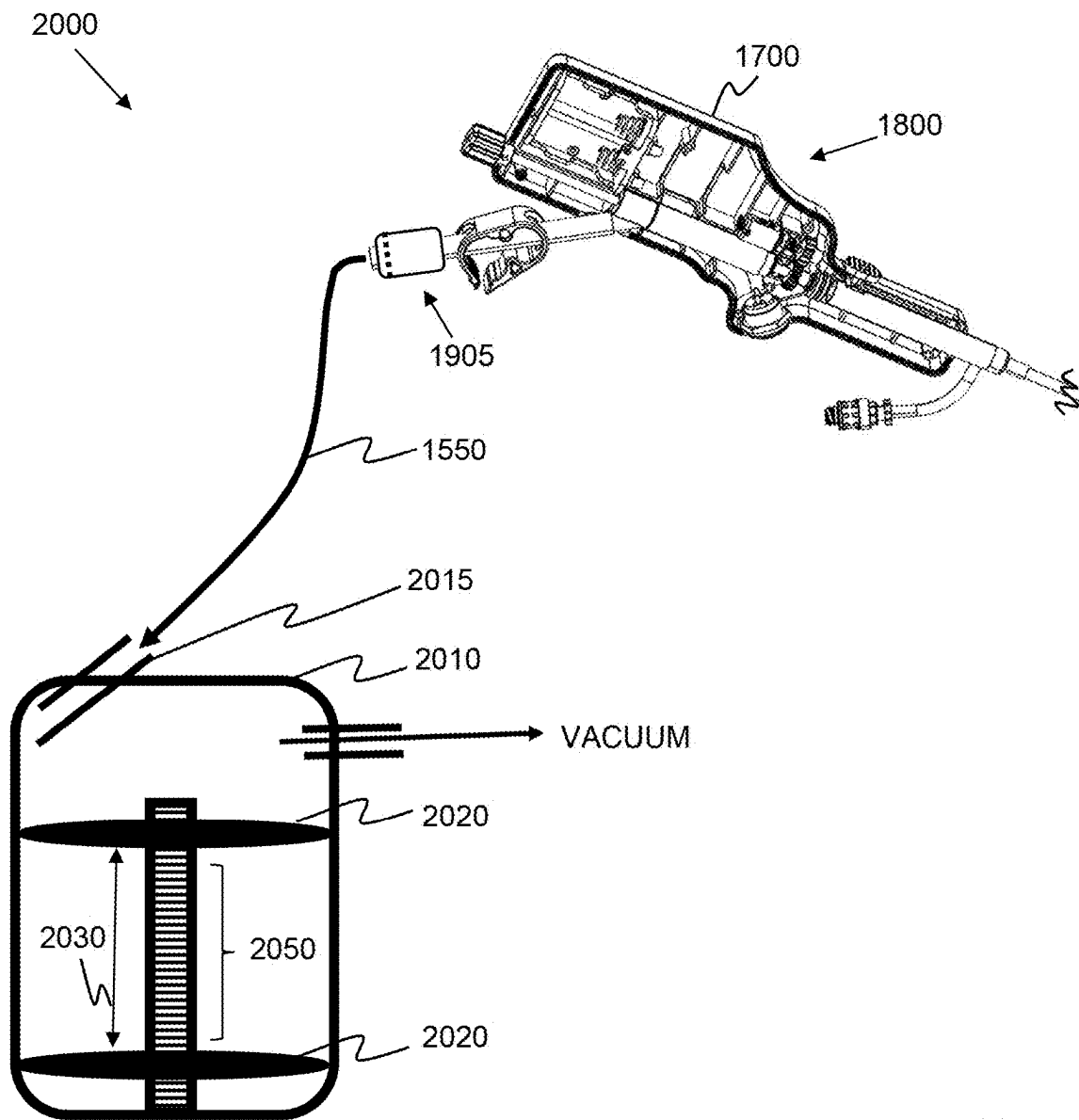


FIG. 20

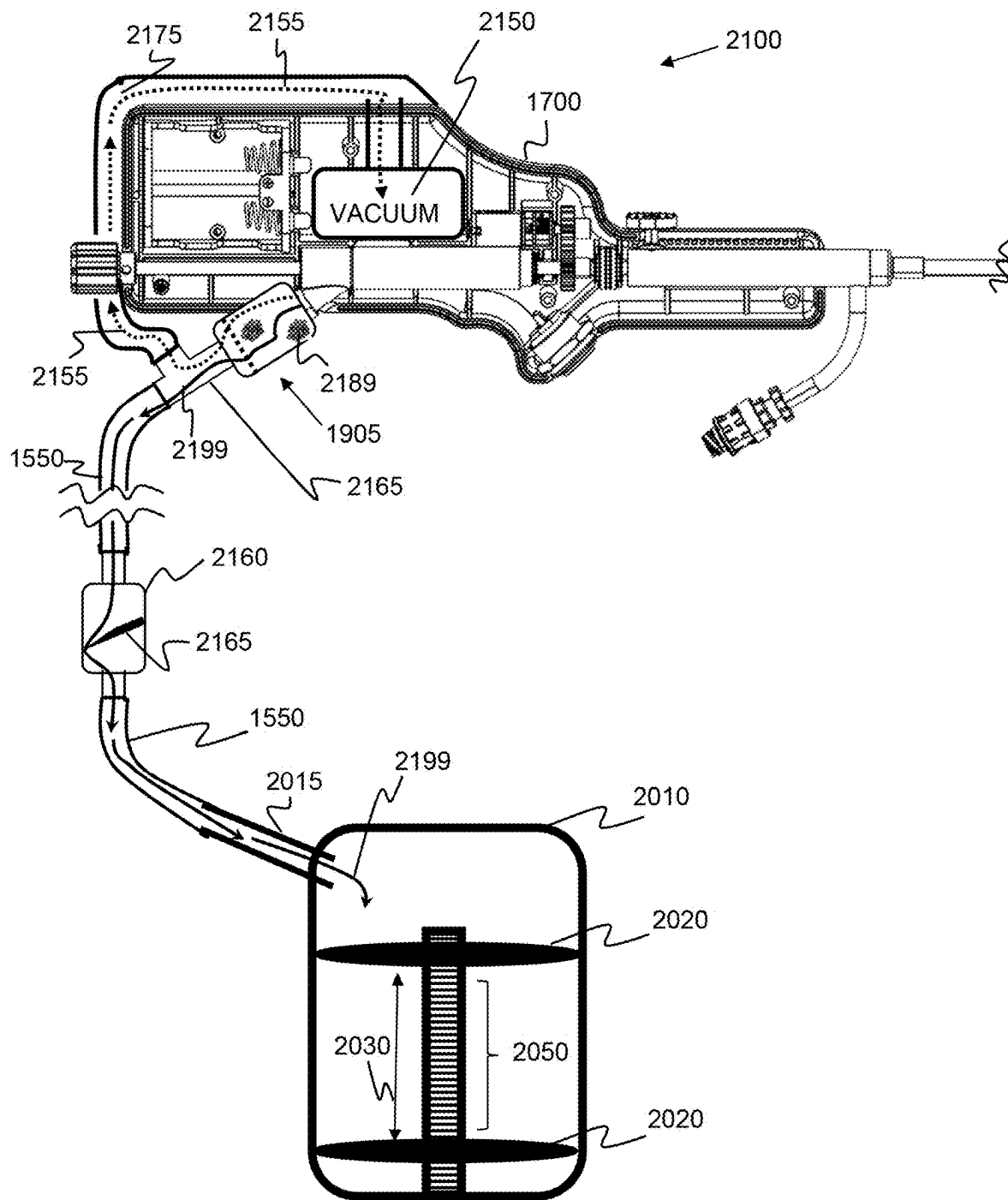


FIG. 21

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CUTTERS WITH PULSATING VACUUM CONTROL

CROSS REFERENCES TO RELATED APPLICATIONS

This application claims no priority benefit.

BACKGROUND

Field of the Invention

The teachings herein are directed generally to medical devices and methods, including devices and methods for performing thrombectomies.

Description of the Related Art

Vascular surgical devices such as vascular cutting devices remove tissue from inside blood vessels and, although they are highly beneficial to patients in need, they currently create a risk of suffering vessel injuries and an undesirable volume of blood loss during the surgical procedures. A vascular cutter procedure can potentially cause blood vessel damage, primarily due to the mechanical interaction of the device used to remove a target tissue from the inner lining of the blood vessel. Such damage can include tears, abrasions, or even perforations, particularly when dealing with delicate or smaller vessels. Complications associated with vessel damage include bleeding due to a vessel tear or perforation, hematoma formation around the puncture site, and vessel narrowing (stenosis) from scarring and narrowing of the blood vessel lumen. Blood loss is associated with the vacuum aspiration used to remove the target tissue.

A thrombectomy, for example, is the removal of a blood clot from a blood vessel. Blood clots are made up of platelets and a meshwork of protein strands called fibrin. Common applications for thrombectomies include any location in a blood vessel where a thrombus may occur. Clots in arteries have a different composition than clots in veins, in which clots in arteries contain mostly platelets and clots in veins contain mostly fibrin. The thrombectomy removes the blood clot from the wall of a blood vessel to help alleviate symptoms of the condition, as well as the downstream complications that can include further vasculature complications, and perhaps even death from the release of an embolus.

State-of-the-art thrombectomy devices leave much to be desired, as many problems remain to be solved in the art, including (i) dragging expanded baskets along veins which can injure the vein walls or valves, (ii) clogging suction in the operation of the devices; (iii) removing too much blood while removing the thrombus; (iv) failing in the removal of tough or mature thrombi; and (v) failing to be versatile enough to handle each of the variety of soft, tough, fibrous, and hard tissue effectively, either not cutting all types of tissue or breaking up the tissue into pieces that can become dangerous emboli that can clog blood vessels downstream and, in some cases, cause death. As discussed above, there are also the problems of (vi) an undesirable volume of blood loss during procedures, and (vii) an undesirable contact between the cutter and untargeted vascular structures causing injury to the vessel during procedures.

A versatile device that can handle each of the variety of soft, tough, fibrous, and hard tissue effectively, can also be useful in removing an atheroma from the lumen of an artery while avoiding an undesirable loss of blood and an unde-

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sirable contact with untargeted vascular structures. Such devices can also be designed for atherectomies, for example, to treat an atherosclerosis, also referred to as plaque on the luminal wall of an artery, composed of deposits of fatty substances, cholesterol, cellular waste products, calcium, and fibrin. Both the atherosclerosis and the thrombus pose a risk of fragmenting into the blood stream, and moving to the heart, brain, or lungs, causing health complications and often proving to be fatal.

One of skill will appreciate the devices and systems taught herein that, at the least, (i) can effectively cut and remove all types of thrombus and atheroma tissue, whether soft, tough, or hard; (ii) can safely collect and remove such while avoiding the release of tissue fragments, such as emboli; (iii) can effectively treat a blood vessel with a reduced risk of suffering vessel injuries that can, for example, lead to increased stenosis, and (iv) can avoid an undesirable volume of blood loss during procedures. Of particular interest herein, among the several indications taught for the technologies, is the removal of a venous thrombus, and the removal of a pulmonary arterial thrombus, also referred to as a pulmonary embolism.

SUMMARY

Thrombectomy systems, devices, and methods of using them are provided. Several embodiments are taught for the technologies herein, and these embodiments include systems, methods, and devices for (i) the removal of arterial and venous thrombus, and (ii) the removal of a pulmonary arterial thrombus, also referred to as a pulmonary embolism. In some embodiments, the systems, devices, and methods taught herein can be used in atherectomies. The technologies taught herein can effectively cut and remove all types of thrombus and atheroma tissue, whether soft, tough, or hard, which will be appreciated by those of skill, and particularly those devices, systems, and methods that can do so with a reduced risk of suffering vessel injuries while avoiding an undesirable volume of blood loss during procedures.

Thrombectomy systems, devices, and methods of use are provided with a reduced risk of suffering vessel injuries while avoiding an undesirable volume of blood loss during procedures. In some embodiments, these systems, devices, and methods of use include a pulsating vacuum control. In some embodiments, the pulsating vacuum control can provide a reduced risk of suffering vessel injuries. In some embodiments, the pulsating vacuum control can avoid an undesirable volume of blood loss during procedures. In some embodiments, the pulsating vacuum control can provide both a reduced risk of suffering vessel injuries and avoid an undesirable volume of blood loss during procedures.

The pulsating vacuum control embodiments are their own embodiments and designs, and can be considered stand-alone technologies apart from the vascular cutter devices, systems, and methods taught herein. Likewise, the pulsating vacuum control can also be used in connection with any vascular cutter. In some embodiments, the pulsating vacuum control can be used in systems constructed and designed to control blood loss during a vascular tissue removal procedure. In some embodiments, the systems can comprise:

- a vascular cutter device designed to remove diseased vascular tissue, the vascular cutter device having
- a cutting head with a cutting edge on a mouth, and a lumen operably connected to an aspiration tube for discharging the diseased vascular tissue from the vascular lumen;

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and,

- a pulsating vacuum control assembly in an operable connection with the aspiration tube and having
- a valve configured to open and close at one or more desired frequencies to create a pulsating vacuum during operation of the device; and,
- a port configured for operable communication with a vacuum source to facilitate transport of the tissue out of the subject;

In some embodiments, any one or any combination of frequencies can be used to cycle a vacuum. In some embodiments, the one or more desired frequencies is a constant frequency that ranges from 1-5 Hz. In some embodiments, the one or more desired frequencies is a variable frequency that ranges from 1-5 Hz.

In some embodiments, any one or any combination of vacuum pressures can be used. In some embodiments, the one or more desired frequencies pulses the vacuum from a constant high pressure ranging from 100 mmHg to 800 mmHg to a constant low pressure ranging from 10 mmHg to 100 mmHg. In some embodiments, the one or more desired frequencies pulses the vacuum from a variable high pressure ranging from 100 mmHg to 800 mmHg to a variable low pressure ranging from 10 mmHg to 100 mmHg.

The term "close" can refer to a partial closure of the valve or complete closure of the valve. In some embodiments, a complete closure of the valve can provide a low pressure greater than 0 mmHg. In some embodiments, a complete closure of the valve can provide a low pressure ranging from 0 mmHg to 100 mmHg, or any pressure therein in increments of 1 mmHg, in embodiments where the pressure does not have sufficient time to drop to 0 mmHg. In some embodiments, a partial closure of the valve can provide a low pressure ranging from 0 mmHg to 100 mmHg, 100 mmHg to 800 mmHg, or any pressure therein in increments of 1 mmHg. The degree to which a valve is closed over the course of a vascular tissue removal procedure can be random, variable, or constant, or any combination thereof, ranging from 100% closed to 99% open, or any percent therein in increments of 1%.

The term "open" can refer to a partial opening of the valve or complete opening of the valve. In some embodiments, the pressure. In some embodiments, a complete opening of the valve can provide a high pressure of greater than 100 mmHg. In some embodiments, a complete opening of the valve can provide a high pressure ranging from 10 mmHg to 100 mmHg, 100 mmHg to 800 mmHg, or any pressure therein in increments of 1 mmHg. The degree to which a valve is opened over the course of a vascular tissue removal procedure can be random, variable, or constant, or any combination thereof, ranging from 99% closed to 100% open, or any percent therein in increments of 1%.

The systems can include a vascular cutter driven by a motor that drives the vascular cutter. In some embodiments, the motor that drives the vascular cutter can be located in a handle for ease of control by a user of the device. As such, in some embodiments, the systems can further comprise a handle that includes

- a motor operably connected to a drive assembly that includes a flexible rotating shaft with a proximal end, a distal end, and a lumen, the cutting head operably connected to the distal end of the flexibly rotating shaft, and the motor configured to provide a rotational movement to the flexible, rotating shaft for rotating the cutting head; and,

the pulsating vacuum control assembly.

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The control of the motor can be independent, or shared with, the control of the pulsating vacuum. In some embodiments, the systems can include a motor actuator and a pulsating vacuum actuator. In some embodiments, the systems can include a combination actuator for actuating the motor and the pulsating vacuum control assembly to create the vacuum simultaneously with actuation of the motor at an on-position and stop the vacuum and the motor simultaneously upon actuation to an off-position.

Any known method of creating a pulsating vacuum can be used. In some embodiments, the pulsating vacuum is created by pulsing power electronically to the vacuum. In some embodiments, the pulsating vacuum is created by mechanically opening and closing a valve. In some embodiments, the flexible rotating shaft has a wall with an aspiration port; and, the pulsating vacuum control assembly includes

- a sleeve bearing having a proximal end, a distal end, and a wall with an aspiration outlet, wherein the flexible rotating shaft is rotatably translatable within the sleeve bearing; and,
- an aspiration manifold having a proximal end, a distal end and a lumen, the aspiration manifold configured to house the sleeve bearing and operably connect with the vacuum source to facilitate the transport of the diseased tissue out of the subject when the aspiration port of the flexible rotating shaft aligns with the aspiration outlet of the sleeve bearing.

In some embodiments, the sleeve bearing is housed at the distal end of the lumen of the aspiration manifold; and, the aspiration outlet of the sleeve bearing is an elongate opening that extends to the proximal end of the sleeve bearing to direct the diseased tissue out of the subject through the aspiration manifold.

In view of the systems and devices taught herein, methods of performing a vascular procedure in a subject are also provided. In some embodiments, the method is a thrombectomy. In some embodiments, the methods can comprise:

- creating a point of entry in a vascular lumen of the subject;
- inserting the cutting head of the thrombectomy device into the vascular lumen;
- delivering the cutting head to the target site having the tissue for removal;
- cutting tissue from the vascular lumen with the cutting head;
- discharging the tissue from the vascular lumen with the pulsating vacuum;
- and,
- removing the cutting head of the thrombectomy device from the vascular lumen of the subject.

The vascular cutter device can be any device, including a preferred device as taught herein. For example, the vascular cutter device can be a preferred thrombectomy device that includes

- a flexible rotating shaft with a proximal end, a distal end, and a lumen;
- a curved, tubular cutter operably connected to the distal end of the flexible rotating shaft, the tubular cutter including a head with an axis with a mouth having a z-axis, a neck connected to the head and having an axis of rotation, and an angle, θ_z , ranging from 90° to 135°, between the z-axis of the mouth and the axis of rotation, the mouth the configured with
 - a leading edge having a highest height, h-leading;
 - a trailing edge having a lowest height, h-trailing;
 - a vertical bias ratio of h-leading/h-trailing ranging from 1.05 to 2.0;

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a distance, B_L , between the axis of rotation and the leading edge;
 a distance, B_T , between the axis of rotation and the trailing edge; and,
 a lateral bias ratio of B_L/B_T ranging from 1.05 to 2.0;
 a delivery sheath for delivering the cutter to a target site in a subject having a tissue for removal; and,
 a vacuum port configured for operable communication with a vacuum source to facilitate transport of the tissue out of the subject.

In some embodiments, the devices include a delivery sheath for a safe delivery of the cutter to the target site for the tissue removal. The proviso, however, is that in some embodiments, the devices do not include a delivery sheath by preference or by design.

In some embodiments, the cutter is deflected from the axis of rotation to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the axis of the head is at a vertical angle, ϕ , with the axis of rotation, wherein ϕ ranges from 0° to 90° to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the axis of the head is at a lateral angle, θ_0 , with the axis of rotation, wherein θ_0 ranges from 0° to 90° to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the flexible rotating shaft is deflected at an angle, θ_m , with the axis of rotation to form a curve region in the flexible rotating shaft, wherein θ_m ranges from 0-90 degrees to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the mouth has a cross-sectional area that is equal to, or less than, the lumen of the flexible, rotating shaft.

In some embodiments, the z-axis of the mouth is an angle, θ_r , from a plane containing the curve region to open the mouth toward the target site for the tissue for removal.

In some embodiments, the device further includes a straightening sheath.

Systems are also provided. In some embodiments, a system includes the devices taught herein, along with a handle that includes a motor operably connected to a drive assembly that includes the flexible rotating shaft and the cutter, the motor configured to provide a rotational movement to the flexible, rotating shaft for rotating the cutter.

In some embodiments, a system includes the devices taught herein, and the delivery sheath is a straightening sheath.

In some embodiments, a system includes the devices taught herein, wherein the z-axis of the mouth is an angle, θ_r , from a plane containing the curve region to open the mouth toward the target site for the tissue for removal.

In some embodiments, a system includes the devices taught herein, along with a vacuum source, and wherein the vacuum port is in operable connection with the vacuum for removal of the tissue from the system.

In some embodiments, the systems include a delivery sheath for a safe delivery of the cutter to the target site for the tissue removal. The proviso, however, is that in some embodiments, the systems do not include a delivery sheath by preference or by design.

In some embodiments, a system includes the devices taught herein, wherein the delivery sheath is a straightening sheath, the system further comprising a vacuum source; wherein the vacuum port is in operable connection with the vacuum for removal of the tissue from the system.

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Systems, devices, and methods are also provided in view of all of the features presented herein. In some embodiments, the thrombectomy systems can include

a flexible rotating shaft with a proximal end, a distal end, and a lumen;

a curved, tubular cutter operably connected to the distal end of the flexible rotating shaft, the tubular cutter including a head with an axis with a mouth having a z-axis, a neck connected to the head and having an axis of rotation, and an angle, θ_z , ranging from 90° to 135° , between the z-axis of the mouth and the axis of rotation, the mouth the configured with

a leading edge having a highest height, h-leading;

a trailing edge having a lowest height, h-trailing;

a vertical bias ratio of h-leading/h-trailing ranging from 1.05 to 2.0;

a distance, B_L , between the axis of rotation and the leading edge;

a distance, B_T , between the axis of rotation and the trailing edge; and,

a lateral bias ratio of B_L/B_T ranging from 1.05 to 2.0; and,

a pulsating vacuum control assembly having

a valve configured to open and close at one or more desired frequencies to create a pulsating vacuum that applies fluctuates the vacuum pressure applied to the cutter during operation of the device; and,

a port configured for operable communication with a vacuum source to facilitate transport of the tissue out of the subject.

In some embodiments, the valve is configured to operate within a desired range of frequencies, wherein the one or more desired frequencies have a frequency ranging from 1 Hz to 5 Hz.

In some embodiments, the vacuum is controlled to operate within a desired range of vacuum pressures, wherein the vacuum pressure fluctuates between two or more pressures including a high pressure and a low pressure, the high pressure ranging from 100 mmHg to 800 mmHg, and the low pressure ranging from 0 mmHg to 100 mmHg.

In some embodiments, the cutter can be deflected from the axis of rotation to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the axis of the head of the cutter is at a vertical angle, ϕ , with the axis of rotation, wherein β ranges from 0° to 90° to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the axis of the head is at a lateral angle, θ_0 , with the axis of rotation, wherein θ_0 ranges from 0° to 90° to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the flexible rotating shaft is deflected at an angle, θ_m , with the axis of rotation to form a curved region in the flexible rotating shaft, wherein θ_m ranges from 0-90 degrees to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the system further comprises a straightening sheath.

The systems can include a cutter driven by a motor that drives the cutter. In some embodiments, the motor that drives the cutter can be located in a handle for ease of control by the user. As such, in some embodiments, the systems can further comprise a handle that includes

a motor operably connected to a drive assembly that includes a flexible rotating shaft with a proximal end, a distal end, and a lumen, the cutting head operably

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connected to the distal end of the flexibly rotating shaft, and the motor configured to provide a rotational movement to the flexible, rotating shaft for rotating the cutting head; and,

the pulsating vacuum control assembly.

The control of the motor can be independent, or shared with, the control of the pulsating vacuum. In some embodiments, the systems can include a motor actuator and a pulsating vacuum actuator. In some embodiments, the systems can include a combination actuator for actuating the motor and the pulsating vacuum control assembly to create the vacuum simultaneously with actuation of the motor at an on-position and stop the vacuum and the motor simultaneously upon actuation to an off-position.

Any known system for creating a pulsating vacuum can be used. In some embodiments, the pulsating vacuum is created by pulsing power electronically to the vacuum. In some embodiments, the pulsating vacuum is created by mechanically opening and closing a valve. In some embodiments, the flexible rotating shaft has a wall with an aspiration port; and, the pulsating vacuum control assembly includes

a sleeve bearing having a proximal end, a distal end, and a wall with an aspiration outlet, wherein the flexible rotating shaft is rotatably translatable within the sleeve bearing; and,

an aspiration manifold having a proximal end, a distal end and a lumen, the aspiration manifold configured to house the sleeve bearing and operably connect with the vacuum source to facilitate the transport of the diseased tissue out of the subject when the aspiration port of the flexible rotating shaft aligns with the aspiration outlet of the sleeve bearing.

In some embodiments, the sleeve bearing is housed at the distal end of the lumen of the aspiration manifold; and, the aspiration outlet of the sleeve bearing is an elongate opening that extends to the proximal end of the sleeve bearing to direct the diseased tissue out of the subject through the aspiration manifold.

In some embodiments, the flexible rotating shaft has a wall with an aspiration port; and, the pulsating vacuum control assembly includes

a sleeve bearing having a proximal end, a distal end, and a wall with an aspiration outlet, wherein the flexible rotating shaft is rotatably translatable within the sleeve bearing; and,

an aspiration manifold having a proximal end and a distal end and a lumen, the aspiration manifold configured to house the sleeve bearing and operably connect with the vacuum source to facilitate the transport of the diseased tissue out of the subject when the aspiration port of the flexible rotating shaft aligns with the aspiration outlet of the sleeve bearing.

In some embodiments, a system can have the sleeve bearing housed at the distal end of the lumen of the aspiration manifold; and, the aspiration outlet of the sleeve bearing can be an elongate opening that extends to the proximal end of the sleeve bearing to direct the diseased tissue out of the subject through the aspiration manifold.

Blood loss can also be monitored and controlled using a system designed to help control blood loss during a procedure. In some embodiments, a system taught herein can include a dynamic response monitor, wherein the valve closes dynamically at a maximum tolerable level of a blood loss measured using a real-time monitoring. In some embodiments, the system can include a blood aspiration

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canister with a float valve, wherein the maximum tolerable level of the blood loss is measured using the height of the float valve.

Lost blood can be filtered and, in some embodiments, administered to the subject. In some embodiments, the filtered blood can be redirected back to a subject during a procedure. In some embodiments, a system taught herein can include a blood filter configured to filter an aspirated blood from the aspiration tube and create a filtered blood. In some embodiments, a system can include a blood return catheter for delivering the filtered blood back to the subject.

Methods of using such devices and systems are also provided, and any blood vessel, venous or arterial, can be treated using the devices and systems taught herein. In some embodiments, a method of removing a vascular tissue with a vascular cutter device is provided, the method including creating a point of entry in a vascular lumen of the subject;

inserting the cutter head of a vascular cutter device into the vascular lumen;

delivering the vascular cutter device to the target site, the target site having the tissue for removal;

cutting the tissue from the vascular lumen with the cutter head of the vascular cutter device;

and,

removing the cutter head of the vascular cutter device from the vascular lumen of the subject.

In some embodiments, methods of performing a thrombectomy using a thrombectomy device is provided. The method can include:

creating a point of entry in a vascular lumen of the subject;

inserting the cutter head of a thrombectomy device into the vascular lumen;

delivering the cutter head to the target site in the delivery sheath, the target site having the tissue for removal;

cutting the tissue from the vascular lumen with the cutter head of the thrombectomy device;

and,

removing the cutter head of the thrombectomy device from the vascular lumen of the subject.

In some embodiments, a method of removing a vascular tissue with a vascular cutter device having a deflection distance of the cutter head is provided, the method including creating a point of entry in a vascular lumen of the subject;

inserting a cutter head of the vascular cutter device into the vascular lumen, the vascular cutter device having a deflection distance;

delivering the cutter head to the target site, the target site having the tissue for removal;

cutting tissue from the vascular lumen with the cutter head of the vascular cutter device;

and,

removing the cutter head of the vascular cutter device from the vascular lumen of the subject.

In some embodiments, a method of removing a vascular tissue with a thrombectomy device having a deflection distance of the cutter head is provided, the method including creating a point of entry in a vascular lumen of the subject;

inserting a cutter head of the thrombectomy device into the vascular lumen, the cutter head of the thrombectomy device having a deflection distance;

delivering the cutter head to the target site, the target site having the tissue for removal;

cutting tissue from the vascular lumen with the cutter head of the thrombectomy device;
and,
removing the cutter head of the vascular cutter device from the vascular lumen of the subject.
removing the cutter head of the thrombectomy device from the vascular lumen of the subject.

In some embodiments, the method further includes discharging a cut tissue from the vascular lumen with a vacuum. In some embodiments, the vacuum can be a pulsating vacuum.

In some embodiments, a method of removing a vascular tissue with a vascular cutter device having a deflection distance of the cutter head is provided, the method including creating a point of entry in a vascular lumen of the subject;

inserting a cutter head of the vascular cutter device into the vascular lumen, the vascular cutter device having a deflection distance;

delivering the cutter head to the target site, the target site having the tissue for removal;

cutting tissue from the vascular lumen with the cutter head of the vascular cutter device;

discharging the tissue from the vascular lumen with the pulsating vacuum;

and,

removing the cutter head of the vascular cutter device from the vascular lumen of the subject.

In some embodiments, a method of removing a vascular tissue with a thrombectomy device having a deflection distance of the cutter head is provided, the method including creating a point of entry in a vascular lumen of the subject;

inserting a cutter head of the thrombectomy device into the vascular lumen, the cutter head of the thrombectomy device having a deflection distance;

delivering the cutter head to the target site, the target site having the tissue for removal;

cutting tissue from the vascular lumen with the cutter head of the thrombectomy device;

discharging the tissue from the vascular lumen with the pulsating vacuum;

and,

removing the cutter head of the vascular cutter device from the vascular lumen of the subject.

In some embodiments, the methods include using a delivery sheath for a safe delivery of the cutter to the target site for the tissue removal. The proviso, however, is that in some embodiments, the methods do not include using a delivery sheath, either by preference or by design.

In some embodiments, the method further includes inserting a guidewire in the point of entry and delivering the guidewire to a target tissue location, such as the location of a thrombus, and guiding the cutter head of the vascular cutter device to the target tissue location on the guidewire.

The devices, systems, and methods taught herein can be used in the treatment of any vascular condition. In some embodiments, the treatment includes the treatment of a deep vein thrombosis. In some embodiments, the treatment includes the treatment of a pulmonary embolism. In some embodiments, the treatment includes the treatment of a peripheral vascular disease. In some embodiments, the treatment includes the treatment of a saphenous vein graft.

In the course of the teachings of the above in the following detailed description, methods of making thrombectomy tubular cutters, thrombectomy devices, and thrombectomy systems are also provided. Methods of treating any vein or

artery is provided including, for example, (i) the removal of a venous thrombus, (ii) the removal of a pulmonary arterial thrombus, also referred to as a pulmonary embolism, and (iii) the removal of arterial plaque. And, of particular interest are those devices, systems, and methods that can do so with a reduced risk of suffering vessel injuries while avoiding an undesirable volume of blood loss during procedures.

BRIEF DESCRIPTION OF THE FIGURES

FIGS. 1A-1E illustrate a human leg containing a thrombus, a partial blockage of a blood vessel from the thrombus, the release of an embolus, a total blockage of the blood vessel from the thrombus, and an accumulation of platelets behind a thrombus.

FIGS. 2A-2H illustrate a thrombectomy tubular cutter, according to some embodiments.

FIGS. 3A-3I illustrate thrombectomy tubular cutter systems, according to some embodiments.

FIGS. 4A-4D illustrate the configuration of a rotating thrombectomy tubular cutter system for removing a thrombus, according to some embodiments.

FIGS. 5A-5L illustrate a variety of mouth shapes and sweep offsets, according to some embodiments.

FIGS. 6A-6F illustrate a variety of rim and cutting edge shapes, according to some embodiments.

FIGS. 7A and 7B illustrate the size of a tubular cutter of thrombus relative to the size of the lumen in the neck of the thrombectomy tubular cutter, showing how the size of the mouth of the tubular cutter can affect the passage of the thrombus through the lumen, according to some embodiments.

FIG. 8 illustrates a thrombectomy system having a motorized handle for rotating the thrombectomy tubular cutter, according to some embodiments.

FIGS. 9A, 9B, 9D, 9E, and 9F illustrate blood filtration ports in the thrombectomy tubular cutter, according to some embodiments.

FIGS. 10A and 10B illustrate a rotational shaft of a thrombectomy tubular cutter system having a positive displacement pump, according to some embodiments.

FIG. 11 is a flowchart of a thrombectomy method, according to some embodiments.

FIGS. 12A, 12B1-12B3, and 12C illustrate a mechanical valve assembly that includes a cam for opening a valve, according to some embodiments.

FIG. 13 illustrates an electromechanical valve assembly that includes a solenoid that receives an electric signal to apply a mechanical pressure for opening a valve, according to some embodiments.

FIG. 14 illustrates a computer system that can be used open a valve electrically, according to some embodiments.

FIGS. 15A-15D illustrate a vacuum manifold assembly for creating a pulsating vacuum with a mechanical valve configuration that includes a rotating shaft having an aspiration port that rotates within a sleeve bearing having an aspiration outlet, according to some embodiments.

FIGS. 16A-16F illustrate a variety of sleeve bearing configurations that can be designed and used in a vacuum manifold to obtain a desired vacuum pulse profile, according to some embodiments.

FIG. 17 illustrates the placement of an aspiration manifold inside the handle of a vascular cutting device.

FIG. 18 illustrates a preferred vascular cutter device, according to some embodiments.

FIGS. 19A and 19B illustrate blood filter configurations, according to some embodiments.

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FIG. 20 illustrates a vascular cutter system that includes a component for monitoring blood loss from a subject during a vascular cutting procedure, according to some embodiments.

FIG. 21 illustrates a vascular cutter system that includes a vacuum source inside the handle, according to some embodiments.

DETAILED DESCRIPTION

The devices, systems, and methods taught herein can be used in the treatment of any vascular condition. In some embodiments, the treatment includes the treatment of a deep vein thrombosis. In some embodiments, the treatment includes the treatment of a pulmonary embolism. In some

embodiments, the treatment includes the treatment of a peripheral vascular disease. In some embodiments, the treatment includes the treatment of a saphenous vein graft. One of skill will appreciate, for example, having vascular cutter technologies that (i) can effectively cut and remove many different types of tissue, include hard and soft thrombus tissue; (ii) can safely collect and remove plaque particles to avoid release of emboli; (iii) can effectively treat a blood vessel with a reduced risk of suffering vessel injuries; and (iv) can also handle tight or tough lesions having little to no luminal opening in the lesion. And, of particular interest are those devices, systems, and methods that can provide treatments with (v) a reduced risk of suffering vessel injuries, in some embodiments, (vi) a way of avoiding an undesirable volume of blood loss in some embodiments, and (vii) a way of providing both a reduced risk of suffering vessel injuries while avoiding an undesirable volume of blood loss in some embodiments.

Several embodiments are taught for the technologies herein, and these embodiments include systems, methods, and devices for (i) the removal of a venous thrombus, and (ii) the removal of a pulmonary arterial thrombus, also referred to as a pulmonary embolism.

In the course of the teachings of the above in the following detailed description, methods of making thrombectomy tubular cutters, thrombectomy devices, and thrombectomy systems are also provided. Methods of treating any vein or artery is provided including, for example, (i) the removal of a venous thrombus, (ii) the removal of a pulmonary arterial thrombus, also referred to as a pulmonary embolism, and (iii) the removal of arterial plaque. Pulsating vacuum systems, blood filtering systems, and blood recirculation systems are also provided.

One of skill will appreciate that a “subject” receiving a vascular cutting procedure can be a “patient” that is receiving the procedure. As such, the term “subject” and “patient” can be used interchangeably and refer to an animal such as a mammal including, but not limited to, non-primates such as, for example, a cow, pig, horse, cat, dog, rabbit, rat and mouse; and primates such as, for example, a monkey or a human. The subject can also be a cadaver, in some embodiments, or a portion of a cadaver. It should be appreciated that some uses of the devices, systems, and methods can be in vitro uses, animal models, and the like. As such, the subject can include a “subject tissue” used in in vitro testing, animal models, and the like.

The devices, systems, and methods taught herein can remove a target tissue from arteries or veins, for example, in your heart, brain, lungs, abdomen, arms, and legs. In some embodiments, the target tissue can have a venous location. In some embodiments, the target tissue can have an arterial location. Arterial clots can also take root in your kidneys,

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intestines, or eyes, although considered rare. In some embodiments, the blood clot is stationary, which is a “thrombosis” that can block blood flow, and the devices, systems, and methods taught herein can remove the thrombosis. If the blood clot breaks loose and moves in the blood vessel, it becomes an “embolism” which can travel to other parts of the body and become dangerous and even fatal.

Arterial thrombi can be referred to as white thrombi, and are characterized by predominance of platelets. Venous thrombi can be referred to as red thrombi, and are characterized by predominance of red blood cells. A thrombus can start soft and become more fibrous and harder over time. Thrombus formation can play a major role in the formation and histopathologic disorders of in arteries and old saphenous vein grafts (SVGs). A non-occlusive thrombus can be formed by silent plaque rupture or erosion. As the thrombus increases in size to become flow occlusive, a provisional matrix forms at healed lesions as an organized thrombus which may contain fibrin and be infiltrated by smooth muscle cell granulation tissue with proteoglycans and type III collagen. The thrombus proximal and distal to the site of a plaque rupture, for example, can be replaced by fibrous plaque in some embodiments. In some embodiments, the devices, systems, and methods can be used to treat SVGs. In some embodiments, the devices, systems, and methods can be used to inhibit the onset of plaque formation that would follow the development of the thrombus.

The devices, systems, and methods taught herein can be used, for example, to treat, or inhibit the onset of, symptoms by removing the blood clots from arteries or veins of a subject. As such, the devices, systems, and methods taught herein can remove arterial clots or venous clots. In some embodiments, the devices, systems, and methods treat, or inhibit the onset of, symptoms from a clot in the arms or legs including redness and warmth with swelling, tenderness, intense cramping, or any combination thereof. In some embodiments, the devices, systems, and methods treat, or inhibit the onset of, symptoms from a clot in the abdomen including stomach pain, diarrhea, vomiting, or any combination thereof. In some embodiments, the devices, systems, and methods treat, or inhibit the onset of, symptoms from a clot in the heart including breathing problems, nausea, dizziness, sweating, pain and heaviness in the chest, or any combination thereof. In some embodiments, the devices, systems, and methods treat, or inhibit the onset of, symptoms from a clot in the lungs including coughing-up of blood and a racing heart, shortness of breath, sweating, fever, sharp chest pain, or any combination thereof. In some embodiments, the devices, systems, and methods treat, or inhibit the onset of, symptoms from a clot in the brain including a headache; dizziness; difficulty talking; difficulty seeing clearly; weakness in the subject’s face, arms, or legs; or any combination thereof.

The devices, systems, and methods taught herein can be used, for example, to remove three types of blood clots that form in the veins. In some embodiments, the devices, systems, and methods taught herein can be used, to remove (i) a superficial venous thrombosis close to the surface of skin; (ii) a deep vein thrombosis (DVT) that forms deep in the body including in the lower leg, thigh, pelvis, arm, intestines, liver, kidney, or brain; and, (iii) a pulmonary embolism (PE) in the lungs.

FIGS. 1A-1E illustrate a human leg containing a thrombus, a partial blockage of a blood vessel from the thrombus, the release of an embolus, a total blockage of the blood vessel from the thrombus, an accumulation of platelets behind a thrombus. FIG. 1A illustrates a subject’s leg 105

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with symptomatic pain **110** from a thrombus **100** in a blood vessel **115** in the subject's leg **105**. FIG. 1B illustrates a partial blockage of the blood vessel **115** from the thrombus **100** that has deposited on the wall of the blood vessel **115**. FIG. 1C illustrates the release of an emboli **101** within the blood vessel **115** which, one of skill will appreciate, can result in a more serious set of symptoms that can include death. FIG. 1D illustrates a total blockage of the blood vessel **115** from the thrombus **100** that results in backflow, or pooling, **125** behind the thrombus **100** in the blood vessel **115**. FIG. 1E is a more anatomically correct illustration of the blood vessel **115** containing the thrombus **100**, in which there is an accumulation of platelets behind the thrombus **100**.

The thrombectomy devices taught herein can include a flexible rotating shaft with a proximal end, a distal end, and a lumen; a tubular cutter at the distal end of the rotating shaft, the tubular cutter having a proximal end and a distal end, and a top and a bottom, the tubular cutter configured with a mouth having a rim with a cutting edge, the mouth having (i) an inlet; and (ii) an outlet in communication with the lumen of the flexible rotating shaft; and, a neck having a central axis, the neck configured to operably connect and communicate with the lumen of the flexible rotating shaft. The systems can also include a vacuum port configured for operable communication with a vacuum source.

In some embodiments, methods of treating a thrombosis are provided. These methods can include creating a point of entry into the vasculature of the subject, advancing a thrombectomy device taught herein to the site of the thrombus through the point of entry, removing the thrombus with the thrombectomy device, and removing the thrombectomy device from the subject. In some embodiments, the methods include advancing a guidewire to the site of the thrombus, advancing the thrombectomy device to the thrombus over the guidewire, and removing the guidewire from the subject. In some embodiments, the method is used to treat a venous thrombosis. In some embodiments, the method is used to treat a superficial vein thrombosis. In some embodiments, the method is used to treat a deep vein thrombosis (DVT). In some embodiments, the method is used to treat a saphenous vein thrombosis. In some embodiments, the method is used to treat a renal vein thrombosis. In some embodiments, the method is used to treat an arterial thrombosis.

The thrombectomy tubular cutter can be referred to as a "tubular cutter", "cutter", "cutting head", or the like, in some embodiments. FIGS. 2A-2H illustrate various views showing features of a thrombectomy tubular cutter, according to some embodiments. FIG. 2A is a top-down view of the thrombectomy tubular cutter. As shown in FIG. 2A, the tubular cutter **200** can have a head with mouth **220** and a neck **225** having a lumen or throat **230** with a central axis **270**, which can also be referred to as an "axis of rotation" in some embodiments. The head or mouth **220** of the thrombectomy tubular cutter **200** can have a rim **221** that has a cutting edge **205**. In some embodiments, the thrombectomy tubular cutter **200** can have a guidewire port **215** located to guide the thrombectomy tubular cutter **200** on a guidewire (not shown) into a vascular lumen to a target site of a thrombus **100**. In some embodiments, the thrombectomy tubular cutter **200** does not have a guidewire port **215**. In some embodiments, the rim **221** can have a curvature.

In some embodiments, the rim **221** can have a trailing edge **210**, noting that the relationship between the cutting edge **205** and the trailing edge **210** is a design feature that can vary in the cutters taught herein. For example, the cutting edge **205** can be referred to as "the leading edge", the

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leading edge **205** configured to reach the target site of tissue excision first, before the trailing edge **210**, upon a rotation of the cutting head **200**. The top and bottom of the cutter are shown in FIG. 2B, the axis of the head and the axis of the neck (axis of rotation), where the axis of the head is concentric with the axis of the neck, along with a description of how the term "vertical" describes a top-to-bottom or bottom-to-top orientation. From a viewpoint of the user, viewing the cutting head **200** from the proximal end with the top side of the cutter oriented upward, a left-hand, or counter clockwise rotation of the cutting head **200** would require the leading edge **205** to be on the right side. Likewise, from the viewpoint of the user, viewing the cutting head **200** from the proximal end with the top side of the cutter oriented upward, a right-hand, or clockwise rotation of the cutting head **200** would require the leading edge **205** to be on the left side.

A desirable design feature that can be included in the configuration of the cutter is a "bias" of the cutting surface. The "bias" can be defined as an imbalance of distances between the leading edge **205** from the axis of rotation **270**, referred to as B_L and the trailing edge **210** from the axis of rotation **270**, referred to as B_T . In some embodiments B_L is greater than B_T . B_L is measured at the furthest point away from and normal to the axis of rotation on the leading edge. B_T is measured at the furthest point away from and normal to the axis of rotation on the trailing edge. Since the actual relative differences between B_L and B_T can depend on the size of the cutting head, it should be appreciated that the bias, a can be expressed as a ratio, which can be referred to as a "lateral bias" ratio in some embodiments. In some embodiments, the ratio of B_L/B_T can range from 1.00 to 2.00, 1.05 to 2.00, 1.10 to 2.00, 1.15 to 2.00, 1.20 to 2.00, 1.25 to 2.00, 1.30 to 2.00, 1.35 to 2.00, 1.40 to 2.00, 1.45 to 2.00, 1.50 to 2.00, from 1.00 to 1.95, from 1.00 to 1.90, from 1.00 to 1.85, from 1.00 to 1.80, from 1.00 to 1.75, from 1.00 to 1.70, from 1.00 to 1.65, from 1.00 to 1.60, from 1.00 to 1.55, from 1.00 to 1.50, from 1.00 to 1.45, from 1.00 to 1.40, from 1.00 to 1.35, from 1.00 to 1.30, from 1.00 to 1.25, from 1.00 to 1.20, from 1.00 to 1.15, from 1.00 to 1.10, from 1.00 to 1.05, or any ratio or range therein in increments of 0.01. In some embodiments, the B_L is greater than B_T by 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 1%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, or any amount or range therein in increments of 0.1%. In some embodiments, the B_L is greater than B_T by 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 100%, or any amount or range therein in increments of 0.1%. In some embodiments, the B_L is greater than B_T by 1.0x, 2.0x, 3.0x, 4.0x, or any amount or range therein in amounts of 0.1x.

FIGS. 2B and 2F are lateral views of the thrombectomy tubular cutter **200** showing the overlap of curvatures of the cutting edge **205** and trailing edge **210** that can be configured to facilitate an efficient removal of the thrombus **100**, in some embodiments. The presence of a smooth, non-cutting surface **250** on the distal end of the head is noted. In these embodiments, the curvature can include the cutting edge **205** and the trailing edge **210**.

FIG. 2C is a view from the bottom that shows the optional guidewire port **215** of the thrombectomy tubular cutter **200**, the cutting edge **205** and the trailing edge **210**, along with a

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description of how the term “lateral” describes a side to side orientation. FIG. 2D is a distal-to-proximal perspective view showing the lumen or throat **230**, as it is configured relative to the optional guidewire port **215** (sharing concentric axes with the throat **230** in this embodiment), the cutting edge **205**, and the trailing edge **210** of the thrombectomy tubular cutter **200**.

A cutting head **200** configured for the right-hand, or clockwise rotation from a proximal viewpoint of the user is illustrated in FIG. 2E, and it should be appreciated that the left-hand, or counterclockwise rotation is the mirror image, which means the leading edge **205** and trailing edge **210** will be switched.

FIG. 2E is a proximal-to-distal perspective view showing the lumen or throat **230**, as it is configured relative to the optional guidewire port **215** (sharing concentric axes with the throat **230** in this embodiment), the cutting edge **205**, and the trailing edge **210** of the tubular cutter **200**. It should be appreciated that the lumen **230** and optional guidewire port **215** do not have to share concentric axes in some embodiments.

FIGS. 2E and 2F provide perspective views of the leading edge **205** and the trailing edge **210**, showing how the leading edge **205** can be configured to lead the trailing edge. FIG. 2E is a top down perspective, having 0° rotation from the median plane **227** of the neck **225**, in which the height, h-leading, of leading edge **205** is more than the height, h-trailing, of trailing edge **210** as measured from the bottom of the neck. The h-leading is measured at the highest point of the leading edge and normal to the bottom of the cutter, and h-trailing is measured at the lowest point of the trailing edge and normal to the bottom of the cutter. Since the relative differences in height, Δh , can depend on the size of the cutting head, it should be appreciated that the relative differences in height, Δh , can be expressed as a ratio, which can be referred to as a “vertical bias” ratio in some embodiments. In some embodiments the ratio of h-leading/h-trailing can range from 1.00 to 2.00, 1.05 to 2.00, 1.10 to 2.00, 1.15 to 2.00, 1.20 to 2.00, 1.25 to 2.00, 1.30 to 2.00, 1.35 to 2.00, 1.40 to 2.00, 1.45 to 2.00, 1.50 to 2.00, from 1.00 to 1.95, from 1.00 to 1.90, from 1.00 to 1.85, from 1.00 to 1.80, from 1.00 to 1.75, from 1.00 to 1.70, from 1.00 to 1.65, from 1.00 to 1.60, from 1.00 to 1.55, from 1.00 to 1.50, from 1.00 to 1.45, from 1.00 to 1.40, from 1.00 to 1.35, from 1.00 to 1.30, from 1.00 to 1.25, from 1.00 to 1.20, from 1.00 to 1.15, from 1.00 to 1.10, from 1.00 to 1.05, or any ratio or range therein in increments of 0.01. In some embodiments, the h-leading is greater than h-trailing by 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, or any amount or range therein in increments of 0.1%. In some embodiments, the h-leading is greater than h-trailing by 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 100%, or any amount or range therein in increments of 0.1%. In some embodiments, the h-leading is greater than h-trailing by 1.00x, 2.00x, 3.00x, 4.00x, or any amount or range therein in amounts of 0.01x

FIG. 2G illustrates how a curved, tubular cutter can be fabricated, in some embodiments, in which a curved tube is cut to form a rim with a desired configuration. For example, the rim can be configured helical, in some embodiments, and

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with a cutting edge **205** and a trailing edge **210**, at least in some embodiments. The curve of the tube can be used to configure an angle, Θ_z , between the z-axis of the mouth and the central axis **270** of the neck or throat **230**, in some embodiments.

The z-axis of the mouth is defined as a line (i) orthogonal to a line connecting the most distal point of the mouth to the most proximal point of the mouth on the y-axis and (ii) orthogonal to the median plane on the x-axis, referring to FIG. 2E.

It should be appreciated that the angle, Θ_z , between the z-axis of the mouth and the central axis **270** of the neck or throat **230** can be configured independent of the angle of the curve of tube, merely by angling the cut of the tube on the dash to form the rim. Such variations in the fabricating methods can be used to form rims and mouths of varying desirable configurations.

The thrombectomy tubular cutter **200** can be made of any medical grade material that is known be useful in the making of cutting devices that are used in vivo in a subject in a sterile surgical environment. In some embodiments, the thrombectomy tubular cutter **200** can be made polymers such as PEEK, polycarbonate, PET, polyacrylamide, PEBAX, polyethylene, or a fluoropolymer such as FEP, PTFE, or PVDF. In some embodiments, the thrombectomy tubular cutter **200** can be made of a surgical grade metal such as stainless steel, titanium alloys, or cobalt alloys. In some embodiments the cutter can be made of ceramic or glass. In some embodiments, the thrombectomy tubular cutter **200** can be coated with a lubricious coating such as silicone oil, parylene, a hydrophilic coating, a diamond coating, a fluoropolymer coating, a ceramic coating, or any combination thereof. Such coatings can facilitate trouble free use of the thrombectomy tubular cutter. For example, such coatings can help to reduce friction in engulfing and conveying clot into shaft.

In some embodiments, the angle, Θ_z , between the central axis of the neck and the z-axis of the mouth can range from 0° to 180°, from 0° to 90°, from 90° to 180°, from 120° to 180°, or any amount or range therein in increments of 1°. In some embodiments, the angle, Θ_z , between the central axis of the neck and the z-axis of the mouth can range from 1° to 30°, from 2° to 30°, from 3° to 30°, from 4° to 30°, from 5° to 30°, 6° to 30°, from 7° to 30°, from 8° to 30°, from 9° to 30°, from 10° to 30°, from 15° to 30°, from 20° to 30°, from 25° to 30°, or any range or amount therein in increments of 1°. In some embodiments, the angle, Θ_z , between the central axis of the neck and the z-axis of the mouth can range from 1° to 45°, from 2° to 45°, from 3° to 45°, from 4° to 45°, from 5° to 45°, 6° to 45°, from 7° to 45°, from 8° to 45°, from 9° to 45°, from 10° to 45°, from 15° to 45°, from 20° to 45°, from 25° to 45°, from 30° to 45°, from 35° to 45°, from 40° to 45°, from, or any range or amount therein in increments of 1°. In some embodiments, the angle, Θ_z , can range from 1° to 60°, from 2° to 60°, from 3° to 60°, from 4° to 60°, from 5° to 60°, 6° to 60°, from 7° to 60°, from 8° to 60°, from 9° to 60°, from 10° to 60°, from 15° to 60°, from 20° to 60°, from 25° to 60°, from 30° to 60°, from 35° to 60°, from 40° to 60°, from 45° to 60°, from 50° to 60°, from 55° to 60°, or any range or amount therein in increments of 1°. In some embodiments, the angle, Θ_z , between the central axis of the neck and the z-axis of the mouth can range from 1° to 90°, from 2° to 90°, from 3° to 90°, from 4° to 90°, from 5° to 90°, 6° to 90°, from 7° to 90°, from 8° to 90°, from 9° to 90°, from 10° to 90°, from 15° to 90°, from 20° to 90°, from 25° to 90°, from 30° to 90°, from 35° to 90°, from 40° to 90°, from 45° to 90°, from 50° to 90°, from 55° to 90°,

from 60° to 90°, from 65° to 90°, from 70° to 90°, from 75° to 90°, from 80° to 90°, from 85° to 90°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_z , between the central axis of the neck and the z-axis of the mouth can range from 1° to 120°, from 2° to 120°, from 3° to 120°, from 4° to 120°, from 5° to 120°, from 6° to 120°, from 7° to 120°, from 8° to 120°, from 9° to 120°, from 10° to 120°, from 15° to 120°, from 20° to 120°, from 25° to 120°, from 30° to 120°, from 35° to 120°, from 40° to 120°, from 45° to 120°, from 50° to 120°, from 55° to 120°, from 60° to 120°, from 65° to 120°, from 70° to 120°, from 75° to 120°, from 80° to 120°, from 85° to 120°, from 90° to 120°, from 95° to 120°, from 100° to 120°, from 110° to 120°, from 115° to 120°, or any range or amount therein in increments of 1°.

In some embodiments, the angle, θ_z , between the central axis of the neck and the z-axis of the mouth can range from 25° to 55°, from 10° to 110°, from 20° to 110°, from 25° to 110°, from 30° to 110°, from 35° to 110°, from 40° to 110°, from 45° to 110°, from 10° to 100°, from 20° to 100°, from 25° to 100°, from 30° to 100°, from 35° to 100°, from 40° to 100°, from 45° to 100°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_z , between the central axis of the neck and the z-axis of the mouth can range from 2° to 40°, from 3° to 35°, from 4° to 30°, from 5° to 25°, from 5° to 40°, from 5° to 35°, from 10° to 40°, from 10° to 35°, from 10° to 25°, from 15° to 40°, from 15° to 35°, from 15° to 25°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_z , between the central axis of the neck and the z-axis of the mouth can be 0°, 1.0°, 2.0°, 3.0°, 4.0°, 5.0°, 6.0°, 7.0°, 8.0°, 9.0°, 10.0°, 11.0°, 12.0°, 13.0°, 14.0°, 15.0°, 16.0°, 17.0°, 18.0°, 19.0°, 20.0°, 21.0°, 22.0°, 23.0°, 24.0°, 25.0°, 26.0°, 27.0°, 28.0°, 29.0°, 30.0°, 31.0°, 32.0°, 33.0°, 34.0°, 35.0°, 36.0°, 37.0°, 38.0°, 39.0°, 40.0°, 41.0°, 42.0°, 43.0°, 44.0°, 50.0°, 55.0°, 60.0°, 65.0°, 70.0°, 75.0°, 80.0°, 85.0°, 90.0°, 95.0°, 100.0°, 105.0°, 110.0°, 115.0°, 120.0°, 125.0°, 130.0°, 135.0°, 140.0°, 145.0°, 150.0°, 155.0°, 160.0°, 165.0°, 170.0°, 175.0°, 180.0°, or any angle, θ_z , or range therein in 1° increments.

Blood vessel lumen diameters range considerably in size, becoming quite large, perhaps larger than practical for vascular entry to a subject in some embodiments. In larger vessels, the skilled artisan can opt for eccentric cutting to remove a larger region than the diameter of the cutter assembly without adding or exchanging the cutter for other larger tools for removal. The cutter can be biased off-center within the larger blood vessels to create added sweep. Inducing a deflection into the systems and devices can be used to induce a “sweep”, and this produces an effective “sweep diameter” to help the cutter **200** reach the lumen walls of large blood vessels and increase the centrifugal force realized by an excised tissue against the wall of the cutter, helping to ensure the tissue remains in the cutter head **220**. As the deflection increases, the sweep increases and centrifugal force increases. Likewise, as the deflection decreases, the sweep decreases and centrifugal force decreases. As such, the sweep can be selected and designed into the systems and devices by the user to obtain the desired function of increasing reach and centrifugal force.

In some embodiments, the length of deflection, or sweep, can be referred to as the “deflection distance” and can be used to define the sweep that is induced in the thrombectomy device. The sweep can be obtained in any manner devisable by the skilled artisan and can be defined, in some embodiments, as the distance of the cutter head from the axis of rotation **270**. The sweep can be defined, in some embodi-

ments, as the deflection distance, which is the point on the leading edge **205** that is at the greatest distance from the axis of rotation **270**.

In some embodiments, the sweep can be adjustable, such that the systems and devices can be equipped with a gauge on a handle of the systems and devices that indicate the sweep diameter obtained through an angle induced on a component of the system to create sweep. The inducing of a curve on a component of the system can be done in variety of methods, including those known to one of skill, such as through the use of a tendon to tension a component to induce the curve, or through the use of a material having a memory that contains the curve as a built-in shape during its formation.

15 Cutting Heads can be Configured to Induce Sweep and Centrifugal Force

Sweep, S_o , can be designed into a cutting head by selecting an angle, ϕ , between the head **220** and neck **225** of the cutter **200**, providing a desired deflection distance from axis of rotation **270**. The sweep increases the reach of the device within a vessel, or the diameter that can be reached by the cutting edge **205** of the tubular cutting head **220** of the device **200**. In addition, the sweep can add centrifugal force to the cutting head, so that excised tissue receives additional force to aid in its retention in the cutter head **220**. In some embodiments, the sweep, S_o , can be easily estimated using the equation $S_o = L_D \sin \phi$. The sweep adds a “deflection distance” of the cutting edge **205** from the axis of rotation **270**, where the sweep is 0 where $\phi = 0^\circ$, and the sweep is a maximum of L_D , the length of the deflected portion of the cutting device, where $\phi = 90^\circ$.

FIG. 2H illustrates how sweep can be introduced to a cutter **200** by inducing an angle, ϕ , between the neck **225** and the head **220**, in some embodiments. Introduction of sweep increases the reach of the cutting surface within a vessel, as well as introduce an addition of centrifugal force to help retain an excised tissue in the head **220** of the tubular cutter **200** after excision of the excised tissue from a target site in a subject. The angle, θ_z , can be selected to increase the reach of the cutter **200** and create a centrifugal force inside the mouth of the cutter to increase retention of the excised tissue in the head **220** after excision of the tissue from the subject, for example, excision of a thrombus from a vasculature. The leading edge **205** of the cutter **200** is further distanced from the axis of rotation **270**, resulting in an increased reach of the leading edge **205**, and increase in centrifugal force created during the rotation. The sweep in this embodiment can be defined as the deflection distance, S_o , which is the distance from the axis of rotation **270** to the point **205d**, which is at the greatest distance from the axis of rotation **270**.

The angle, θ_z , correlates with an angle, ϕ , which creates the sweep in the head **220** of the cutter **200**. In some embodiments, the angle, ϕ , can range from 0° to 90°, from 1° to 90°, from 2° to 90°, from 3° to 90°, from 4° to 90°, from 5° to 90°, from 6° to 90°, from 7° to 90°, from 8° to 90°, from 9° to 90°, from 10° to 90°, from 15° to 90°, from 20° to 90°, from 25° to 90°, from 30° to 90°, from 35° to 90°, from 40° to 90°, from 45° to 90°, from 50° to 90°, or any amount or range therein in increments of 1°. In some embodiments, the angle, ϕ , can range from 60° to 90°, from 70° to 90°, from 80° to 90°, or any amount or range therein in increments of 1°. In some embodiments, the angle, ϕ , can range from 0° to 60°, from 5° to 60°, from 10° to 60°, from 15° to 60°, from 20° to 60°, from 25° to 60°, from 30° to 60°, from 35° to 60°, from 40° to 60°, from 45° to 60°, from 50° to 60°, from 55° to 60°, or any amount or range therein in increments of 1°. In some embodiments, the angle, ϕ , can

range from 0° to 45°, from 5° to 45°, from 10° to 45°, from 15° to 45°, from 20° to 45°, from 25° to 45°, from 30° to 45°, from 35° to 45°, from 40° to 45°, or any amount or range therein in increments of 1°. In some embodiments, the angle, ϕ , can range from 5° to 30°, from 10° to 30°, from 15° to 30°, from 20° to 30°, from 25° to 30°, or any amount or range therein in increments of 1°. In some embodiments, the angle, ϕ , can be 0°. In some embodiments, the angle, ϕ , can be 0°, 1.0°, 2.0°, 3.0°, 4.0°, 5.0°, 6.0°, 7.0°, 8.0°, 9.0°, 10.0°, 11.0°, 12.0°, 13.0°, 14.0°, 15.0°, 16.0°, 17.0°, 18.0°, 19.0°, 20.0°, 21.0°, 22.0°, 23.0°, 24.0°, 25.0°, 26.0°, 27.0°, 28.0°, 29.0°, 30.0°, 31.0°, 32.0°, 33.0°, 34.0°, 35.0°, 36.0°, 37.0°, 38.0°, 39.0°, 40.0°, 41.0°, 42.0°, 43.0°, 44.0°, 50.0°, 55.0°, 60.0°, 65.0°, 70.0°, 75.0°, 80.0°, 85.0°, 90.0°, 95.0°, 100.0°, 105.0°, 110.0°, 115.0°, 120.0°, 125.0°, 130.0°, 135.0°, 140.0°, 145.0°, 150.0°, 155.0°, 160.0°, 165.0°, 170.0°, 175.0°, 180.0°, or any angle or range therein in 1° increments.

The thrombectomy devices drive the cutters with a flexible rotating shaft having a proximal end, a distal end, and a lumen, the tubular cutter operably connected with the distal end of the flexible rotating shaft. In some embodiments, the tubular cutter can be configured with

- a head having a distal end with a smooth, non-cutting surface, and a mouth with a rim and a z-axis, the rim having a cutting edge configured for removing a tissue from a target site within a blood vessel in a subject when rotating the tubular cutter within the blood vessel of the subject; and,
- a neck operably connected to the head, the neck having a throat with a central axis at an angle, θ_z , ranging from 0° to 180° from the z-axis of the mouth, the throat in communication with the lumen of the flexible rotating shaft for transport of the tissue away from the target site out of the subject.

The cutter **200** is attached to a flexible rotating shaft **305**. In some embodiments, the cutter can be operably attached to the flexible rotating shaft using a friction fitting. In some embodiments, the flexible rotating shaft can be allowed slip on the base of the cutter when engaged with a thrombus and meeting a maximum torque limit.

In some embodiments, an outer sheath **307** or outer catheter **307** can be used for safely delivering the flexible rotating shaft **305** and cutting head **200** to a target site in a subject for tissue removal. In some embodiments, any outer sheath used can be used to deliver the flexible rotating shaft and cutting head and can be referred to as a “delivery sheath”, “delivery catheter”, “guide sheath”, “guide catheter”, and the like. In some embodiments, the delivery catheter has a bending stiffness that is greater than the bending stiffness of the flexible rotating shaft. In some embodiments, the combined stiffness of the flexible rotating shaft and the delivery catheter are minimized to help guide the flexible rotating shaft to the target site of the thrombus in the blood vessel.

The devices, systems, and methods taught herein can also include an annulus between the outer sheath and the flexible rotating shaft to allow for a step of introducing a fluid including, for example, a contrast agent, a drug, saline, or a combination thereof, to visualize the procedure, remove air, aid in clot removal, lubricate the spinning of the flexible rotating shaft in the outer sheath, or a combination thereof. Flexible Rotating Shafts can be Configured to Induce Sweep and Centrifugal Force

Another way to create sweep, or a deflection distance, is to induce an angle, θ_m , on the flexible rotating shaft. The angle, θ_m , can be induced on the flexibly rotating shaft **305**

using a tendon, in some embodiments, or perhaps a memory curve, in some embodiments, and can range from 0° to 180°, or any amount or range therein in increments of 1°.

FIGS. 3A-3I illustrate thrombectomy tubular cutter systems, according to some embodiments. FIGS. 3A and 3B show thrombectomy tubular cutter systems **300,310** that do not include the optional guidewire port **215**. FIG. 3A shows thrombectomy tubular cutter system **300** including the thrombectomy tubular cutter **200** operatively connected to the distal end of a flexible rotating shaft **305**. FIG. 3B shows system **310** which includes system **300** with a delivery catheter **307**. FIG. 3C shows thrombectomy tubular cutter system **320** including the thrombectomy tubular cutter **200** operatively connected to the distal end of a flexible rotating shaft **305**, the thrombectomy tubular cutter **200** having guidewire port **215**. FIG. 3D shows system **330** which includes system **320** with a delivery catheter **307**. The neck of the thrombectomy tubular cutter **225** and the flexible rotating shaft **305** can share a central axis **315**.

As shown in FIG. 3A, the tubular cutter **200** can be configured to have a talon-like shape, in some embodiments, in which the distal end of the rim extends a distance, D_T , from the proximal end of the rim at the top of the tubular cutter **200**. In some embodiments, the distance, D_T , can range from 0.1 mm to 3.0 mm, from 0.2 mm to 3.0 mm, from 0.3 mm to 3.0 mm, from 0.3 mm to 3.0 mm, from 0.4 mm to 3.0 mm, from 0.5 mm to 3.0 mm, from 0.6 mm to 3.0 mm, from 0.7 mm to 3.0 mm, from 0.8 mm to 3.0 mm, from 0.9 mm to 3.0 mm, from 1.0 mm to 3.0 mm, from 1.1 mm to 3.0 mm, from 1.2 mm to 3.0 mm, from 1.3 mm to 3.0 mm, from 1.4 mm to 3.0 mm, from 1.5 mm to 3.0 mm, from 1.6 mm to 3.0 mm, from 1.7 mm to 3.0 mm, from 1.8 mm to 3.0 mm, from 1.9 mm to 3.0 mm, from 2.0 mm to 3.0 mm, from 2.1 mm to 3.0 mm, from 2.2 mm to 3.0 mm, from 2.3 mm to 3.0 mm, from 2.4 mm to 3.0 mm, from 2.5 mm to 3.0 mm, from 2.6 mm to 3.0 mm, from 2.7 mm to 3.0 mm, from 2.8 mm to 3.0 mm, from 2.9 mm to 3.0 mm, or any range or amount therein in increments of 0.01 mm. In some embodiments, the distance, D_T , can range from 0.1 mm to 2.0 mm, from 0.2 mm to 2.0 mm, from 0.3 mm to 2.0 mm, from 0.3 mm to 2.0 mm, from 0.4 mm to 2.0 mm, from 0.5 mm to 2.0 mm, from 0.6 mm to 2.0 mm, from 0.7 mm to 2.0 mm, from 0.8 mm to 2.0 mm, from 0.9 mm to 2.0 mm, from 1.0 mm to 2.0 mm, from 1.1 mm to 2.0 mm, from 1.2 mm to 2.0 mm, from 1.3 mm to 2.0 mm, from 1.4 mm to 2.0 mm, from 1.5 mm to 2.0 mm, from 1.6 mm to 2.0 mm, from 1.9 mm to 2.0 mm, from 1.8 mm to 2.0 mm, from 1.9 mm to 2.0 mm, or any range or amount therein in increments of 0.01 mm. In some embodiments, the distance, D_T , can range from 0.1 mm to 1.0 mm, from 0.2 mm to 1.0 mm, from 0.3 mm to 1.0 mm, from 0.3 mm to 1.0 mm, from 0.4 mm to 1.0 mm, from 0.5 mm to 1.0 mm, from 0.6 mm to 1.0 mm, from 0.7 mm to 1.0 mm, from 0.8 mm to 1.0 mm, from 0.9 mm to 1.0 mm, or any range or amount therein in increments of 0.01 mm.

In some embodiments, the angle, θ_m , induced on the flexible rotating shaft can range from 0° to 180°, from 0° to 170°, from 0° to 160°, from 0° to 150°, from 0° to 140°, from 0° to 130°, from 0° to 120°, from 0° to 110°, from 0° to 100°, from 0° to 90°, from 0° to 45°, or any amount or range therein in increments of 1°. In some embodiments, the angle, θ_m , induced on the flexibly rotating shaft **305** can range from 0° to 30°, from 1° to 30°, from 2° to 30°, from 3° to 30°, from 4° to 30°, from 5° to 30°, from 6° to 30°, from 7° to 30°, from 8° to 30°, from 9° to 30°, from 10° to 30°, from 15° to 30°, from 20° to 30°, from 25° to 30°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_m , induced on the flexibly rotating shaft **305** can

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range from 0° to 45°, from 1° to 45°, from 2° to 45°, from 3° to 45°, from 4° to 45°, from 5° to 45°, 6° to 45°, from 7° to 45°, from 8° to 45°, from 9° to 45°, from 10° to 45°, from 15° to 45°, from 20° to 45°, from 25° to 45°, from 30° to 45°, from 35° to 45°, from 40° to 45°, from, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_m , induced on the flexibly rotating shaft **305** can range from 0° to 60°, from 1° to 60°, from 2° to 60°, from 3° to 60°, from 4° to 60°, from 5° to 60°, 6° to 60°, from 7° to 60°, from 8° to 60°, from 9° to 60°, from 10° to 60°, from 15° to 60°, from 20° to 60°, from 25° to 60°, from 30° to 60°, from 35° to 60°, from 40° to 60°, from 45° to 60°, from 50° to 60°, from 55° to 60°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_m , induced on the flexibly rotating shaft **305** can range from 0° to 90°, from 1° to 90°, from 2° to 90°, from 3° to 90°, from 4° to 90°, from 5° to 90°, 6° to 90°, from 7° to 90°, from 8° to 90°, from 9° to 90°, from 10° to 90°, from 15° to 90°, from 20° to 90°, from 25° to 90°, from 30° to 90°, from 35° to 90°, from 40° to 90°, from 45° to 90°, from 50° to 90°, from 55° to 90°, from 60° to 90°, from 65° to 90°, from 70° to 90°, from 75° to 90°, from 80° to 90°, from 85° to 90°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_m , induced on the flexibly rotating shaft **305** can be 0°, 1.0°, 2.0°, 3.0°, 4.0°, 5.0°, 6.0°, 7.0°, 8.0°, 9.0°, 10.0°, 11.0°, 12.0°, 13.0°, 14.0°, 15.0°, 16.0°, 17.0°, 18.0°, 19.0°, 20.0°, 21.0°, 22.0°, 23.0°, 24.0°, 25.0°, 26.0°, 27.0°, 28.0°, 29.0°, 30.0°, 31.0°, 32.0°, 33.0°, 34.0°, 35.0°, 36.0°, 37.0°, 38.0°, 39.0°, 40.0°, 41.0°, 42.0°, 43.0°, 44.0°, 50.0°, 55.0°, 60.0°, 65.0°, 70.0°, 75.0°, 80.0°, 85.0°, 90.0°, 95.0°, 100.0°, 105.0°, 110.0°, 115.0°, 120.0°, 125.0°, 130.0°, 135.0°, 140.0°, 145.0°, 150.0°, 155.0°, 160.0°, 165.0°, 170.0°, 175.0°, 180.0°, or any angle or range therein in 1° increments.

As such, where there is an angle induced in the flexible rotating shaft, the outer sheath can be designed to straighten the flexible rotating shaft for delivery into a subject. In some embodiments, the outer sheath can be referred to as a “straightening-sheath”, “straightening-catheter”, and the like, as well as “delivery sheath”, “delivery catheter”, “guiding sheath”, “guiding catheter”, and the like. In some embodiments, the straightening catheter can have a bending stiffness that is the same, or similar to, the bending stiffness of the flexible rotating shaft.

FIG. 3E shows a memory-shaft system **340** which includes a straightening-catheter **307** that is slidably translational over a memory-shaped flexible rotating shaft **305C**. In this embodiment the flexible rotating shaft has a memory C-curve with an angle, θ_m , which is pre-shaped and elastic, and curved axis **370C** that lies on a plane shared by the curve in the curved axis **370C**. Due to the placement of the cutting edge **205**, the device shown in FIG. 3E requires a counter-clockwise rotation to cut, from the proximal perspective of the person using the device.

The straightening-sheath has the function of straightening the memory C-curve of the memory-shaped rotating shaft **305C** for delivery into a blood vessel of a subject. The straightening-sheath/catheter **307** has a proximal end (not shown) and a distal end. A distal slide of the straightening-sheath/catheter **307** from a position in which the distal end of the straightening-sheath/catheter **307** is proximal to the induced angle to gradually removes curvature from the C-curve. Likewise, a proximal slide of the straightening-sheath/catheter **307** from a position in which the distal end of the straightening-sheath/catheter **307** is distal to the memory C-curve gradually re-establishes curvature to the C-curve.

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A braid-pattern construction can be used, in some embodiments, to achieve a desired material strength in the delivery catheter, flexible rotating shaft, or a combination thereof. In some embodiments, a spiral pattern construction can be used to achieve a desired material strength in the delivery catheter, flexible rotating shaft, or a combination thereof. In some embodiments, the straightening-sheath can include a braid-pattern construction which can be fabricated using any material known to be suitable to the skilled artisan. The braid-pattern construction, for example, can be selected to include any sort of braiding material deemed acceptable by one of skill for the intended use. For example, the braid-pattern or the spiral pattern can be include use of a metal, a polymer braiding, a fiber, or a combination thereof. An suitable medical grade material known to be suitable to the skilled artisan can be used.

FIG. 3F shows a tendon-shaft system **350** which includes inducing a curve with angle, θ_m , on the flexible rotating shaft by applying tension to a tendon that integrated with the flexible rotating shaft, according to some embodiments. In some embodiments, the angle, θ_m , is induced on the flexible rotating shaft by the tendon can be straightened by releasing the tension on the tendon. In some embodiments, the angle, θ_m , is induced on the flexible rotating shaft can be straightened by pushing the tendon distally. In some embodiments, the angle, θ_m , is induced on the flexible rotating shaft can be straightened using a straightening catheter **307** that is slidably translational over the flexible rotating shaft **305C** which, in this embodiment has a curve on curved axis **370C** that lies on a plane shared by the curve.

Regardless of how the angle, θ_m , may be induced on the flexible rotating shaft, the tubular cutter **200** can have it's z-axis **260** rotated at an angle, θ_r , from an orthogonal position of the z-axis with respect to the plane, maintaining an orthogonal relationship with curved axis **370C**, such that the cutting edge **205** leads the trailing edge **210** in the amount of the angle, θ_r , relative to the plane upon which the curved axis **370C** lies. To clarify, the rotation of the cutter **200** to open the mouth to the target site is a rotating of the median plane **227** away from the plane shared by the curve in the deflection, where $\theta_r=0$ is where the median plane **227** is coincident with the plane shared by the axis of the curve formed by the deflection. A rotation of $\theta_r>0$ is a rotation of the median plane away from it's coincident relationship with the plane shared by the axis of the curve formed by the deflection, and the rotation of $\theta_r>0$ functions to open the mouth of the head **220** toward the target site.

The amount of the angle θ_r is 0° where the position of the z-axis **260** is orthogonal to the plane upon which the curved axis **370C** lies. The angle increases as the z-axis **260** is rotated relative to the 0° orthogonal position of the z-axis with respect to the plane, while maintaining an orthogonal relationship with curved axis **370C**. As the angle increases the cutting edge **205** leads the trailing edge **210** in the amount of the angle θ_r . In some embodiments, the angle θ_r can range from 1° to 90°, from 2° to 90°, from 3° to 90°, from 4° to 90°, from 5° to 90°, 6° to 90°, from 7° to 90°, from 8° to 90°, from 9° to 90°, from 10° to 90°, from 15° to 90°, from 20° to 90°, from 25° to 90°, from 30° to 90°, from 35° to 90°, from 40° to 90°, from 45° to 90°, from 50° to 90°, from 55° to 90°, from 60° to 90°, or any range or amount therein in increments of 1°. In some embodiments, the angle θ_r can range from 1° to 60°, from 2° to 60°, from 3° to 60°, from 4° to 60°, from 5° to 60°, 6° to 60°, from 7° to 60°, from 8° to 60°, from 9° to 60°, from 10° to 60°, from 15° to 60°, from 20° to 60°, from 25° to 60°, from 30° to 60°, from 35° to 60°, from 40° to 60°, from 45° to 60°, from 50° to 60°,

from 55° to 60°, or any range or amount therein in increments of 1°. In some embodiments, the angle θ_r can range from 1° to 45°, from 2° to 45°, from 3° to 45°, from 4° to 45°, from 5° to 45°, from 6° to 45°, from 7° to 45°, from 8° to 45°, from 9° to 45°, from 10° to 45°, from 15° to 45°, from 20° to 45°, from 25° to 45°, from 30° to 45°, from 35° to 45°, from 40° to 45°, or any range or amount therein in increments of 1°. In some embodiments, the angle θ_r can range from 1° to 45°, from 2° to 40°, from 3° to 35°, from 4° to 30°, from 5° to 25°, from 5° to 45°, from 5° to 40°, from 5° to 35°, from 5° to 30°, from 10° to 45°, from 10° to 40°, from 10° to 35°, from 10° to 30°, from 10° to 25°, from 15° to 45°, from 15° to 40°, from 15° to 35°, from 15° to 30°, from 15° to 25°, or any range or amount therein in increments of 1°. In some embodiments, the angle θ_r can be 1.0°, 2.0°, 3.0°, 4.0°, 5.0°, 6.0°, 7.0°, 8.0°, 9.0°, 10.0°, 11.0°, 12.0°, 13.0°, 14.0°, 15.0°, 16.0°, 17.0°, 18.0°, 19.0°, 20.0°, 21.0°, 22.0°, 23.0°, 24.0°, 25.0°, 26.0°, 27.0°, 28.0°, 29.0°, 30.0°, 31.0°, 32.0°, 33.0°, 34.0°, 35.0°, 36.0°, 37.0°, 38.0°, 39.0°, 40.0°, 41.0°, 42.0°, 43.0°, 44.0°, 45.0°, 55°, 60°, 65°, 70°, 75°, 80°, 85°, 90°, or any range or amount therein in increments of 1°.

FIGS. 3G and 3H show the angle, θ_m , induced as the memory curve on the flexibly rotating shaft 305 memory-shaft system 340, in some embodiments. Due to the placement of the cutting edge 205, the device shown in FIG. 3G requires a counterclockwise rotation to cut, from the perspective of the person using the device. In FIG. 3G, the angle, θ_m , induced as the memory curve on the flexibly rotating shaft 305 memory-shaft system 340 can range from 0° to 180°, from 0° to 90°, from 90° to 180°, from 5° to 45°, from 10° to 45°, from 15° to 30°, from 20° to 60°, from 25° to 55°, from 30° to 45°, from 10° to 120°, from 20° to 120°, from 25° to 120°, from 30° to 120°, from 35° to 120°, from 40° to 120°, from 45° to 120°, from 10° to 110°, from 20° to 110°, from 25° to 110°, from 30° to 110°, from 35° to 110°, from 40° to 110°, from 45° to 110°, from 10° to 100°, from 20° to 100°, from 25° to 100°, from 30° to 100°, from 35° to 100°, from 40° to 100°, from 45° to 100°, from 10° to 90°, from 20° to 90°, from 25° to 90°, from 30° to 90°, from 35° to 90°, from 40° to 90°, from 45° to 90°, from 50° to 90°, from 55° to 90°, from 60° to 90°, from 65° to 90°, from 70° to 90°, from 75° to 90°, from 80° to 90°, or any range therein in increments of 1°. In some embodiments, the angle, θ_m , between the central axis of the neck and the z-axis of the mouth can be 10°, 15°, 20°, 25°, 30°, 35°, 40°, 45°, 50°, 55°, 60°, 65°, 70°, 75°, 80°, 85°, 90°, 95°, 100°, 105°, 110°, 115°, 120°, 125°, 130°, 135°, 140°, 145°, 150°, 155°, 160°, 165°, 170°, 175°, 180°, or any angle, θ_m , therein in 1° increments.

The angle, θ_m , deflects a distal portion of the thrombectomy device, the distal portion beginning at the apex of the angle formed by the memory C-curve. As can be seen from FIGS. 3G and 3H, the angle, θ_m , in some embodiments, can be used to provide a “sweep” or a “deflection distance” of the cutting edge 205 from the axis of rotation 270. The sweep, So , can be designed by selection of the angle, θ_m , of the memory C-curve, or by selection of the angle, θ_m , of the angle induced by a tendon, to provide a desired deflection distance from axis of rotation 270. The sweep increases the reach of the device within a vessel, or the diameter that can be reached by the cutting edge 205 of the tubular cutting head 220 of the device 200. The sweep in FIGS. 3G and 3H is the deflection distance, So , which is measured from the axis of rotation 270 to the point 205d on the leading edge 205, because the point 205d is at the greatest distance from the axis of rotation 270. The sweep can be easily estimated using $So=L_D \sin \theta_m$, where the sweep is 0 where $\theta_m=0^\circ$,

and the sweep is a maximum of L_D , the length of the deflected portion of the cutting device, where $\theta_m=90^\circ$.

It should be appreciated that the sweep facilitates the use of the thrombectomy devices in larger vasculature. As such, any method of obtaining sweep to extend the reach of the cutter and add centrifugal force is contemplated in the embodiments. For example, more than one angle, θ_m , can be used to deflect a flexible rotating shaft. In some embodiments, the memory curve can be S-shaped, including an angle, θ_{m1} , and an angle, θ_{m2} , and the total sweep, So , is the total effect of the angle, θ_{m1} , and the angle, θ_{m2} , on the deflection distance. FIG. 3I provides an illustration of such an S-curve. It should be appreciated that although estimated deflection distances using $So=L_D \sin \theta_m$ can be fairly accurate where θ_m is 90° or less, these estimates start to become less accurate as θ_m increases above 90°, because the curve in the angle due to the physicality of the flexible rotating shaft introduces error. Moreover, the calculation is not useful at all for FIG. 3I, as the calculation of $[So1=L_D \sin \theta_{m1}]+[So2=L_D \sin \theta_{m2}]=0$ where θ_{m1} and θ_{m2} are both 180°, and this is clearly not the case, as the curve in the combined angles due to the physicality of the flexible rotating shaft actually creates the deflection distance.

The teachings provided herein include methods of creating a deflection distance, in which the configuration of a device taught herein deflects the leading edge of the cutter away from the axis of rotation. As can be seen, the deflection distance can be created using any manner devised by one of skill to deflect the leading edge away from the axis of rotation, so that a sweep that increases the cutting diameter of the cutter results. Just as B_L is measured at the furthest point away from and normal to the axis of rotation on the leading edge, the deflection distance can be measured as $\Delta B_L=B_L$ (after deflection, or B_{LAD})- B_L (before deflection, or B_{LBD}). Referring to FIG. 3I, for example, it can be seen that $So=\Delta B_L$, in this embodiment. As such, in some embodiments, it can be seen that $So=\Delta B_L$ =sweep=deflection distance.

In some embodiments, the deflection distance can range from 0.1 mm-30.0 mm, from 0.2 mm-30.0 mm, from 0.3 mm-30.0 mm, from 0.4 mm-30.0 mm, from 0.5 mm-30.0 mm, from 1.0 mm-30.0 mm, from 2.0 mm-30.0 mm, from 3.0 mm-30.0 mm, from 4.0 mm-30.0 mm, from 5.0 mm-30.0 mm, from 6.0 mm-30.0 mm, from 7.0 mm-30.0 mm, from 8.0 mm-30.0 mm, from 9.0 mm-30.0 mm, from 10.0 mm-30.0 mm, from 11.0 mm-30.0 mm, from 12.0 mm-30.0 mm, from 13.0 mm-30.0 mm, from 14.0 mm-30.0 mm, from 15.0 mm-30.0 mm, from 16.0 mm-30.0 mm, from 17.0 mm-30.0 mm, from 18.0 mm-30.0 mm, from 19.0 mm-30.0 mm, or any range or amount therein in increments of 0.1 mm. In some embodiments, the deflection distance can range from 0.1 mm-20.0 mm, from 0.2 mm-20.0 mm, from 0.3 mm-20.0 mm, from 0.4 mm-20.0 mm, from 0.5 mm-20.0 mm, from 1.0 mm-20.0 mm, from 2.0 mm-20.0 mm, from 3.0 mm-20.0 mm, from 4.0 mm-20.0 mm, from 5.0 mm-20.0 mm, from 6.0 mm-20.0 mm, from 7.0 mm-20.0 mm, from 8.0 mm-20.0 mm, from 9.0 mm-20.0 mm, from 10.0 mm-20.0 mm, from 11.0 mm-20.0 mm, from 12.0 mm-20.0 mm, from 13.0 mm-20.0 mm, from 14.0 mm-20.0 mm, from 15.0 mm-20.0 mm, from 16.0 mm-20.0 mm, from 17.0 mm-20.0 mm, from 18.0 mm-20.0 mm, from 19.0 mm-20.0 mm, or any range or amount therein in increments of 0.1 mm. In some embodiments, the deflection distance can range from 0.5 mm-19.0 mm, from 1.0 mm-18.0 mm, from 2.0 mm-17.0 mm, from 3.0 mm-16.0 mm, from 4.0 mm-15.0 mm, from 5.0 mm-14.0 mm, from 6.0 mm-13.0 mm, from 7.0 mm-12.0 mm, from 8.0 mm-10.0 mm, or any range or

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amount therein in increments of 0.1 mm. In some embodiments, the deflection distance can be 0.1 mm, 0.2 mm, 0.3 mm, 0.4 mm, 0.5 mm, 1.0 mm, 2.0 mm, 3.0 mm, 4.0 mm, 5.0 mm, 6.0 mm, 7.0 mm, 8.0 mm, 9.0 mm, 10.0 mm, 11.0 mm, 12.0 mm, 13.0 mm, 14.0 mm, 15.0 mm, 16.0 mm, 17.0 mm, 18.0 mm, 19.0 mm, 20.0 mm, 21.0 mm, 22.0 mm, 23.0 mm, 24.0 mm, 25.0 mm, 26.0 mm, 27.0 mm, 28.0 mm, 29.0 mm, 30.0 mm, or any amount or range therein in increments of 0.1 mm.

Component Characteristics

The components of the devices and systems can be designed to have desirable characteristics. The desirable characteristics can be obtained from material selection and the design of the materials to obtain a desired range of functions.

In some embodiments, the flexible rotating shaft may be fabricated from a polymer tube, a metal tube with slots for flexibility (for example, a type 304 stainless steel tube or the like), or a composite of polymer and metal. For a composite material, a metal coil, mesh, or braid can be laminated, using perhaps a polymer coating or molecular coating, for increased corrosion resistance, contamination control, and heat stability. In some embodiments, the polymer may be PEBAX, nylon, polyimide, PEEK, polyethylene, fluoropolymer (PTFE, ETFE, PVDF, FEP), or any combination thereof. The single molecule may be fluoro silane, in some embodiments. The dimensions of the tube may depend at least partially on the intended use of the thrombectomy device/apparatus/system.

It should be appreciated that, in some embodiments, the flexible, rotating shaft 305 can have a bending stiffness that is suitable for directing the cutting head to the target sight of the thrombus in a blood vessel. In some embodiments, the flexible, rotating shaft 305 can have an axial stiffness that is sufficient to transfer an axial force to the tubular cutter during operation of the thrombectomy device. Likewise, in some embodiments, the flexible, rotating shaft 305 can have a torsional stiffness that is sufficient to transfer torque from the proximal end of the flexible rotating shaft to the distal end of the flexible rotating shaft to rotate the tubular cutter during operation of the thrombectomy device.

The sheath/delivery catheter 307 can be "relatively stiff" compared to the flexible rotating shaft, in some embodiments, in that it can be designed to be substantially stiffer than the flexible rotating shaft 305C. As such, the sheath/delivery catheter 307 is relatively stiffer than the flexible rotating shaft 305C, such that a tendon (bold dashed line) can be pulled proximally to induce the memory curve region. The sheath/delivery catheter 307 can have the same or similar stiffness compared to the flexible rotating shaft.

One of skill will appreciate that control of the flexural stiffness of the device will allow additional control of the cutter to the user of the device. As such, in some embodiments, any device taught herein can be designed to obtain a desired amount of deflection to direct the cutter a desired surface of the vascular lumen wall. One of skill will appreciate that there are methods to vary the flexural stiffness of the flexible rotating shaft, the delivery sheath, or a combination thereof. The stiffness or flexibility of the components may be adjusted, for example, by varying the composition and/or structure of materials. In some embodiments, to increase flexural stiffness, the filaments on the flexible rotating shaft, the delivery sheath, or a combination thereof, can be bound together with stiffer material, or the flexural stiffness can be increased (or reduced) by increasing (or decreasing) the filament size.

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In addition to providing the proper flexural stiffness and torsional stiffness for maneuverability and function, the devices also need the required axial tensile stiffness for a better response of the thrombectomy devices to push and pull. Axial tensile stiffness is the resistance to stretch or contraction of along the length of the component under axial loading and is in units of N/mm. Key components in the design, in some embodiments, include a desired axial tensile stiffness in the delivery sheath and the flexible rotating shaft.

The flexural stiffness of the delivery catheter could be similar to, equal to, or greater than the flexible rotating shaft, in some embodiments. The ratio of flexural stiffness of the flexible rotating shaft to the flexural stiffness of the delivery catheter can range from 0.03 to 1.0 in some embodiments, 0.03 to 0.30 in some embodiments, 0.05 to 0.25 in some embodiments, 0.06 to 0.30 in some embodiments, from 0.5 to 1.0, from 0.6 to 1.0, from 0.7 to 1.0, from 0.8 to 1.0, from 0.9 to 1.0, or any range or amount therein in increments of 0.01. In some embodiments, the ratio of flexural stiffness of the flexible rotating shaft to the flexural stiffness of the delivery catheter can be 0.10, 0.20, 0.30, 0.40, 0.40, 0.40, 0.40, 0.40 or any ratio or range therein in increments of 0.01. In some embodiments, however, the flexural stiffness of the delivery catheter is 2×, 3×, or 4× greater than the flexural stiffness of the flexible rotating shaft, or any range or amount therein in increments of 0.1×.

In some embodiments, a straightening-sheath is merely configured to have a minimal stiffness suitable to straighten the flexible rotating shaft for delivery to a target site in a subject. In some embodiments, the force required to bend the straightening-sheath is greater than the force required to bend the flexible rotating shaft, for example, perhaps to bend memory-shaped C-curve portion of the flexible rotating shaft. In some embodiments, the force required to bend the straightening-sheath is greater than the force required to bend the flexible rotating shaft, and in some embodiments the memory C-curve, by 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, or any amount or range therein in amounts of 0.1%. In some embodiments, the force required to bend the straightening-sheath is greater than the force required to bend the flexible rotating shaft, and in some embodiments the memory C-curve, by a range of 1%-50%, 2%-50%, 3%-50%, 4%-50%, 5%-50%, 6%-50%, 7%, 8%-50%, 9%-50%, 10%-50%, 11%-50%, 12%-50%, 13%-50%, 14%-50%, 15%-50%, 16%-50%, 17%-50%, 18%-50%, 19%-50%, 20%-50%, 21%-50%, 22%-50%, 23%-50%, 24%-50%, 25%-50%, or any amount or range therein in amounts of 0.1. In some embodiments, the force required to bend the straightening-sheath is greater than the force required to bend the flexible rotating shaft, and in some embodiments the memory C-curve, can be 2×, 3×, or 4× greater than the force required to bend the memory-shaped C-curve, or any amount or range therein in amounts of 0.1×.

The stiffness of the straightening-sheath can affect the ability of the straightening-sheath to remove the memory-C-curve of the flexible rotating shaft, but it should be appreciated that the stiffness needs to be limited due to the adverse effects of the stiffness of the combination of device components on the friction applied to the vascular lumen wall. As such, it is desirable to minimize device stiffness to target a tissue location for treatment and direct the tubular head to the target area.

In some embodiments, a desirable stiffness of the straightening-sheath can range from 15 Nmm² to 900 Nmm², 30 Nmm² to 800 Nmm², from 20 Nmm² to 200 Nmm², 400

Nmm² to 800 Nmm², or any amount or range therein in increments of 1 Nmm². In some embodiments, a desirable stiffness of the straightening-sheath can be dependent on the location of the stiffness measurement along the sheath, for example, to add flexibility at a distal region of the sheath and a relative stiffness at a proximal region of the sheath, where the border between distal and proximal is at the midpoint of the sheath as defined by the length of the sheath in the subject during an interventional procedure. In some embodiments, a desired stiffness in a distal region of a straightening sheath can range from 20 Nmm² to 200 Nmm², from 30 Nmm² to 100 Nmm², or any amount or range therein in increments of 1 Nmm². In some embodiments, a desired stiffness in a proximal region of a sheath can range from 300 Nmm² to 900 Nmm², from 400 Nmm² to 800 Nmm², or any amount or range therein in increments of 1 Nmm². In some embodiments, a desired combined stiffness at any point on the thrombectomy device, whether in proximal or distal portion of the device, can be 10 Nmm², 20 Nmm², 30 Nmm², 40 Nmm², 50 Nmm², 60 Nmm², 70 Nmm², 80 Nmm², 90 Nmm², 100 Nmm², 200 Nmm², 300 Nmm², 400 Nmm², 500 Nmm², 600 Nmm², 700 Nmm², 800 Nmm², 900 Nmm², or any amount or range therein in increments of 1 Nmm².

The axial strength of the flexible rotating shaft, delivery catheter, or the combination thereof, is also a design feature. In some embodiments, the flexible rotating shaft, delivery catheter, or the combination thereof, can be configured to have a column strength suitable to withstand pushing into a subject without increasing the diameter of In some embodiments, the flexible rotating shaft, delivery catheter, or the combination thereof, to an undesirable amount. As such, In some embodiments, the flexible rotating shaft, delivery catheter, or the combination thereof, can be configured to have a column strength suitable to withstand pulling out of a subject without reducing the diameter of the flexible rotating shaft, delivery catheter, or the combination thereof, to an undesirable amount. In some embodiments, the undesirable amount of increase in diameter of the flexible rotating shaft, delivery catheter, or the combination thereof, upon pushing into a subject is less than 1%, less than 2%, less than 3%, less than 4%, less than 5%, less than 6%, less than 7%, less than 8%, less than 9%, less than 10%, less than 11%, less than 12%, less than 13%, less than 14%, less than 15%, less than 16%, less than 17%, less than 18%, less than 19%, less than 20%, or any amount or range therein in amounts of 0.1%; and, in some embodiments, the increase in diameter of the flexible rotating shaft, delivery catheter, or the combination thereof, upon pushing into a subject can range from 0%-20%, 1%-20%, 2%-20%, 3%-20%, 4%-20%, 5%-20%, 6%-20%, 7%-20%, 8%-20%, 9%-20%, 10%-20%, or any amount or range therein in amounts of 0.1%. In some embodiments, the undesirable amount of decrease in diameter of the flexible rotating shaft, delivery catheter, or the combination thereof, upon pulling out of a subject is less than 1%, less than 2%, less than 3%, less than 4%, less than 5%, less than 6%, less than 7%, less than 8%, less than 9%, less than 10%, less than 11%, less than 12%, less than 13%, less than 14%, less than 15%, less than 16%, less than 17%, less than 18%, less than 19%, less than 20%, or any amount or range therein in amounts of 0.1%; and, in some embodiments, the decrease in diameter of the flexible rotating shaft, delivery catheter, or the combination thereof, upon pulling out of a subject can range from 0%-20%, 1%-20%, 2%-20%, 3%-20%, 4%-20%, 5%-20%, 6%-20%, 7%-20%, 8%-20%, 9%-20%, 10%-20%, or any amount or range therein in amounts of 0.1%.

Generally speaking, in some embodiments, the outer diameter of the tubular cutter and/or flexible rotating shaft can range from 2.0 mm-10.0 mm, from 2.5 mm-9.5 mm, from 3.0 mm-9.0 mm, from 3.5 mm-8.5 mm, from 4.0 mm-8.0 mm, from 4.5 mm-7.5 mm, from 5.0 mm-7.0 mm, or any amount or range therein in increments of 0.1 mm. In some embodiments, the outer diameter of the tubular cutter can be 2.0 mm, 2.2 mm, 2.4 mm, 2.6 mm, 2.8 mm, 3.0 mm, 3.2 mm, 3.4 mm, 3.6 mm, 3.8 mm, 4.0 mm, 4.2 mm, 4.4 mm, 4.6 mm, 4.8 mm, 5.0 mm, 5.2 mm, 5.4 mm, 5.6 mm, 5.8 mm, 6.0 mm, 6.2 mm, 6.4 mm, 6.6 mm, 6.8 mm, 7.0 mm, 7.2 mm, 7.4 mm, 7.6 mm, 7.8 mm, 8.0 mm, 8.2 mm, 8.4 mm, 8.6 mm, 8.8 mm, 9.0 mm, 9.2 mm, 9.4 mm, 9.6 mm, 9.8 mm, 10.0 mm, or any amount or range therein in increments of 0.1 mm.

For example, in some variations the outer diameter of the tube may be limited to about 2.2 mm, while in other variations the outer diameter of the tube may be limited to about 1.6 mm, and others can range from as much as 2.0 mm to 10.0 mm, 3.0 mm to 9.0 mm, 4.0 mm to 8.0 mm, 5.0 mm to 7.0 mm, or any amount or range therein in increments of 0.1 mm. In some embodiments, the wall thickness of the tube may be limited to about 0.05 to 1 mm. In some embodiments, the wall thickness of the tube may be limited to 0.1 mm to 0.2 mm. In some embodiments, the overall length of the tube of the flexible rotating shaft may range from about 500 mm to 1500 mm (about 20 inches to about 60 inches), from about 500 mm to 1400 mm, from about 500 mm to 1300 mm, from about 500 mm to 1200 mm, from about 500 mm to 1100 mm, from about 500 mm to 1000 mm, or any amount or range therein in increments of 1 mm.

FIGS. 4A-4D illustrate the configuration of a rotating thrombectomy tubular cutter system for removing a thrombus, according to some embodiments. FIG. 4A shows system 300 approaching a target site 402 to rotate 400 the head of the thrombectomy tubular cutter 200 to remove thrombus 100 from within blood vessel 115. The rotating 400 of the head of the thrombectomy tubular cutter 200 can be in a clockwise direction, or counterclockwise direction, when viewed from a position that is proximal to the thrombus 100. The rotating 400 separates the thrombus 100 from the blood vessel 115, for example by tubular cutting the cutting edge 205 through the interface between the thrombus 100 and the luminal wall of the blood vessel 115. In some embodiments, the cutting edge 205 removes a portion of the thrombus 100 without contacting the luminal wall of the blood vessel 115 by cutting through the thrombus 100 to remove just a portion of the thrombus. After the rotating 400 and tubular cutting of the thrombus 100, the thrombus 100 is captured in the mouth 220 of the thrombectomy tubular cutter 200 for removal from the blood vessel 115. FIG. 4B shows the shape of the rim 221 of the mouth 220, having a diameter D_s, which can be any shape that will facilitate the tubular cutting of a thrombus within a blood vessel. The cutting edge 205 is positioned for a clockwise rotation, as seen from a user's position proximal to the head of the cutter, the cutting edge highlighted by the dashed-line in FIG. 4B.

One of skill should appreciate that the rim 221 of the mouth 220 can have any shape that facilitates thrombus removal, and often the shape can be undulating, helical, serrated, sinusoidal, straight, or some combination of such features. In some embodiments, all surfaces of the rim 221 can contact the same plane. However, in some embodiments, some surfaces of the rim 221 do not contact the same plane as other surfaces of the rim 221. In some embodiments only a select few surfaces of the rim 221 of the mouth 220 contact

the same plane. For example, in some embodiments, only 2, 3, or 4 surfaces of the rim **221** of the mouth **220** contact the same plane.

FIG. 4B, for example, shows a rim **221** of the mouth **220** having an elliptical shape having an axis that is rotated counter clockwise from the axis of the lumen of the neck of the thrombectomy tubular cutter when viewed from a top-down perspective to the thrombectomy tubular cutter. Cutters with a rim **221** such as FIG. 4G can have only 2 surfaces that contact a plane that is parallel to a horizontal plane that dissects the thrombectomy device. This configuration results from the “helical” nature of the relative shapes of the cutting edge **205** and trailing edge **210**. The cutting edge **205** is also designed for a clockwise rotation using the right-hand helical shape of the cutting edge **205**, again viewed from a user’s position proximal to the head of the cutter, the cutting edge shown by the dashed-line in FIG. 4B. On the contrary, a cutting edge for a counterclockwise rotation when viewed from a position proximal to the head of the cutter would have a left-hand helical shape.

FIG. 4C shows how a right-hand helical drill bit can be used, in some embodiments, to create a thrombectomy tubular cutter **200**. A drill bit can be created by selecting a surgical grade metal using well-known drill bit manufacturing techniques. The bit will be sized such that the distance between opposite flute edges matches the desired width, or diameter of the tubular cutter, D_s , of the thrombectomy tubular cutter **200**. The point **405** of the bit can serve as the distal end of the thrombectomy tubular cutter **200**, and the body of the bit can be ground to form a mouth **220** for the thrombectomy tubular cutter **200**. The helical shapes of the flutes **410** can be ground to form the rim **221** of the mouth **220**, where cutting edge **205** and trailing edge **210** are formed by the flutes of the bit. The shank (not shown) of the bit can be bored-out to create a lumen **230** having a desired diameter, D_L , in the neck of the thrombectomy tubular cutter **200**. FIG. 4D illustrates the transformation of the drill bit shown in FIG. 4C. As noted above, a right-hand helical bit would be used for a clockwise rotation when viewed from a position proximal to the head of the cutter, whereas a left-hand helical bit (not shown) would be used for a counterclockwise rotation when viewed from a position proximal to the head of the cutter.

Mouth Orientation can Add to Sweep by Providing Bias

The inlet of the mouth has a major axis that can be oriented at an angle, θ_B , from the orientation of the central axis of the neck of the tubular cutter. The angle, θ_B , can be measured on a horizontal plane that bisects the central axis of the neck of the tubular cutter to separate the top of the tubular cutter from the bottom of the tubular cutter.

FIGS. 5A-5L illustrate a variety of mouth shapes and sweep offsets, according to some embodiments. It should be appreciated that any mouth shape that facilitates the removal of a thrombus from the luminal wall of a vessel can be used. As shown in FIG. 5A, the rim **221** of the mouth **220** can be elliptical with an axis (a major axis) and a mathematical center point.

In some embodiments, the major axis of the ellipse can be rotated about the mathematical center point. In some embodiments, the inlet of the mouth has a major axis that is oriented at an angle, θ_B , from the orientation of the central axis of the neck of the tubular cutter, wherein θ_B is measured on a horizontal plane that bisects the central axis of the neck of the tubular cutter to separate the top of the tubular cutter from the bottom of the tubular cutter.

In some embodiments, the angle or rotation of the mouth, θ_B , can range from 1° to 90°, from 2° to 90°, from 3° to 90°,

from 4° to 90°, from 5° to 90°, 6° to 90°, from 7° to 90°, from 8° to 90°, from 9° to 90°, from 10° to 90°, from 15° to 90°, from 20° to 90°, from 25° to 90°, from 30° to 90°, from 35° to 90°, from 40° to 90°, from 45° to 90°, from 50° to 90°, from 55° to 90°, from 60° to 90°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_B , can range from 1° to 60°, from 2° to 60°, from 3° to 60°, from 4° to 60°, from 5° to 60°, 6° to 60°, from 7° to 60°, from 8° to 60°, from 9° to 60°, from 10° to 60°, from 15° to 60°, from 20° to 60°, from 25° to 60°, from 30° to 60°, from 35° to 60°, from 40° to 60°, from 45° to 60°, from 50° to 60°, from 55° to 60°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_B , can range from 1° to 45°, from 2° to 45°, from 3° to 45°, from 4° to 45°, from 5° to 45°, 6° to 45°, from 7° to 45°, from 8° to 45°, from 9° to 45°, from 10° to 45°, from 15° to 45°, from 20° to 45°, from 25° to 45°, from 30° to 45°, from 35° to 45°, from 40° to 45°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_B , can range from 1° to 45°, from 2° to 40°, from 3° to 35°, from 4° to 30°, from 5° to 25°, 5° to 45°, from 5° to 40°, from 5° to 35°, from 5° to 30°, from 10° to 45°, 10° to 40°, from 10° to 35°, from 10° to 30°, from 10° to 25°, from 15° to 45°, 15° to 40°, from 15° to 35°, from 15° to 30°, from 15° to 25°, or any range or amount therein in increments of 1°. In some embodiments, the angle, θ_B , can be 1.0°, 2.0°, 3.0°, 4.0°, 5.0°, 6.0°, 7.0°, 8.0°, 9.0°, 10.0°, 11.0°, 12.0°, 13.0°, 14.0°, 15.0°, 16.0°, 17.0°, 18.0°, 19.0°, 20.0°, 21.0°, 22.0°, 23.0°, 24.0°, 25.0°, 26.0°, 27.0°, 28.0°, 29.0°, 30.0°, 31.0°, 32.0°, 33.0°, 34.0°, 35.0°, 36.0°, 37.0°, 38.0°, 39.0°, 40.0°, 41.0°, 42.0°, 43.0°, 44.0°, 45.0°, 55.0°, 60.0°, 65.0°, 70.0°, 75.0°, 80.0°, 85.0°, 90.0°, or any range or amount therein in increments of 1°. The angle, θ_B , alters the mouth orientation and can add to sweep by providing bias as described herein. As the angle, θ_B , is increased, bias increases and sweep increases.

In some embodiments, as shown in FIG. 5B, the rim **221** of the mouth **220** can be circular with a mathematical center point to serve as a point of reference for mouth orientation relative to the neck of the thrombectomy device. In some embodiments, as shown in FIGS. 5C-5E, the rim **221** of the mouth **220** can be pear-shaped with an axis (a major axis) and a mathematical center point, wherein the wider part of the pear-shape can be located in proximal or distal location with no rotation of the major axis about the mathematical center point. In some embodiments, the major axis of the pear-shape can be rotated about the mathematical center point.

Head Tilt to Add Sweep

FIGS. 5F-5L show how to design a sweep offset, S_O , by selecting an offset angle, θ_O , which adds “head tilt” in the cutter to offset the central axis of the head **220** of the tubular cutter **200** from the central axis of the neck **225** of the tubular cutter. The sweep offset allows the thrombectomy tubular cutter **200** to sweep beyond the diameter of the tubular cutter, D_s . Using the length of the head, L_H , of the thrombectomy tubular cutter **200** as a constant, the selection of the offset angle, θ_O , provides the desired sweep offset, S_O to increasing the cutting diameter inside the blood vessel for removal of a blood clot from a subject. In some embodiments, the diameter of the tubular cutter, D_s , is limited to 4 mm maximum to limit the size of the point of entry of the thrombectomy tubular cutter into the blood vessel of the subject. If the blood vessel is the iliac vein, the diameter can be, for example, about 10 mm in diameter. As such, it would be beneficial to offset the sweep of the head of the throm-

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bectomy tubular cutter **200** to gain a much larger cutting diameter of the 4 mm thrombectomy tubular cutter in the iliac vein. For example,

Using $S_O = L_H \sin \theta_O$, where D_S is 4 mm, and L_H is 6 mm, then

$S_O = 6 \text{ mm} \sin 5 = 0.5 \text{ mm}$, where the offset angle, θ_O , is selected as 5 degrees in FIG. 5F.

$S_O = 6 \text{ mm} \sin 10 = 1.0 \text{ mm}$, where the offset angle, θ_O , is selected as 10 degrees in FIG. 5G.

$S_O = 6 \text{ mm} \sin 15 = 1.6 \text{ mm}$, where the offset angle, θ_O , is selected as 15 degrees in FIG. 5H.

$S_O = 6 \text{ mm} \sin 20 = 2.1 \text{ mm}$, where the offset angle, θ_O , is selected as 20 degrees in FIG. 5I.

$S_O = 6 \text{ mm} \sin 25 = 2.5 \text{ mm}$, where the offset angle, θ_O , is selected as 25 degrees in FIG. 5J.

$S_O = 6 \text{ mm} \sin 30 = 3.0 \text{ mm}$, where the offset angle, θ_O , is selected as 30 degrees in FIG. 5K.

$S_O = 6 \text{ mm} \sin 35 = 3.4 \text{ mm}$, where the offset angle, θ_O , is selected as 35 degrees (not shown).

A protractor is shown in FIG. 5L illustrating angles ranging from 0 degrees to 45 degrees for convenience of reference. Using the above as an example of a design choice, an offset angle, θ_O , which can be referred to as a lateral angle, lateral offset angle, or head tilt, of 25 degrees provides a sweep offset of 2.8 mm, which means that the thrombectomy tubular cutter **200** will have a cutting diameter of 4 mm+2.8 mm+2.8 mm=9.6 mm on a full rotation which will more effectively tubular cutter a thrombus **100** out of the iliac vein of the subject. As can be seen, the skilled artisan can control the sweep of the thrombectomy tubular cutter **200** by selecting the diameter of the tubular cutter, D_S , and the length of the head, L_H , and the offset angle, θ_O .

In some embodiments, the lateral offset angle, θ_O , can range from 0° to 90°, from 1° to 90°, from 2° to 90°, from 3° to 90°, from 4° to 90°, from 5° to 90°, from 6° to 90°, from 7° to 90°, from 8° to 90°, from 9° to 90°, from 10° to 90°, from 15° to 90°, from 20° to 90°, from 25° to 90°, from 30° to 90°, from 35° to 90°, from 40° to 90°, from 45° to 90°, from 50° to 90°, or any amount or range therein in increments of 1°. In some embodiments, the lateral offset angle, θ_O , can range from 60° to 90°, from 70° to 90°, from 80° to 90°, or any amount or range therein in increments of 1°. In some embodiments, the lateral offset angle, θ_O , can range from 0° to 60°, from 5° to 60°, from 10° to 60°, from 15° to 60°, from 20° to 60°, from 25° to 60°, from 30° to 60°, from 35° to 60°, from 40° to 60°, from 45° to 60°, from 50° to 60°, from 55° to 60°, or any amount or range therein in increments of 1°. In some embodiments, the lateral offset angle, θ_O , can range from 0° to 45°, from 5° to 45°, from 10° to 45°, from 15° to 45°, from 20° to 45°, from 25° to 45°, from 30° to 45°, from 35° to 45°, from 40° to 45°, or any amount or range therein in increments of 1°. In some embodiments, the lateral offset angle, θ_O , can range from 5° to 30°, from 10° to 30°, from 15° to 30°, from 20° to 30°, from 25° to 30°, or any amount or range therein in increments of 1°. In some embodiments, the lateral offset angle, θ_O , can be 0°, 1 degree, 2 degrees, 3 degrees, 4 degrees, 5 degrees, 6 degrees, 7 degrees, 8 degrees, 9 degrees, 10 degrees, 11 degrees, 12 degrees, 13 degrees, 14 degrees, 15 degrees, 16 degrees, 17 degrees, 18 degrees, 19 degrees, 20 degrees, 21 degrees, 22 degrees, 23 degrees, 24 degrees, 25 degrees, 26 degrees, 27 degrees, 28 degrees, 29 degrees, 30 degrees, 31 degrees, 32 degrees, 33 degrees, 34 degrees, 35 degrees, 36 degrees, 37 degrees, 38 degrees, 39 degrees, 40 degrees, 41 degrees, 42 degrees, 43 degrees, 44 degrees, 45 degrees, or any amount or range therein in increments of 0.1 degrees. In some embodiments, the offset angle, θ_O , can be selected to range

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from 1 degree to 45 degrees, from 1 degree to 40 degrees, from 1 degree to 35 degrees, from 1 degree to 30 degrees, from 1 degree to 25 degrees, from 1 degree to 20 degrees, from 1 degree to 15 degrees, from 1 degree to 10 degrees, from 1 degree to 5 degrees, or any amount or range therein in increments of 0.1 degrees.

Design of the Cutting Edge

FIGS. 6A-6F illustrate a variety of rim and cutting edge shapes, according to some embodiments. In some embodiments, the cutting edge **205** and the trailing edge **210** each follow a parabolic equation of the general form $y=a(x-h)^2+k$, where a “+a” indicates that the parabola opens up, and a “-a” indicates that the parabola opens downward, and (h,k) indicates the position of the parabola’s vertex, and the value of “a” dictates the steepness the parabola where, as the absolute value of “a” increases, the steepness of the parabola increases. The steepness of the parabola is the maximum slope possible, regardless of whether the slope is positive or negative, on a line tangent to the parabola. If the cutting edge **205** and the trailing edge **210** align, or at least “substantially” align, both parabola have the same or similar coordinates for the vertex which tells us whether the cutting edge, and both have the same or similar coefficient “a”. The parabola “substantially” align when the alignment is close enough to believe that they may be the same, and there is little-to-no difference seen by a user of the device in removing the same or similar thrombus type from subjects. FIG. 6A illustrates equal parabolic, upward-opening curves that are aligned between the cutting edge **205** and the trailing edge **210** on the rim **221** of the mouth **220**, meaning both parabola have the same or similar coordinates for the vertex, and both have the same or similar coefficient “a”. FIG. 6B illustrates slightly unequal parabolic, upward-opening curves that when aligning the cutting edge **205** and the trailing edge **210** of the rim **221** of the mouth **220**, appearing as though the k values are the same or similar because the vertical aligning is the same or similar, but the h values are significantly different because the parabola do not align horizontally, shifted left or right, of each other; the “a” coefficient appears the same or similar as the parabola have similar steepness. FIG. 6C can be explained similar to FIG. 6B, with the difference being that the difference in h values for the vertices are even more different than in FIG. 6B. As such, the shapes of the rims, namely the cutting edge and trailing edge, can be described and varied using the parabolic equation $y=a(x-h)^2+k$, varying the relative values of a, h, and/or k between the parabolic shapes of the cutting edge **205** and the trailing edge **210**.

FIG. 6D illustrates how the cutting edge **205** can be linear in some embodiments, and the linear cutting edge **205** can lie on a plane having a major axis that is parallel to the axis **270** of the lumen of the neck of the thrombectomy device. In some embodiments, the linear cutting edge **205** can lie on a plane having a major axis that is at an angle, θ , relative to the axis **270** of the lumen of the neck of the thrombectomy device, where θ can have an angle that ranges from 1 degree to 20 degrees, from 1 degree to 18 degrees, from 1 degree to 16 degrees, from 1 degree to 14 degrees, from 1 degree to 12 degrees, from 1 degree to 10 degrees, from 1 degree to 8 degrees, from 1 degree to 6 degrees, from 1 degree to 4 degrees, or any angle or range therein in amounts of 1 degree. In some embodiments, the angle can be selected from the group consisting of 1 degree, 2 degrees, 3 degrees, 4 degrees, 5 degrees, 6 degrees, 7 degrees, 8 degrees, 9 degrees, 10 degrees, 11 degrees, 12 degrees, 13 degrees, 14 degrees, 15 degrees, 16 degrees, 17 degrees, 18 degrees, 19 degrees, and 20 degrees.

FIG. 6E illustrates how the cutting edge 205 can be undulating in some embodiments, and each undulation can be the same as the others, following the same or similar parabolic equation, or each undulation can have a parabolic equation that is independently selected, such that only some of the undulations share the same or substantially similar parabolic equation, or none of the undulations share the same or similar parabolic equation. Here we note that “substantially similar” parabolic equations can be close enough to visually conclude that they might be the same and would not be expected to perform differently in a thrombectomy. FIG. 6F illustrates how the cutting edge 205 can be serrated in some embodiments, and each serration can be the same or substantially similar to the others. Here we note that “substantially similar” serrations can be close enough to visually conclude that they might be the same and would not be expected to perform differently in a thrombectomy.

FIGS. 7A and 7B illustrate the size of a tubular cutter of thrombus relative to the size of the lumen in the neck of the thrombectomy tubular cutter, showing how the size of the mouth of the tubular cutter can affect the passage of the thrombus through the lumen, according to some embodiments. Referring back to FIG. 4B for detail, the mouth of the cutter can be designed to have an inlet diameter, D_M , that is smaller than the lumen D_L of the flexible rotating shaft 305, to help ensure that a blood clot 100 captured by the thrombectomy tubular cutter 200 will not clog the flexible rotating shaft 305 during a thrombus removal. In some embodiments, the inlet of the mouth of the tubular cutter can have a cross-sectional area that is equal to, or less than, cross-sectional area of the lumen of the flexible, rotating shaft. In some embodiments, the inlet of the mouth of the tubular cutter can have a cross-sectional area that is greater than the lumen of the flexible, rotating shaft.

Since the actual relative differences between D_M and D_L can depend on the size of the cutting head, it should be appreciated that the sizing of the mouth and the lumen can be expressed as a ratio, where the ratio of D_M/D_L can range from 0.20 to 1.00, from 0.30 to 1.00, from 0.40 to 1.00, from 0.50 to 1.00, from 0.60 to 1.00, from 0.70 to 1.00, from 1.00 to 2.00, from 1.00 to 1.90, from 1.00 to 1.80, from 1.00 to 1.70, from 1.00 to 1.60, from 1.00 to 1.50, from 1.00 to 1.45, from 1.00 to 1.40, from 1.00 to 1.35, from 1.00 to 1.30, from 1.00 to 1.25, from 1.00 to 1.20, from 1.00 to 1.15, from 1.00 to 1.10, from 1.00 to 1.05, or any ratio or range therein in increments of 0.01. In some embodiments, the D_M is greater than D_L by 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, or any amount or range therein in increments of 0.1%. In some embodiments, the D_M is greater than D_L by 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 100%, or any amount or range therein in increments of 0.1%.

The lumen of the flexible rotating shaft can be approximated by the lumen of the neck of the thrombectomy tubular cutter 200 in some embodiments. It is reasonable that the diameter of the lumen of the neck of the thrombectomy tubular cutter can be, perhaps, smaller than the lumen of the flexible rotating shaft and, as such, can also be limited for thrombus size. As such, the diameter of the lumen D_L of the neck 225 of the thrombectomy tubular cutter 200 can be

sized as equal to or greater than the diameter of the inlet of the thrombectomy tubular cutter D_M . In some embodiments, $D_M < D_L$. In some embodiments, $D_M = D_L$. In some embodiments, $D_M > D_L$.

One of skill will appreciate that the “cleared diameter” of a vessel can be used to describe the diameter of the lumen of the blood vessel after passage of the thrombectomy tubular cutter through the lumen of the vessel. Since the vessel is often elastic, the cleared diameter of the lumen of a blood vessel may or may not be equal to the diameter of the cutter 200, namely the diameter of the head/mouth 220. The cleared diameter of the head/mouth 220 can be greater than the outer diameter of the neck 225 and, in some embodiments, the cleared diameter of the head/mouth 220 can be equal to than the outer diameter of the neck 225. In some embodiments, the cleared diameter of the head/mouth 220 can be less than the outer diameter of the neck 225.

In some embodiments, the relative sizes of the device components can have a significant impact on the movement of the device within a vessel lumen. In some embodiments, the cutter diameter can be configured to be greater than, and in some embodiments at least 10% greater, at least 20% greater, or at least 30% greater than the diameter of the flexible rotating shaft. The ratio of cutter diameter to the diameter of the flexible rotating shaft can range from 1.1 to 1.6 in some embodiments, 1.3 to 1.5 in some embodiments, 1.2 to 1.4 in some embodiments, 1.3 to 1.4 in some embodiments, or any range therein. In some embodiments, the ratio of cutter diameter to the diameter of the flexible rotating shaft can be 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, or any ratio therein in increments of 0.05, in some embodiments. In some embodiments, however, the cutter diameter is 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, or any percent therein in increments of 0.5%, greater than, or less than, the diameter of the flexible rotating shaft.

Systems

It should be appreciated that systems can also be assembled to include any devices taught herein. In some embodiments, a system can include a thrombectomy device taught herein and a delivery catheter. In some embodiments, a system can include a thrombectomy device taught herein and a guidewire. In some embodiments, a system can include a thrombectomy device taught herein, a guidewire, and a delivery catheter.

In some embodiments, a system can include a thrombectomy device taught herein and a vacuum port for connecting to a vacuum source. In some embodiments, a system can include a thrombectomy device taught herein, a guidewire, a delivery catheter, a vacuum port for connecting to a vacuum source, and the vacuum source. In some embodiments, the thrombectomy devices can include a vacuum port configured for operable communication with any vacuum source to facilitate transport of an excised tissue out of the subject. In some embodiments, a system can include a thrombectomy device taught herein having a vacuum port, wherein the vacuum port is in operable connection with an aspiration manifold for removal of a thrombus from the system, the aspiration manifold including a vented shaft with an aspiration hole configured for the removal of the thrombus away from the target site and out of the subject.

In some embodiments, the thrombectomy devices can have a handle that includes a motor operably connected to the flexible rotating shaft to turn the flexible rotating shaft to excise a tissue from a subject. In some embodiments, the

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thrombectomy devices can have a handle that includes a motor operably connected to a positive displacement pump having a flexible driveshaft, the motor configured to provide a rotational torque to the flexible driveshaft that actuates the positive displacement pump to transport thrombus tissue away from the target site and out of the subject.

In some embodiments, the thrombectomy devices can have a handle that includes a motor operably connected to a screw pump having a flexible driveshaft operably connected to a helical screw, the motor configured to provide a rotational torque to the flexible driveshaft that actuates the helical screw to transport thrombus tissue away from the target site and out of the subject.

In some embodiments, the thrombectomy devices can have a handle with a motor; and a flexible driveshaft operably connected to a helical screw; wherein, the motor is operably connected to

a drive assembly that includes the flexible rotating shaft and the tubular cutter, the motor configured to provide a rotational torque to the flexible rotating shaft for rotating the tubular cutter and cutting the thrombus; and,

the flexible driveshaft for rotating the helical screw to transport thrombus tissue away from the target site and out of the subject.

In some embodiments, the thrombectomy device can have a handle that includes a motor operably connected to a drive assembly that includes the flexible rotating shaft and the tubular cutter, the motor can be configured to provide a rotational torque to the flexible, rotating shaft for rotating the tubular cutter. The position of the cutting edge on the tubular cutter will determine whether the cutting head is rotated clockwise or counterclockwise as viewed from a position of the user that is proximal to the cutting head.

FIG. 8 illustrates a thrombectomy system having a motorized handle for rotating the thrombectomy tubular cutter, according to some embodiments. The thrombectomy tubular cutter **200** is operatively connected to a flexible rotating shaft **310** which is operatively connected to a manifold **820** at fitting **825**. The manifold **820** has an aspiration port **830** for discharging thrombus **100** through vacuum discharge port **835** using vacuum. The manifold **820** is operatively connected to a vent tube **810** that is connected to vent **820** in the handle **805**. The handle **805** contains motor, **M**, for rotating **840** the vent tube **810** which in turn rotates the flexible, rotating shaft and the thrombectomy tubular cutter **200** from removal of a thrombus **100**. The motor can operate using any source of power and, in some embodiments, the power is a battery source (not shown) located in the handle **805**. The actuation of the rotation **840** through the motor, **M**, occurs through depression of a switch **845**. In some embodiments, the switch **845** can be actuated using a squeeze trigger **850**. Placing a seal over vent **820** allows vacuum to reach the lumen of the thrombectomy tubular cutter for vacuum removal of the thrombus **100** which can then be discharged through aspiration port **830** via the manifold **820**.

FIGS. 9A, 9B, 9D, 9E, and 9F illustrate blood filtration ports in the thrombectomy tubular cutter, according to some embodiments. The blood filtration ports **905, 910** are a feature that facilitates the collection of thrombus tissue will allowing the blood to be released back into the blood vessel. Such ports can be any shape desired for optimal release of blood while tubular cutting and capturing a thrombus under vacuum. In some embodiments, as shown in FIGS. 9A and 9B, the blood filtration ports are round or elliptical **905**. In some embodiments, as shown in FIGS. 9D and 9E, the blood filtration ports are square or rectangular **910**. Any polygon or

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shape can be used, in some embodiments. Some alternate shapes contemplated for optimal filtration of thrombus from blood are provided in FIG. 9F. The blood filtration ports need to have a minimum size that allows the passage of all blood cells. For example, in some embodiments, the blood filtration ports can have a diameter (or effective diameter) as small as about 20 μm to allow for the passage of monocytes.

In some embodiments, the blood filtration ports can have a diameter ranging from about 20 μm to 2 mm, from about 20 μm to 1 mm, from about 20 μm to 0.5 mm, from about 20 μm to 0.2 mm, from about 20 μm to 0.1 mm, from about 20 μm to 90 μm , from about 20 μm to 80 μm , from about 20 μm to 70 μm , from about 20 μm to 60 μm , from about 20 μm to 50 μm , from about 20 μm to 40 μm , from about 20 μm to 30 μm , or any diameter or range therein in amounts of 1 μm . In some embodiments, the shape of the blood filtration ports are not circular or elliptical, for example, and the minimum dimension is at least 20 μm , the diameter possibly ranging from at least 20 μm to 200 μm , 20 μm to 100 μm , 50 μm to 100 μm , 50 μm to 150 μm , 75 μm to 125 μm , or any amount or range therein in increments of 1 μm .

In some embodiments, the blood filtration ports can have a diameter ranging from about 40 μm to 2 mm, from about 40 μm to 1 mm, from about 40 μm to 0.5 mm, from about 40 μm to 0.2 mm, from about 40 μm to 0.1 mm, from about 40 μm to 90 μm , from about 40 μm to 80 μm , from about 40 μm to 70 μm , from about 40 μm to 60 μm , from about 40 μm to 50 μm , or any diameter or range therein in amounts of 1 μm . In some embodiments, the shape of the blood filtration ports are not circular or elliptical, for example, and the diameter is at least 40 μm and can range from at least 40 μm to 200 μm , 40 μm to 100 μm , or any amount or range therein in increments of 1 μm . In some embodiments, the blood filtration ports can have a minimum dimension of 10 μm , 15 μm , 20 μm , 25 μm , 30 μm , 35 μm , 40 μm , 45 μm , 50 μm , 55 μm , 60 μm , 65 μm , 70 μm , 75 μm , 80 μm , 85 μm , 90 μm , 95 μm , 100 μm , 105 μm , 110 μm , 115 μm , 120 μm , 125 μm , 130 μm , 135 μm , 140 μm , 145 μm , 150 μm , or any dimension or range therein in increments of 1 μm . The maximum dimension of a blood filtration port can be, for example, 100 μm , 150 μm , 200 μm , 250 μm , 300 μm , 350 μm , 400 μm , 450 μm , 500 μm , or any amount therein in increments of 10 μm .

In some embodiments, the blood filtration ports can be placed closer to the cutting edge of the thrombectomy tubular cutter than the trailing edge of the thrombectomy tubular cutter. In some embodiments, the blood filtration ports can be placed closer to the trailing edge of the thrombectomy tubular cutter than the cutting edge of the thrombectomy tubular cutter. In some embodiments, the blood filtration ports can cover the head **220** of the thrombectomy tubular cutter in an amount of 1, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, or any amount or range therein in increments of 0.1%. In some embodiments, the blood filtration ports can cover the head **220** of the thrombectomy tubular cutter in a range of 1% to 10%, 1% to 15%, 1% to 20%, 1% to 25%, 1% to 30%, 1% to 40%, 1% to 50%, or any amount or range therein in increments of 0.1%. In some embodiments, the blood filtration ports can cover the head **220** of the thrombectomy tubular cutter in a range of 3% to 10%, 3% to 15%, 3% to 20%, 3% to 25%, 3% to 30%, 3% to 40%, 3% to 50%, or any amount or range therein in increments of 0.1%. In some embodiments, the blood filtration ports can cover the head **220** of the thrombectomy

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tubular cutter in a range of 5% to 10%, 5% to 15%, 5% to 20%, 5% to 25%, 5% to 30%, 5% to 40%, 5% to 50%, or any amount or range therein in increments of 0.1%. Any number of blood filtration ports can be used. In some embodiments, this percent coverage can translate to 1 port, 2 ports, 3 ports, 4 ports, 5 ports, 6 ports, 7 ports, 8 ports, 9 ports, 10 ports, 11 ports, 12 ports, 13 ports, 14 ports, 15 ports, 16 ports, 17 ports, 18 ports, 19 ports, 20 ports, 25 ports, 30 ports, 35 ports, 40 ports, 45 ports, 50 ports, or more, and any range of number of ports therein.

The blood filtration ports can be open ports in some embodiments, and one-way valves in some embodiments. The one-way valves would be operable to allow blood to leave the lumen of the thrombectomy tubular cutter through the ports, but not enter the lumen of the thrombectomy tubular cutter through the ports. In some embodiments, the one-way valve can be a flap valve, or a butterfly valve. In some embodiments, the one-way valve can be referred to as a non-return valve and can be selected from the group consisting of duckbill valves, cross slit valves, dispensing valves, flange valves, flap valves, and combinations thereof. Any one-way valve known to one of skill to work in such a micro environment can be used.

Those of skill understand that guidewires can be used to locate a diseased region, or target region, in a blood vessel. Also, a guidewire can be used to direct the thrombectomy devices taught herein over the target region. In some embodiments, the guidewire lumen diameter can range in size from 0.01 to 0.20 inches, from 0.01 to 0.18 inches, from 0.01 to 0.15 inches, from 0.01 to 0.10 inches, or any range therein in some embodiments. In some embodiments, the guidewire lumen diameter can range from 0.01 to 0.14 inches. In some embodiments, the guidewire lumen diameter is 0.01 inches (0.254 mm), 0.02 inches (0.508 mm), 0.04 inches (1.016 mm), 0.06 inches (1.524 mm), 0.08 inches (2.032 mm), 0.10 inches (2.540 mm), 0.12 inches (3.048 mm), 0.14 inches (3.556 mm), 0.16 inches (4.064 mm), 0.18 inches (4.572 mm), 0.20 inches (5.080 mm), or any diameter therein in increments of 0.01 inches (0.254 mm).

The tissue removed by the devices taught herein can be removed mechanically, by vacuum displacement, by a positive displacement pump, or some combination thereof. In some embodiments, the thrombectomy device has a handle that includes a motor operably connected to a positive displacement pump having a flexible driveshaft, the motor configured to provide a rotational torque to the flexible rotating shaft that actuates the positive displacement pump to transport thrombus tissue away from the tubular cutter.

In some embodiments, the thrombectomy devices can have a screw pump with a rotating helical screw for removing thrombus tissue from the blood vessel, and the right hand or left hand direction of the helix will determine whether the screw pump is rotated clockwise or counterclockwise as viewed from a proximal position of the user that is proximal to the cutting head. The cutting head and the screw pump can be rotated in the same direction in some embodiments, or opposite directions in other embodiments. A right-hand helical screw could be used for a clockwise rotation of the screw when viewed from a position proximal to the head of the cutter, whereas a left-hand helical screw (not shown) could be used for a counterclockwise rotation of the screw when viewed from a position proximal to the head of the cutter.

FIGS. 10A and 10B illustrate a rotational shaft of a thrombectomy tubular cutter system having a positive displacement pump, according to some embodiments. In some embodiments, thrombectomy tubular cutter system 1000

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includes a drive assembly that includes the flexible, rotating shaft 305 (not shown) that operably connects to the neck 225, for example, by friction fit, mechanical interlock, or adhesion, and turns the head/mouth 220, in a clockwise rotation from the perspective of the user at the proximal end of the device, to cut and remove the thrombus 100 (not shown) from the vascular lumen.

As shown in FIG. 10A, positive displacement pump 1050 is a helical screw 1050 that can be a helical-shaped wire that is fixably attached to the lumen of the neck 225 of the thrombectomy tubular cutter 200, such that a rotation of the thrombectomy tubular cutter 200 is necessary a rotation of the helical screw 1050 to convey thrombus tissue proximally through the thrombectomy tubular cutter 200. The helical-shaped wire can be designed to conform to the lumen of the neck 225 of the thrombectomy device 200 by selecting a radius, r , that matches the radius of the lumen of the neck 225 of the thrombectomy device 200. The pitch of the helix, P , can be selected to obtain an effective transport of thrombus tissue. The positive displacement pump 1050 can be further assisted by configuring the system such that the positive displacement pump is in communication with a vacuum source (not shown) to draw the excised tissue out of the subject.

In some embodiments, the thrombectomy device has a handle that includes a motor operably connected to a screw pump having a drive assembly (not shown) operably connected to the thrombectomy tubular cutter 200 to turn the head 200 to cut the thrombus tissue 100, and concurrently turn the helical screw 1050 that is fixably attached to the lumen of the neck 225 of the thrombectomy tubular cutter 200. The motor can be configured to provide a rotational torque to the thrombectomy tubular cutter that actuates the cutting of the thrombus tissue and the transport of the thrombus tissue proximally with the helical screw. The helical-shaped wire can be a right-handed helix much like a right-handed drillbit, or a left-handed helix, much like a left-handed drillbit, where the right-handed helix is used when turning the thrombectomy tubular cutter 200 clockwise when viewed from a position proximal to the thrombectomy tubular cutter 200, and the left-handed helix is used when turning the thrombectomy tubular cutter 200 counterclockwise when viewed from a position proximal to the thrombectomy tubular cutter 200. As noted, a source of vacuum can be used to facilitate removal of the excised tissue from the subject.

FIG. 10B illustrates a cross-section of the neck and flexible, rotational shaft of a thrombectomy tubular cutter system having a positive displacement pump, according to some embodiments. As shown in FIG. 10B, the positive displacement pump can rotate with the drive assembly, and independent of the thrombectomy tubular cutter 200 in some embodiments. For example, the positive displacement pump can have a rotating screw pump 1050 with a helical screw 1020 to move the thrombus tissue. In some embodiments, the positive displacement pump can rotate independent of the rotation of the drive assembly for the cutting head 200. In some embodiments, the positive displacement pump can be a screw pump 1050 configured with a helical screw 1020 located on a flexible driveshaft 1030 that rotates independent of the rotation of the cutting head 200. As noted, a source of vacuum can be used to facilitate removal of the excised tissue from the subject.

As such, the thrombectomy devices can have a handle with a motor; and a flexible driveshaft operably connected to a helical screw; wherein, the motor is operably connected to a drive assembly that includes the flexible rotating shaft and

the tubular cutter, the motor configured to provide a rotational torque to the flexible rotating shaft for rotating the tubular cutter and cutting the thrombus; and, a flexible driveshaft 1030 within the flexible rotating shaft 305 for rotating the helical screw to transport thrombus tissue away from the tubular cutter.

Indications and Design of Devices

The versatile devices, systems, and methods taught herein can handle each of the variety of soft, tough, fibrous, and hard tissue effectively, and as such, can be used in thrombectomies and atherectomies. A thrombectomy is the removal of a blood clot from a blood vessel. Blood clots are made up of platelets and a meshwork of protein strands called fibrin. Clots in arteries have a different composition than clots in veins, in which clots in arteries contain mostly platelets and clots in veins contain mostly fibrin. Common applications for thrombectomies include any location in a blood vessel where a thrombus may occur. The thrombectomy removes the blood clot from the wall of a blood vessel to help alleviate symptoms of the condition, as well as the downstream complications that can include further vasculature complications, and perhaps even death from the release of an embolus. Atherosclerosis is also referred to as plaque on the luminal wall of an artery, and plaque includes deposits of fatty substances, cholesterol, cellular waste products, calcium, and fibrin. Both the atherosclerosis and thrombus pose a risk of fragmenting into the blood stream, and moving to the heart, brain, or lungs, causing health complications and often proving to be fatal. The devices, systems, and methods taught herein can be used to treat all such indications that involve the removal of tissue from a blood vessel.

The thrombus can be in a vein as a venous thrombosis, or an artery as an arterial thrombosis. An indication of particular importance is the removal of a thrombus from a pulmonary artery, also referred to as a pulmonary embolism. The size of the artery or vein can be used to select the size, or diameter of the head 220, of the thrombectomy tubular cutter 200. Peripheral vascular disease in the legs is an example of a condition that can be treated using the thrombectomy devices taught herein.

In some embodiments, the diameter of a target blood vessel can be used to size the diameter, D_s , of the head of the tubular cutter 200. In some embodiments, the diameter of the target blood vessel can be a reference point as the maximum diameter to use as the diameter of the head 220 of the tubular cutter. However, in some embodiments, the blood vessel is much larger than the diameter, D_s , of the tubular cutter 200.

It should be appreciated blood vessels of a wide range of lumen diameters can be treated. The lumen diameters can range for example, from about 2.0 mm to about 30.0 mm, from about 3.0 mm to about 25.0 mm, from about 4.0 mm to about 24.0 mm, from about 5.0 mm to about 23.0 mm, from about 6.0 mm to about 25.0 mm, from about 7.0 mm to about 21.0 mm, from about 8.0 mm to about 20.0 mm, or any range or amount therein in increments of 0.1 mm. In some embodiments, the lumen diameters of the blood vessels treated can be 2.0 mm, 3.0 mm, 4.0 mm, 5.0 mm, 6.0 mm, 7.0 mm, 8.0 mm, 9.0 mm, 10.0 mm, 11.0 mm, 12.0 mm, 13.0 mm, 14.0 mm, 15.0 mm, 16.0 mm, 17.0 mm, 18.0 mm, 19.0 mm, 20.0 mm, 21.0 mm, 22.0 mm, 23.0 mm, 24.0 mm, 25.0 mm, 26.0 mm, 27.0 mm, 28.0 mm, 29.0 mm, 30.0 mm or any range or amount therein in increments of 1.0 mm.

In some embodiments, the diameter, D_s , of the head 220 of the tubular cutter 200 is selected to be reduced from the diameter of the target vessel in an amount of at least 1%, 5%,

10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, or any amount or range therein in increments of 1%. In some embodiments, the diameter, D_s , of the head 220 of the tubular cutter is selected to be reduced from the diameter of the target vessel in an amount ranging from 1% to 10%, from 1% to 15%, from 1% to 20%, from 1% to 25%, from 1% to 30%, from 1% to 40%, from 1% to 50%, or any amount or range therein in increments of 1%. In some embodiments, the diameter, D_s , of the head 220 of the tubular cutter is selected to be reduced from the diameter of the target vessel in an amount ranging from 5% to 10%, from 5% to 15%, from 5% to 20%, from 5% to 25%, from 5% to 30%, from 5% to 40%, from 5% to 50%, or any amount or range therein in increments of 0.1%.

The main targets and indications for use with the devices, systems, and methods taught herein includes the removal of venous thrombus. In some embodiments, the systems, devices, and methods can be configured for removal of a venous thrombus, and the lumen diameters of the blood vessels treated can range, for example, from about 4.0 mm to about 16.0 mm, or any amount or range therein in increments of 1.0 mm. In these embodiments, the outer diameter of the tubular cutter and/or the flexible rotating tube can range from about 4 mm to about 8 mm, or any amount or range therein in increments of 1.0 mm.

In some embodiments, the systems, devices, and methods taught herein can be configured for removal of a pulmonary arterial thrombus, and the lumen diameters of the blood vessels treated can range, for example, from about 6.0 mm to about 25.0 mm, or any amount or range therein in increments of 0.1 mm. In these embodiments, the outer diameter of the tubular cutter and/or the flexible rotating tube can range from about 4 mm to about 8 mm, or any amount or range therein in increments of 0.1 mm.

Table 1 provides examples of some venous lumen diameters in mm for veins which can be treated with the devices taught herein.

TABLE 1

Examples of some venous lumen diameters in mm.		
Vein	Average (mm)	Std dev (mm)
Common femoral	13.7	2.8
Superficial femoral proximal	9.2	1.8
Superficial femoral half	8.2	1.5
Superficial femoral distal	8.5	1.4
Popliteal	8.9	1.8
Long saphenous proximal	5.0	0.9
Long saphenous half	4.0	1.0
Long saphenous distal	3.0	0.6
Short saphenous	2.9	1.0
Posterior tibial 1	3.3	0.6
Posterior tibial 2	3.5	0.7
Anterior tibial 1	2.4	0.5
Anterior tibial 2	2.3	0.6
Peroneal 1	3.0	0.8
Peroneal 2	3.0	0.9
Gastrocnemial 1	3.2	1.0
Gastrocnemial 2	2.9	0.7
Common Iliac vein (8-10 mm)	13.0	2.4
External Iliac vein	8.0	
Internal Iliac vein	4.0	

In some embodiments, a treated vein can have a lumen diameter ranging from 4 mm to 18 mm. In some embodiments, the diameter of the vein can range from 1 mm to 10 mm, from 2.0 mm to 10 mm, from 3.0 mm to 10 mm, from 4.0 mm to 10 mm, from 5.0 mm to 10 mm, from 6.0 mm to

10 mm, from 7.0 mm to 10.0 mm, from 8.0 mm to 10.0 mm or any range or amount therein in increments of 1.0 mm. In some embodiments, the treated vein can have a lumen diameter selected from the group consisting of 2.0 mm, 3.0 mm, 4.0 mm, 5.0 mm, 6.0 mm, 7.0 mm, 8.0 mm, 9.0 mm, 10.0 mm, or any amount or range therein in increments of 1.0 mm. In some embodiments, the treated vein can have a lumen diameter selected from the group consisting of 11.0 mm, 12.0 mm, 13.0 mm, 14.0 mm, 15.0 mm, 16.0 mm, 17.0 mm, 18.0 mm, 19.0 mm, 20.0 mm, 21.0 mm, 22.0 mm, 23.0 mm, 24.0 mm, 25.0 mm, 26.0 mm, 27.0 mm, 28.0 mm, 29.0 mm, 30.0 mm or any amount or range therein in increments of 1.0 mm.

Clots in the venous system can vary greatly in size, and the thrombectomy tubular cutter will need to be sized accordingly. In some embodiments, a thrombus can have a clot diameter ranging from about 1.0 mm to about 18.0 mm. In embodiments involving deep vein thrombosis, for example, a thrombus can be found in larger veins and have a clot diameter ranging from about 3.0 mm to about 8.0 mm. In some embodiments, a thrombus can be found in the common iliac veins and the inferior vena cava having a clot diameter ranging from about 7.0 mm to about 17.0 mm. In some embodiments, a thrombus can be found in the external iliac veins and common femoral veins having a clot diameter ranging from about 5 mm to about 14 mm. In some embodiments, a thrombus can be found in the superficial femoral, deep femoral and popliteal veins having a clot diameter ranging from about 4 mm to about 10 mm. And, in some embodiments, a thrombus can be found in the calf veins ranging from about 4 mm to about 8 mm.

Any vein can be treated using the devices, systems, and methods taught herein. In some embodiments, the devices, systems, and methods are designed for treating the iliac veins. And, in some embodiments, the devices, systems, and methods are designed for treating the popliteal veins. In some embodiments, the devices, systems, and methods are designed for treating any veins from the popliteal veins to the iliacs.

As noted, the devices, systems, and methods are directed mainly toward thrombectomies. However, in some embodiments, the systems, devices, and methods can be configured for treatment of arteries, or atherectomies. Table 1 provides examples of some arterial lumen diameters in mm for arteries which can be treated with the devices taught herein.

TABLE 2

Artery	Average (mm)
Aorta	17-35
Main pulmonary artery	18.5-29.5
Right pulmonary artery	12.4-19.8
Left pulmonary artery	14.3-22.1
Superior femoral	5-7
Popliteal	3.5-4.5
Anterior tibial	3.0
Tibioperoneal trunk	2.5
Posterior tibial	2.0
Peroneal	2.0

The superior femoral artery, located about mid-femur, generally has a diameter of about 5 to 7 mm, or about 0.2 to 0.25 inch. As the artery descends below the knee, the popliteal artery generally has a diameter of about 4 to 4.5 mm (0.157 inch to 0.177 inch), and then reduces to about 3.5 mm (0.137 inch) as you move in the direct of the subject's foot. The popliteal artery branches again into the anterior tibial artery and the tibioperoneal trunk, reducing further in

diameter to about 3.0 mm and then about 2.5 mm or about 0.118 inch to 0.098 inch. The tibioperoneal trunk further subdivides into the posterior tibial and peroneal arteries, further reducing in diameter to about 2.0 mm (0.078 inch).

In some embodiments, the diameters of the peripheral arteries of the leg can vary, typically, from about 2 mm to about 7 mm. Any blood vessel can contain plaque and be a prospective target area for the thrombectomy devices taught herein. For example, coronary arteries are about 3 mm in size, varies from 2.5-4.5 in diameter, and coronary arteries can be prospective target areas for the devices taught herein if designed for use in an atherectomy.

For smaller clots, the diameter of the cutter can range from about 1.0 mm to about 3.0 mm in some embodiments, 1.5 mm to 4.0 mm in some embodiments, or any range therein in increments of 0.10 mm. In some embodiments, the diameter of the cutter can be about 1.0 mm, about 2.0 mm, about 3.0 mm, about 4.0 mm, about 5.0 mm, or any diameter therein, or range therein, in increments of 0.1 mm.

For larger clots, the diameter of the cutter can range from about 3.0 mm to 9.0 mm in some embodiments, 4.0 mm to 8.0 mm in some embodiments, and 5.0 mm to 7.0 mm in some embodiments, or any amount or range therein in increments of 0.1 mm. In some embodiments, the diameter of a larger cutter can be about 3.0 mm, about 4.0 mm, about 5.0 mm, about 6.0 mm, about 7.0 mm, about 8.0 mm, about 9.0 mm, 10.0 mm or any diameter therein, or range therein, in increments of 0.1 mm.

Although it seems reasonable to simply increase the diameter of the cutter for larger blood vessels, the skilled artisan will realize that the diameter of the cutter can also be limited by physical complications of the patient's anatomy. For example, there can be complications that occur during surgery due to bleeding at the puncture access, tortuous vessels, vessel size variations, and the like. The skilled artisan can use the teachings herein, including cutter size, cutter bias, ratio of leading edge to trailing edge, cutter deflection, selection of stiffness, and any combination thereof, to design a desired thrombectomy device to obtain the desired treatment results.

The skilled artisan will also realize that the length of the cutter has to be limited to have the maneuverability needed. One of skill will realize that the size of the head of the cutter can be any length known to be suitable in the art for the particular treatment. In some embodiments, the length of the head of the cutter can range from about 4.0 mm to about 20.00 mm, from about 4.0 mm to about 18.0 mm, from about 4.0 mm to about 16.0 mm, from about 4.0 mm to about 12.0 mm, or any range or amount therein in increments of 0.10 mm. In some embodiments, the length of the head of the cutter can range from about 6.0 mm to about 20.00 mm, from about 6.0 mm to about 18.0 mm, from about 6.0 mm to about 16.0 mm, from about 6.0 mm to about 12.0 mm, or any range or amount therein in increments of 0.10 mm. In some embodiments, the length of the head of the cutter can range from about 8.0 mm to about 20.00 mm, from about 8.0 mm to about 18.0 mm, from about 8.0 mm to about 16.0 mm, from about 8.0 mm to about 12.0 mm, or any range or amount therein in increments of 0.10 mm. In some embodiments, the length of the head of the cutter can be about 1.0 mm, about 2.0 mm, about 3.0 mm, about 4.0 mm, about 5.0 mm, about 6.0 mm, about 7.0 mm, about 8.0 mm, or any diameter therein, or range therein, in increments of 0.1 mm.

Given the relative sizes of the vessels and devices taught herein, it should be appreciated by the skilled artisan that a thrombectomy device can have a shaft diameter that 2× the size of a shaft used in an atherectomy device. In some

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embodiments, this is why the torsional stiffness of a thrombectomy device can be 16× that of the atherectomy device, and the flexural stiffness of the thrombectomy devices can be 8× that of the atherectomy device. Likewise, the axial stiffness of the thrombectomy device can be about 2× that of the atherectomy device.

As discussed the devices, systems and methods can be used in thrombectomies and atherectomies, although thrombectomies are the focus. As such, methods of using thrombectomy devices and systems, and methods of using atherectomy devices and systems are provided.

In some embodiments, the teachings are directed to a method of removing tissue from a blood vessel in a subject using a device taught herein, the method comprising:

creating a point of entry in a vascular lumen of the subject;

inserting the device into the vascular lumen;

cutting tissue from the vascular lumen with the cutting head of the device;

and,

removing the device from the vascular lumen of the subject.

In some embodiments, the teachings are directed to a method of performing a thrombectomy in a subject using a thrombectomy device taught herein, the method comprising:

creating a point of entry in a vascular lumen of the subject;

inserting the thrombectomy device into the vascular lumen;

cutting thrombus tissue from the vascular lumen with the cutting head of the thrombectomy device;

and,

removing the thrombectomy device from the vascular lumen of the subject.

In some embodiments, the teachings are directed to a method of performing an atherectomy in a subject using a device taught herein, the method comprising:

creating a point of entry in a vascular lumen of the subject;

inserting the device into the vascular lumen;

cutting tissue from the vascular lumen with the cutting head of the device;

and,

removing the device from the vascular lumen of the subject.

In some embodiments, the methods further include inserting a guidewire in the point of entry and delivering the guidewire to a target location in the vascular lumen of a subject, and guiding the device to the target location on the guidewire.

In some embodiments, the methods further include discharging excised tissue from the vascular lumen with a vacuum.

The devices taught herein include several methods of removing a vascular lesion from within a subject. In fact the lesion can be removed from any blood vessel. In some embodiments, the methods can include creating a point of entry in a vascular lumen of the subject; inserting a thrombectomy device taught herein into the vascular lumen; telescoping the flexible rotating shaft; cutting a thrombus from a wall of a vascular lumen with the cutter of the device; discharging the cut tissue from the vascular lumen which in some embodiments can be facilitated with a vacuum pump; and, removing the device from the vascular lumen of the subject.

In some embodiments, the method of removing a thrombus can include obtaining a thrombectomy device taught

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herein, creating an opening into the vasculature of a subject having a thrombus, inserting the device in the vasculature of the subject, moving the distal end of the thrombectomy device to the site of the thrombus, removing the thrombus from the subject with the thrombectomy device, and removing the thrombectomy device from the subject.

In some embodiments, the method includes

inserting a device taught herein into a vascular lumen of a subject, the device having a

a flexible rotating shaft with a proximal end, a distal end, and a lumen;

a cutting head in the shape of, for example, a tubular cutter at the distal end of the flexible rotating shaft, the cutting head having a proximal end, and a distal end, the cutting head and configured with

(i) an open mouth that communicates with the lumen of the rotating shaft, the open mouth having (i) an inlet defined by an asymmetric cutting tip on the mouth, the asymmetric cutting tip having a second longitudinal plane bisecting the cutting tip; and (ii) an outlet in communication with the lumen of the rotating shaft; wherein, the second longitudinal plane is not coincident with the first longitudinal plane to provide the asymmetric cutting tip;

(ii) a vacuum port configured for operable communication with vacuum source;

a distal end, a proximal end, a long axis, and a guidewire lumen passing through the device in the direction of the long axis;

advancing the cutter to a target area in the vascular lumen of the subject, the advancing including

cutting vascular tissue from the wall of the vascular lumen, the cutting including rotating the cutter; and,

removing the vascular tissue from the vascular lumen using the positive displacement pump; and,

removing the thrombectomy device from the subject.

FIG. 11 is a flowchart of a thrombectomy method, according to some embodiments. Method 1150 can be used with the thrombectomy devices taught herein to remove a thrombus from a blood vessel. A guidewire is advanced 1174 through a guiding catheter and across a thrombus in the blood vessel in a target area. Once the guidewire is in place at the target area, the thrombectomy device is advanced 1176 to the thrombus on the guidewire. The thrombectomy device is then in proper position to cut and remove 1178 the thrombus in the target area. To complete the procedure, the thrombectomy device and guidewire are removed 1182 from the patient. The dashed line around the guidewire advancing step 1174 shows that the guidewire is optional. As such, in some embodiments, the use of a guidewire is not necessary. Reducing Risk of Vessel Injuries and Undesirable Blood Loss During Procedures

The technologies taught herein can effectively cut and remove all types of thrombus and atheroma tissue, whether soft, tough, or hard, which will be appreciated by those of skill, and particularly those devices, systems, and methods that can do so with a reduced risk of suffering vessel injuries and/or avoid an undesirable volume of blood loss during procedures.

A vascular cutter procedure can potentially cause blood vessel damage, primarily due to the mechanical interaction of the device used to remove a target tissue from the inner lining of the blood vessel. Such damage can include tears, abrasions, or even perforations, particularly when dealing with delicate or smaller vessels. Complications associated with vessel damage include bleeding due to a vessel tear or

perforation, hematoma formation around the puncture site, and vessel narrowing (stenosis) from scarring and narrowing of the blood vessel lumen.

An amount of blood loss considered acceptable during surgery depends on a patient's individual factors, including their blood volume, medical history, and the type of surgery. The estimated blood volume (EBV) of the patient is used to calculate the maximum allowable blood loss (MABL). The patient's cardiopulmonary and general medical conditions affect how much blood loss they can tolerate. And, in the devices, systems, and methods taught herein, the amount of acceptable blood loss can vary by whether the blood is re-administered to the patient during a procedure. In some embodiments, an acceptable amount of blood loss ranges from 0.1% by volume to up to 14% by volume. In some embodiments, an acceptable amount of blood loss ranges from 0.1% by volume to up to 30% by volume. In some embodiments, an acceptable amount of blood loss in % by volume is 0.1%, 0.5%, 1.0%, 2.0%, 3.0%, 4.0%, 5.0%, 6.0%, 7.0%, 8.0%, 9.0%, 10.0%, 11.0%, 12.0%, 13.0%, 14.0%, 15.0%, 16.0%, 17.0%, 18.0%, 19.0%, 20.0%, 21.0%, 22.0%, 23.0%, 24.0%, 25.0%, 26.0%, 27.0%, 28.0%, 29.0%, 30.0%, 31.0%, 32.0%, 33.0%, 34.0%, 35.0%, or any amount or range therein in amounts of 0.1% by volume.

In some embodiments, vascular cutter systems, devices, and methods of use are provided with a reduced risk of suffering vessel injuries while avoiding an undesirable volume of blood loss during procedures. In some embodiments, these systems, devices, and methods include a pulsating vacuum control. In some embodiments, the pulsating vacuum control can provide a reduced risk of suffering vessel injuries. In some embodiments, the pulsating vacuum control can avoid an undesirable volume of blood loss during procedures. In some embodiments, the pulsating vacuum control can provide both a reduced risk of suffering vessel injuries and avoid an undesirable volume of blood loss during procedures.

The pulsating vacuum control embodiments are their own embodiments and designs, and can be considered standalone technologies apart from the vascular cutter devices, systems, and methods taught herein. Likewise, the pulsating vacuum control can also be used in connection with any vascular cutter. In some embodiments, the pulsating vacuum control can be used in systems constructed and designed to control blood loss during a vascular tissue removal procedure.

In some embodiments, the systems can comprise:

- a vascular cutter device designed to remove diseased vascular tissue, the vascular cutter device having a cutting head with a cutting edge on a mouth, and a lumen operably connected to an aspiration tube for discharging the diseased vascular tissue from the vascular lumen;

and,

- a pulsating vacuum control assembly in an operable connection with the aspiration tube and having a valve configured to open and close at one or more desired frequencies to create a pulsating vacuum during operation of the device; and,
- a port configured for operable communication with a vacuum source to facilitate transport of the tissue out of the subject.

In some embodiments, any one or any combination of vacuum pressures can be used. In some embodiments, the one or more desired frequencies pulses the vacuum from a constant high pressure ranging from 100 mmHg to 800

mmHg to a constant low pressure ranging from 10 mmHg to 100 mmHg. In some embodiments, the one or more desired frequencies pulses the vacuum from a variable high pressure ranging from 100 mmHg to 800 mmHg to a variable low pressure ranging from 10 mmHg to 100 mmHg. Any pressure regulator configuration can be used to control the pressure. In some embodiments, the low pressure or pressures, whether constant or variable, can be selected from the group consisting of 10 mmHg, 20 mmHg, 30 mmHg, 40 mmHg, 50 mmHg, 60 mmHg, 70 mmHg, 80 mmHg, 90 mmHg, 100 mmHg, and any pressure therein in increments of 1 mmHg, and combinations thereof. In some embodiments, the high pressure or pressures, whether constant or variable, can be selected from the group consisting of 100 mmHg, 110 mmHg, 120 mmHg, 130 mmHg, 140 mmHg, 150 mmHg, 160 mmHg, 170 mmHg, 180 mmHg, 190 mmHg, 200 mmHg, 210 mmHg, 220 mmHg, 230 mmHg, 240 mmHg, 250 mmHg, 260 mmHg, 270 mmHg, 280 mmHg, 290 mmHg, 300 mmHg, 310 mmHg, 320 mmHg, 330 mmHg, 340 mmHg, 350 mmHg, 360 mmHg, 370 mmHg, 380 mmHg, 390 mmHg, 400 mmHg, 410 mmHg, 420 mmHg, 430 mmHg, 440 mmHg, 450 mmHg, 460 mmHg, 470 mmHg, 480 mmHg, 490 mmHg, 500 mmHg, 510 mmHg, 520 mmHg, 530 mmHg, 540 mmHg, 550 mmHg, 560 mmHg, 570 mmHg, 580 mmHg, 590 mmHg, 600 mmHg, 610 mmHg, 620 mmHg, 630 mmHg, 640 mmHg, 650 mmHg, 660 mmHg, 670 mmHg, 680 mmHg, 690 mmHg, 700 mmHg, 710 mmHg, 720 mmHg, 730 mmHg, 740 mmHg, 750 mmHg, 760 mmHg, 770 mmHg, 780 mmHg, 790 mmHg, 800 mmHg, and any pressure therein in increments of 1 mmHg, and combinations thereof.

The term "open" can refer to a partial opening of the valve or complete opening of the valve. In some embodiments, the pressure. In some embodiments, a complete opening of the valve can provide a high pressure of greater than 100 mmHg. In some embodiments, a complete opening of the valve can provide a high pressure ranging from 10 mmHg to 100 mmHg, 100 mmHg to 800 mmHg, or any pressure therein in increments of 1 mmHg. The degree to which a valve is opened over the course of a vascular tissue removal procedure can be random, variable, or constant, or any combination thereof, ranging from 99% closed to 100% open, or any percent therein in increments of 1%.

The valve can be controlled to open and close in any amount designed, and to follow any pattern of opening and closing desired. The term "close" can refer to a partial closure of the valve or complete closure of the valve. In some embodiments, a complete closure of the valve can provide a low pressure greater than 0 mmHg. In some embodiments, a complete closure of the valve can provide a low pressure ranging from 0 mmHg to 100 mmHg, or any pressure therein in increments of 1 mmHg, in embodiments where the pressure does not have sufficient time to drop to 0 mmHg. In some embodiments, a partial closure of the valve can provide a low pressure ranging from 0 mmHg to 100 mmHg, 100 mmHg to 800 mmHg, or any pressure therein in increments of 1 mmHg. The degree to which a valve is closed over the course of a vascular tissue removal procedure can be random, variable, or constant, or any combination thereof, ranging from 100% closed to 99% open, or any percent therein in increments of 1%. In some embodiments, the valve can be open 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%,

57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 100%, and any % therein in increments of 0.1%. In some embodiments, the valve can be closed 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 100%, and any % therein in increments of 0.1%. In some embodiments, the % open and % closed can be constant or variable during a treatment. In some embodiments, the valve oscillates from 100% closed to 100% open. In some embodiments, the valve oscillates from 50% closed to 100% open. In some embodiments, the valve oscillates from 25% closed to 100% open, from 50% closed to 100% open, from 100% closed to 90% open, or any amount closed or open desired for a particular indication, or any combination thereof. In some embodiments, the vacuum pressures provided by the device can be any one or any combination of pressures and frequencies, constant, variable, or random.

Any one or any combination of vacuum pulse frequencies can be used to pulse, or cycle, a vacuum, in either a constant or variable rhythm. In some embodiments, the one or more desired frequencies is a constant frequency that ranges from 1-5 Hz. In some embodiments, the one or more desired frequencies is a variable frequency that ranges from 1-5 Hz. In some embodiments, the one or more desired vacuum pulse frequencies can range from 0.5 Hz to 50 Hz, from 0.5 Hz to 40 Hz, from 0.5 Hz to 30 Hz, from 0.5 Hz to 20 Hz, from 0.5 Hz to 10 Hz, from 0.5 Hz to 5.0 Hz, or any amount or range therein in increments of 0.1 Hz. In some embodiments, the one or more desired vacuum pulse frequencies can be 0.1 Hz, 0.2 Hz, 0.3 Hz, 0.4 Hz, 0.5 Hz, 0.6 Hz, 0.7 Hz, 0.8 Hz, 0.9 Hz, 1.0 Hz, 2.0 Hz, 3.0 Hz, 4.0 Hz, 5.0 Hz, 6.0 Hz, 7.0 Hz, 8.0 Hz, 9.0 Hz, 10.0 Hz, 11 Hz, 12 Hz, 13 Hz, 14 Hz, 15 Hz, 16 Hz, 17 Hz, 18 Hz, 19 Hz, 20 Hz, or any amount or range therein in amounts of 0.1 Hz.

The "dwell" or duration of the pulse, expressed in milliseconds per pulse (ms/pulse), is also a feature of the vacuum pulse that can be controlled. In some embodiments, the dwell of a vacuum pulse can range from 100 ms/pulse to 5000 ms/pulse, from 250 ms/pulse to 5000 ms/pulse, from 500 ms/pulse to 5000 ms/pulse, from 750 ms/pulse to 5000 ms/pulse, from 1000 ms/pulse to 5000 ms/pulse, from 1250 ms/pulse to 5000 ms/pulse, from 1500 ms/pulse to 5000 ms/pulse, from 1750 ms/pulse to 5000 ms/pulse, from 2000 ms/pulse to 5000 ms/pulse, from 2500 ms/pulse to 5000 ms/pulse, from 3000 ms/pulse to 5000 ms/pulse, from 3500 ms/pulse to 5000 ms/pulse, from 4000 ms/pulse to 5000 ms/pulse, from 4500 ms/pulse to 5000 ms/pulse, 250 ms/pulse to 4000 ms/pulse, from 500 ms/pulse to 4000 ms/pulse, from 750 ms/pulse to 4000 ms/pulse, from 1000 ms/pulse to 4000 ms/pulse, from 1250 ms/pulse to 4000 ms/pulse, from 1500 ms/pulse to 4000 ms/pulse, from 1750 ms/pulse to 4000 ms/pulse, from 2000 ms/pulse to 4000 ms/pulse, from 2500 ms/pulse to 4000 ms/pulse, from 3000 ms/pulse to 4000 ms/pulse, from 3500 ms/pulse to 4000 ms/pulse, 250 ms/pulse to 3000 ms/pulse, from 500 ms/pulse to 3000 ms/pulse, from 750 ms/pulse to 3000

ms/pulse, from 1000 ms/pulse to 3000 ms/pulse, from 1250 ms/pulse to 3000 ms/pulse, from 1500 ms/pulse to 3000 ms/pulse, from 1750 ms/pulse to 3000 ms/pulse, from 3000 ms/pulse to 4000 ms/pulse, from 2500 ms/pulse to 3000 ms/pulse, or any amount or range therein in increments of 100 ms/pulse. In some embodiments, the dwell of a vacuum pulse can be 100 ms/pulse, 200 ms/pulse, 300 ms/pulse, 400 ms/pulse, 500 ms/pulse, 600 ms/pulse, 700 ms/pulse, 800 ms/pulse, 900 ms/pulse, 1000 ms/pulse, 1100 ms/pulse, 1200 ms/pulse, 1300 ms/pulse, 1400 ms/pulse, 1500 ms/pulse, 1600 ms/pulse, 1700 ms/pulse, 1800 ms/pulse, 1900 ms/pulse, 2000 ms/pulse, 2100 ms/pulse, 2200 ms/pulse, 2300 ms/pulse, 2400 ms/pulse, 2500 ms/pulse, 2600 ms/pulse, 2700 ms/pulse, 2800 ms/pulse, 2900 ms/pulse, 3000 ms/pulse, 3100 ms/pulse, 3200 ms/pulse, 3300 ms/pulse, 3400 ms/pulse, 3500 ms/pulse, 3600 ms/pulse, 3700 ms/pulse, 3800 ms/pulse, 3900 ms/pulse, 4000 ms/pulse, 4100 ms/pulse, 4200 ms/pulse, 4300 ms/pulse, 4400 ms/pulse, 4500 ms/pulse, 4600 ms/pulse, 4700 ms/pulse, 4800 ms/pulse, 4900 ms/pulse, or any amount or range therein in increments of 50 ms/pulse.

The frequency, dwell, and amount of vacuum can be controlled manually by the user, or they can be controlled automatically by the devices, systems, and methods taught herein. In some embodiments, the automatic control can be mechanically driven or computer driven. In some embodiments, a vacuum valve can be opened mechanically, electromechanically, or electrically using a computer control system that actuates a valve through a processor and a memory.

A mechanically driven configuration of controlling the frequency, dwell, and/or amount of vacuum can be accomplished using any mechanical valve configuration known to one of skill to pulse, or cycle, the vacuum with a desired frequency and/or dwell.

In some embodiments, a cam can be used to open a valve to control the frequency and dwell of a vacuum. Any known valve and cam configuration can be used. FIGS. 12A, 12B1-12B3, and 12C illustrate a mechanical valve assembly that includes a cam for opening a valve, according to some embodiments. FIG. 12A shows a portion of a vacuum manifold 1205 present in a valve assembly 1200 with a valve 1210, valve stem 1215, valve spring 1220, valve seat 1225, and vacuum port 1250. When the valve 1210 is in contact with the valve seat 1225, the vacuum can draw through the vacuum port 1250 to aspirate from the subject through the inside of the vacuum manifold 1200. FIGS. 12B1-12B3 show how the rotation of cam 1260, as shown by the rotation arrow, can be used to open the valve by applying a force to rod 1270, which can be the valve stem 1215 itself, or an intermediary that then applies a force to the valve stem 1215, such intermediary being, for example, a push rod, a follower, or the like. A second intermediary, such as a rocker arm (not shown) can also be used to apply pressure indirectly to the valve stem 1215. FIG. 12C shows how the cam 1260 has a body 1263 and a lobe 1265. The lobe 1265 starts where the curve of the body 1263 begins to leave the circular curve and enter the shape of the lobe 1265 on each side of the lobe 1265. The dwell time is proportional to the width of the lobe 1265 and depends on the revolutions per minute (rpm) of the cam. The rpm of the cam can be adjusted using any means known to the skilled artisan, including gear ratios, motor speed, combinations thereof, and the like. In some embodiments, the valve is opened cyclically by a cam having a lobe designed for a desired dwell and vacuum profile. In some embodiments, the valve is opened cyclically by a cam having a lobe designed for a desired dwell and vacuum

profile, and the frequency can be adjusted by speeding up or slowing down the rpm of the rotating shaft.

An electromechanically driven configuration of controlling the frequency, dwell, and/or amount of vacuum can be to pulse, or cycle, the vacuum with a desired frequency and/or dwell.

In some embodiments, a solenoid can be used to open a valve to control the frequency and dwell of a vacuum. Any known valve and solenoid configuration can be used. FIG. 13 illustrates an electromechanical valve assembly that includes a solenoid that receives an electric signal to apply a mechanical pressure for opening a valve, according to some embodiments. Vacuum assembly 1300 shows a portion of a vacuum manifold 1205. Vacuum assembly 1300 includes a valve 1210, valve stem 1215, valve spring 1220, valve seat 1225, and vacuum port 1250. Valve assembly 1300 can open the vacuum to draw aspirate from a subject by applying a force from a rod 1340 on the valve stem 1215, compressing valve spring 1220 to separate valve 1210 from the valve seat 1225. The direction of the arrow is the direction of the force applied to valve stem 1215 from the rod 1340. The rod can be a push rod apply the force to the valve stem 1215, or the rod can be the valve stem 1215 itself, in some embodiments. The force is generated by solenoid 1310 which includes a charge applied to solenoid coil 1320 to generate magnetic field 1330 which creates the force on the valve stem 1215 in the direction of the arrow.

A valve can also be opened electrically through a mechanically driven or electromechanically driven configuration of controlling the frequency, dwell, and/or amount of vacuum can be to pulse, or cycle, the vacuum with a desired frequency and/or dwell. A computer control system can be used to program the pulse, or cycle, of the vacuum with a desired frequency and/or dwell.

FIG. 14 illustrates a computer system that can be used to open a valve electrically, according to some embodiments. The computer system 1400 can be used to control any valve assembly in a vacuum manifold. A processor 1410 and memory 1420 can be configured for use with the devices, systems, and methods taught herein to control one or a combination of the frequency, dwell, amount of vacuum, or a combination thereof, of a vacuum pulse.

Robotic systems are included in the teachings herein. Any robotic system configured with a pulsating vacuum can benefit from the technologies herein. In some embodiments, a robotic vascular cutter system can include a device, system, or method taught herein for a vascular cutter procedure with the processor 1410 and the memory 1420, the memory programmed with instructions configured to instruct the processor to control one or a combination of the frequency, dwell, and amount of vacuum, of one or more vacuum pulses. Any configuration taught herein can be used to facilitate the operation of a robotic system.

Examples of processors that can be used include, but are not limited to, processors, field-programmable gate arrays (FPGAs), graphics processing units (GPUs), central processing units (CPUs), systems on a chip (SoCs), programmable logic controllers (PLCs), and programmable I/Os. Examples of medical grade processors include, but are not limited to, Intel Core i5, i7, and Celeron, as well as the Intel Atom E3900 series, each of which can be used, in some embodiments. In some embodiments, the processor can include an Intel Core i5/i7 13th generation processor. In some embodiments, the processor can include an Intel Atom E3900 series. In some embodiments, the processor can include an 11th generation Intel Core Tiger Lake i5/i7 & Celeron processor. In some embodiments, the processor can

include an Intel Skylake Celeron/Core i5 processor. In some embodiments, the processor can include a Samsung Electronics' Exynos i T100 processor.

The memory in any embodiment can be on a non-transitory computer readable medium for use as an engine or a module, for example, that can be individually programmed to instruct the processor 1410 to control of the frequency, dwell, amount of vacuum, or a combination thereof, of one or more vacuum pulses. In some embodiments, a pulsating vacuum engine can be programmed to instruct the processor 1410 to control the frequency, dwell, amount of vacuum, or a combination thereof, of one or more vacuum pulses. In some embodiments, a frequency module 1440 can be programmed to instruct the processor 1410 to control the frequency of one or more vacuum pulses. In some embodiments, a dwell module 1450 can be programmed to instruct the processor 1410 to control the dwell time of one or more vacuum pulses. In some embodiments, a vacuum module 1460 can be programmed to instruct the processor 1410 to control the strength, or amount of vacuum, of one or more vacuum pulses. In some embodiments, any combination of at least 2 of the modules, the frequency module 1440, dwell module 1450, and vacuum module 1460, can be programmed to instruct the processor 1410 to control the frequency, dwell time, and amount of vacuum, of one or more vacuum pulses. In some embodiments, a default module can be programmed to a default setting for any one or any combination of the frequency, dwell time, and amount of vacuum, of one or more vacuum pulses, wherein the default setting can provide a starting point for a custom configuration for the devices, systems, and methods taught herein.

In some embodiments, the devices, systems, and methods can include a frequency engine and/or module with a vacuum engine and/or module, wherein the dwell is constant. In some embodiments, the devices, systems, and methods can include a dwell engine and/or module with a vacuum engine and/or module, wherein the frequency is constant. In some embodiments, the devices, systems, and methods can include a frequency engine and/or module with a dwell engine and/or module, wherein the vacuum is constant. In some embodiments, the devices, systems, and methods can include a frequency engine and/or module with a dwell engine and/or module and a vacuum engine and/or module, wherein each of the frequency, dwell, and vacuum can be programmed to be constant or variable, in any combination.

Of course, in some embodiments, a user can manually open a valve by simply applying a hand or finger pressure to a manual actuator that mechanically actuates the opening and closing of a valve assembly, for example. It should be appreciated that most any valve assembly can be designed to be manually operated. For example, the solenoid assembly 1310 can be replaced with the hand of a user, such that the force 1340 can be applied from a hand or finger of a user on the valve stem 1215.

As described herein, any known valve design, and/or method of creating a pulsating vacuum can be used. In some embodiments, the pulsating vacuum can include pulsing power electronically to the vacuum. In some embodiments, the pulsing of the power can be an on/off or variable power, the variable power controlled via a rheostat in some embodiments, and the variable rheostat can be controlled using a computer system as taught herein. In some embodiments, the pulsating vacuum can include opening and closing a valve mechanically. In some embodiments, the pulsating vacuum can include opening and closing a valve electromechanically.

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chanically. And, in some embodiments, the pulsating power can control a vacuum manifold component. The terms “manifold assembly”, “vacuum manifold”, “aspiration manifold”, and “pulsating vacuum control assembly” can be used interchangeably in some embodiments.

A vacuum manifold can be configured to house a mechanical valve assembly designed to pulse the vacuum at a desired frequency and dwell. FIGS. 15A-15D illustrate a vacuum manifold assembly for creating a pulsating vacuum with a mechanical valve configuration that includes a rotating shaft having an aspiration port that rotates within a sleeve bearing having an aspiration outlet, according to some embodiments. In FIGS. 15A-15D, the vacuum manifold 1500 is configured to create the pulsating vacuum 1505 by mechanically opening and closing a valve assembly 1510.

The valve assembly, in some embodiments, can include a rotating shaft 1515 with a wall having an aspiration port 1520; and,

a sleeve bearing 1530 having a proximal end, a distal end, and a wall with an aspiration outlet 1525, wherein the rotating shaft 1515 is rotatably translatable within the sleeve bearing 1530.

The aspiration manifold can have a proximal end, a distal end and a lumen 1501, the aspiration manifold 1500 configured to house the sleeve bearing 1530 and operably connect with the vacuum source through an aspiration tube 1550 to facilitate the transport of the cut diseased tissue out of the subject when the aspiration port 1520 of the flexible rotating shaft aligns with the aspiration outlet 1525 of the sleeve bearing 1530;

wherein, the aspiration port 1520 is configured for aligning with an aspiration outlet 1525 of the sleeve bearing 1530.

The aspiration tube 1550 can be closed with a clamp 1551, in some embodiments. The aspiration port 1520 can be configured with a width 1521 to align with the width 1522 of the aspiration outlet. The lumen 1501 is enclosed by shell casing 1502, a distal endcap 1503, and a proximal fitting 1504, the combination of which can be configured to provide a seal for the vacuum. In some embodiments, the shell casing 1502 can be opaque. In some embodiments, the shell casing 1502 can be either transparent or translucent for ease of inspection. The sleeve bearing 1530 and/or the distal endcap 1503 can be sealed around the rotating shaft 1515 to sustain a desired vacuum during operation of the valve assembly 1510. Proximal fitting 1504 can also be sealed around the rotating shaft 1515 to sustain a desired vacuum during operation of the valve assembly 1510 and is a fitting open to couple at a coupling port 1506 with the aspiration tube 1550 and communication with the vacuum source. In some embodiments, the rotating shaft 1515 can be flexible and referred to as a flexible rotating shaft 1515. In some embodiments, a portion of the rotating shaft 1515 can be rigid in the vacuum manifold and become flexible at some point after it passes through the distal endcap 1503, so that the flexibility of the rotating shaft 1515 can provide maneuverability in a vascular cutting procedure. In some embodiments, the vascular cutting procedure can be a thrombectomy.

In some embodiments, the sleeve bearing 1530 can be housed at the distal end of the lumen 1501 of the aspiration manifold 1500; and, the aspiration outlet 1525 of the sleeve bearing 1530 is an elongate opening that extends to the proximal end of the sleeve bearing 1530 to direct the diseased tissue out of the subject through the aspiration manifold 1500.

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The relative shapes of the aspiration port and aspiration outlet can be varied to shape the profile of the vacuum pulses that are obtained from the valve assembly. The difference in relative shapes change the size of the opening in which the vacuum passes during a revolution of the rotating shaft within the sleeve bearing. In some embodiments, the shape of the aspiration port in the rotating shaft can be constant, and the shape of the aspiration outlet in the sleeve bearing can be varied to obtain a desired vacuum profile. In some embodiments, the shape of the aspiration port in the rotating shaft can be varied, and the shape of the aspiration outlet in the sleeve bearing can be constant to obtain a desired vacuum profile. In some embodiments, the shape of the aspiration port in the rotating shaft and the shape of the aspiration outlet in the sleeve bearing can be varied to obtain a desired vacuum profile.

FIGS. 16A-16F illustrate a variety of sleeve bearing configurations that can be designed and used in a vacuum manifold to obtain a desired vacuum pulse profile, according to some embodiments. In FIG. 16A-16F, the aspiration port 1520 of the rotating shaft 1515 is an elongate rectangle, width, w, and height, h, of each rectangle is configured to align with the aspiration outlet 1525 to create a maximum flow at a maximum alignment. The exception is in FIG. 16D, in which the aspiration port 1520 of the rotating shaft 1515 has a width that aligns with the smaller width of the two openings that are cut into the sleeve bearing 1530 to create two different vacuum pulse lengths in one revolution of the rotating shaft 1515.

As shown in FIG. 16A, the original aspiration outlet 1525 has a size that matches the size of the aspiration port 1520, and the solid line in the line graph represents a pulse profile of the vacuum that may be obtained from a matched relationship between the aspiration outlet 1525 and the aspiration port 1520. The dashed representation of the aspiration outlet 1525 in FIG. 16A allows the vacuum to start pulling earlier and to continue to pull later in the revolution of the rotating shaft 1515, providing the dashed line in the line graph representing a modified pulse profile of the vacuum. FIG. 16B provides a “dog bone” shape of the aspiration outlet 1525 to again allow the vacuum to start pulling earlier and to continue to pull later in the revolution of the rotating shaft 1515, providing the dashed line in the line graph representing a modified pulse profile of the vacuum. It appears that FIG. 16A may provide a wider vacuum pulse profile, and thus a longer dwell at the same revolution speed, or rpm, of the rotating shaft 1515. One difference between FIGS. 16A and 16B is the location of the vacuum pull in the valve assembly, where the aspiration outlet 1525 FIG. 16A distributes the expanded pull across the entire length of the aspiration outlet 1525, and the aspiration outlet 1525 of FIG. 16B localizes the early vacuum draw at the two ends of the aspiration outlet 1525. Compare this to FIG. 16C which appears to have a similar vacuum profile as FIG. 16B, but the early vacuum draw is localized at the center of the aspiration outlet 1525. The distribution and/or localization of the vacuum draw can be a design variable used to help facilitate a clean vacuum pull without plugging, in some embodiments.

The valve assembly 1510 can be configured to provide more than one vacuum pulse per revolution of the rotating shaft 1515. FIG. 16D shows a configuration that provides a two pulses of vacuum on one revolution of the rotating shaft 1515, each of the vacuum pulses having a different profile, namely a different dwell, or dwell time. FIG. 16E provides an even longer dwell time of a vacuum pulse, the aspiration outlet 1525 widened. The aspiration port 1520 of the rotating

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shaft **1515** can further control the amount of vacuum and dwell by controlling its height and width wherein, in this embodiment, the height and width are designed to match the aspiration outlet **1525** to maximize the draw and dwell of the vacuum. In FIG. **16F**, aspiration outlets **1525** of equal size is placed on opposite sides of the sleeve bearing to create equally spaced and timed pulses in a single revolution of the rotating shaft. In embodiments having more than a single aspiration outlet **1525**, a sleeve hub **1527** can be included in the cut-out of the sleeve bearing **1530** to provide a structure that can maintain the shape and integrity of the sleeve bearing **1530** during operation.

In some embodiments, the sleeve bearing has a plurality of aspiration outlets. In some embodiments, the rotating shaft has a plurality of aspiration ports. In some embodiments, the sleeve bearing has a plurality of aspiration outlets and the rotating shaft has a plurality of aspiration ports. In some embodiments, the sleeve bearing has more aspiration outlets than the rotating shaft has aspiration ports. In some embodiments, the sleeve bearing has less aspiration outlets than the rotating shaft has aspiration ports.

In some embodiments, the aspiration manifold can be operably placed in a handle of a vascular cutting device. In some embodiments, the aspiration manifold can be operable placed outside the handle of a vascular cutting device.

FIG. **17** illustrates the placement of an aspiration manifold inside the handle of a vascular cutting device. Aspiration manifold **1500** is placed inside handle **1700** with a rotating shaft **1515** having a lumen that is in communication a vacuum source for vacuum flow ("Vac flow") through an aspiration tube **1550** that draws a vacuum through the lumen **1517** of the rotating shaft **1515** by operation of the valve assembly **1510** in aspiration manifold **1500**.

The systems can include a motor that drives the vascular cutter. In some embodiments, the motor configured to drive the vascular cutter can be located in a handle for ease of control by a user of the device. In some embodiments, the motor configured to drive the vascular cutter can be located outside a handle for control by a user of the device.

As shown in FIG. **17**, the systems can further comprise a handle that includes

a motor **1725** operably connected to a drive assembly that includes a rotating shaft **1515** with a proximal end, a distal end, and a lumen **1517**, a cutting head **1750** operably connected to the distal end of the rotating shaft **1515**, and the motor configured to provide a rotational movement to the rotating shaft **1515** for rotating the cutting head **1750**; and, the pulsating vacuum control assembly **1500, 1510**.

The motor **1725** can be powered by battery power from a battery assembly **1740**. The motor can power the rotating shaft through a gear box **1760** having a driveshaft gear **1763** directly driven by the motor that translates the power to a rotating shaft gear **1765** that directly drives the rotating shaft **1515**. The rotating shaft **1515** can be supported by a rotating shaft bearing **1770**, in some embodiments. In some embodiments, the handle **1700** can include a manual rotation knob **1717** that can be used to rotate the rotating shaft **1515** manually.

An actuator control of the motor can be independent, or shared with, the control of the pulsating vacuum. In some embodiments, the systems can include a motor actuator and a pulsating vacuum actuator. In some embodiments, the systems can include a combination actuator for actuating the motor and the pulsating vacuum control assembly to create the vacuum simultaneously with actuation of the motor at an on-position and stop the vacuum and the motor simultane-

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ously upon actuation to an off-position. In some embodiments, the combination actuator actuates a valve that opens to create the vacuum simultaneously with actuation of the cutter to an on-position and closes the vacuum upon actuation of the cutter to an off-position. In some embodiments, the combination actuator actuates a solenoid that opens a valve to open the vacuum simultaneously with actuation of the cutter to an on-position and closes the vacuum upon actuation of the cutter to an off-position.

FIG. **17** illustrates an actuator **1780** that simultaneously controls the motor and the pulsating vacuum. The distal end of the rotating shaft is attached to the cutter head **1750** and, in some embodiments, the cutter head **1750** can have a deflection distance provides that a sweep of the cutter head **1750** inside a blood vessel during a vascular cutting procedure. Embodiments of this type are discussed in more detail herein. In some embodiments, the deflection of the cutting head **1750** can be retained by a guiding sheath **1790** and, in some embodiments, the guiding sheath **1790** is a slidable translatable guiding sheath assembly **1795** that includes guiding sheath **1790**, sheath connector **1791**, slider bushing **1792**, thumb-slider **1793**, and locking mechanism **1794**. In some embodiments, the slidably translatable guiding sheath **1790** translates **1799** proximally to create the deflection distance and create the sweep of the cutter head **1750**, and translates **1799** distally to straighten the deflection and remove the sweep of the cutter head **1750**. In some embodiments, thumb-slider **1793** is in a friction or fixed contact with slider bushing **1792**, so that when thumb-slider **1793** is translated distally, slider bushing **1792** is translated distally to push sheath connector **1791** distally which pushes guiding sheath **1790** distally, and the distal position of the guiding sheath **1790** can be reversibly locked in place by locking mechanism **1794**. In some embodiments, thumb-slider **1793** is in a friction or fixed contact with slider bushing **1792**, so that when thumb-slider **1793** is translated proximally, slider bushing **1792** is translated proximally to pull sheath connector **1791** proximally which pulls guiding sheath **1790** proximally, and the distal position of the guiding sheath **1790** can be reversibly locked in place by locking mechanism **1794**.

In some embodiments, a positive pressure assembly can be added to the system to provide a way to flush the system with saline, for example, or perhaps introduce a contrast material for imaging of a vessel, to determine the location of the cutter head, and the like. In FIG. **17**, positive pressure port **1797** is operably attached to positive pressure tubing **1798**, both of which are in communication with the guiding sheath. The terms "guiding sheath" and "straightening sheath" can be used interchangeably in some embodiments.

FIG. **18** illustrates a preferred vascular cutter device, according to some embodiments. The vascular cutter device **1800** can have a handle **1700** that includes a vacuum manifold **1500** with a vacuum assembly **1510**, and a cutter head **1750** with a deflection distance, So, and a slidably translatable guiding sheath assembly **1795**, according to some embodiments.

In some embodiments, the vascular cutter device can be a thrombectomy device, including

- a flexible rotating shaft with a proximal end, a distal end, and a lumen;
- a curved, tubular cutter **1750** operably connected to the distal end of a flexible rotating shaft, the tubular cutter including a head with an axis with a mouth having a z-axis, a neck connected to the head and having an axis of rotation, and an angle, θ_z , ranging from 90° to 135° ,

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between the z-axis of the mouth and the axis of rotation, the mouth the configured with
 a leading edge having a highest height, h-leading;
 a trailing edge having a lowest height, h-trailing;
 a vertical bias ratio of h-leading/h-trailing ranging from 1.05 to 2.0;
 a distance, BL, between the axis of rotation and the leading edge;
 a distance, BT, between the axis of rotation and the trailing edge; and,
 a lateral bias ratio of BL/BT ranging from 1.05 to 2.0;
 and,
 a pulsating vacuum control assembly having
 a valve configured to open and close at one or more desired frequencies to create a pulsating vacuum that applies fluctuates the vacuum pressure applied to the cutter during operation of the device; and,
 a port configured for operable communication with a vacuum source to facilitate transport of the tissue out of the subject.

In some embodiments, the valve is configured to operate within a desired range of frequencies, wherein the one or more desired frequencies have a frequency ranging from 1 Hz to 5 Hz.

In some embodiments, the vacuum is controlled to operate within a desired range of vacuum pressures, wherein the vacuum pressure fluctuates between two or more pressures including a high pressure and a low pressure, the high pressure ranging from 100 mmHg to 800 mmHg, and the low pressure ranging from 0 mmHg to 100 mmHg.

In some embodiments, the cutter can be deflected from the axis of rotation to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the axis of the head of the cutter is at a vertical angle, ϕ , with the axis of rotation, wherein ϕ ranges from 0° to 90° to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the axis of the head is at a lateral angle, θ_0 , with the axis of rotation, wherein θ_0 ranges from 0° to 90° to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the flexible rotating shaft is deflected at an angle, θ_m , with the axis of rotation to form a curved region in the flexible rotating shaft, wherein θ_m ranges from 0-90 degrees to create a deflection distance between the leading edge of the cutter and the axis of rotation.

In some embodiments, the system further comprises a straightening sheath, the straightening sheath slidably translatable over the flexible rotating shaft. In some embodiments, the straightening sheath can be translated proximally to create the deflection distance by allowing the flexible rotating shaft to exhibit the deflection. And, in some embodiments, the straightening sheath can be translated distally to remove the deflection from the flexible rotating shaft.

The systems can include a cutter driven by a motor that drives the cutter. In some embodiments, the motor that drives the cutter can be located in a handle for ease of control by the user. As such, in some embodiments, the systems can further comprise a handle that includes

a motor operably connected to a drive assembly that includes a flexible rotating shaft with a proximal end, a distal end, and a lumen, the cutting head operably connected to the distal end of the flexibly rotating shaft, and the motor configured to provide a rotational movement to the flexible, rotating shaft for rotating the cutting head; and,

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the pulsating vacuum control assembly.

The control of the motor can be independent, or shared with, the control of the pulsating vacuum. In some embodiments, the systems can include a motor actuator and a pulsating vacuum actuator. In some embodiments, the systems can include a combination actuator for actuating the motor and the pulsating vacuum control assembly to create the vacuum simultaneously with actuation of the motor at an on-position and stop the vacuum and the motor simultaneously upon actuation to an off-position.

Any known system for creating a pulsating vacuum can be used. In some embodiments, the pulsating vacuum is created by pulsing power electronically to the vacuum. In some embodiments, the pulsating vacuum is created by mechanically opening and closing a valve. In some embodiments, the flexible rotating shaft has a wall with an aspiration port; and, the pulsating vacuum control assembly includes

a sleeve bearing having a proximal end, a distal end, and a wall with an aspiration outlet, wherein the flexible rotating shaft is rotatably translatable within the sleeve bearing; and,

an aspiration manifold having a proximal end, a distal end and a lumen, the aspiration manifold configured to house the sleeve bearing and operably connect with the vacuum source to facilitate the transport of the diseased tissue out of the subject when the aspiration port of the flexible rotating shaft aligns with the aspiration outlet of the sleeve bearing.

In some embodiments, the sleeve bearing is housed at the distal end of the lumen of the aspiration manifold; and, the aspiration outlet of the sleeve bearing is an elongate opening that extends to the proximal end of the sleeve bearing to direct the diseased tissue out of the subject through the aspiration manifold.

In some embodiments, the flexible rotating shaft has a wall with an aspiration port; and, the pulsating vacuum control assembly includes

a sleeve bearing having a proximal end, a distal end, and a wall with an aspiration outlet, wherein the flexible rotating shaft is rotatably translatable within the sleeve bearing; and,

an aspiration manifold having a proximal end and a distal end and a lumen, the aspiration manifold configured to house the sleeve bearing and operably connect with the vacuum source to facilitate the transport of the diseased tissue out of the subject when the aspiration port of the flexible rotating shaft aligns with the aspiration outlet of the sleeve bearing.

In some embodiments, a system can have the sleeve bearing housed at the distal end of the lumen of the aspiration manifold; and, the aspiration outlet of the sleeve bearing can be an elongate opening that extends to the proximal end of the sleeve bearing to direct the diseased tissue out of the subject through the aspiration manifold.

Blood Filtration Feature

Lost blood can be filtered and, in some embodiments, administered to the subject. In some embodiments, the filtered blood can be redirected back to a subject during a procedure. In some embodiments, a system taught herein can include a blood filter configured to filter an aspirated blood from the aspiration tube and create a filtered blood. In some embodiments, a system can include a blood return catheter for delivering the filtered blood back to the subject. Any device taught herein can have this blood filtration feature.

FIGS. 19A and 19B illustrate blood filter configurations, according to some embodiments. FIG. 19A shows a blood filter located inside the handle. It should be appreciated that any number of configurations can be used. In some embodiments, the blood filter can be reversibly attached to be replaceable using a method that includes removing the blood filter from inside the handle, and replacing the blood filter with a replacement blood filter.

FIG. 19B shows a blood filter located outside the handle. It should be appreciated that any number of configurations can be used. In some embodiments, the blood filter can be reversibly attached to be replaceable using a method that includes removing the blood filter from outside the handle, and replacing the blood filter with a replacement blood filter.

A blood filter can have any known configuration to filter the aspirated blood for reuse. In some embodiments, the blood filter 1905 can be figured to have a body 1910 and filtration element 1915 located inside the body for filtering cut tissue and/or other foreign bodies from aspirated blood for reuse.

In some embodiments, the pore size of a blood filter can range from 100 microns to 300 microns, from 120 microns to 300 microns, from 140 microns to 300 microns, from 160 microns to 300 microns, from 180 microns to 300 microns, or any size or range therein in increments of 1 micron. In some embodiments, the pore size of a blood filter can range from 170-260 microns, or any amount or range therein in increments of 1 micron. In some embodiments, a filtration element can have or more pore sizes that include 150 microns, 160 microns, 170 microns, 180 microns, 190 microns, 200 microns, 210 microns, 220 microns, 230 microns, 240 microns, 250 microns, 260 microns, 270 microns, 280 microns, 290 microns, 300 microns, or any size or combination of sizes therein in increments of 1 micron. the pore size. The filtration element can be configured to trap cell fragments, clots, and other particulate matter that can develop during aspiration. In some embodiments, a blood filter element can include a woven mesh having a material that includes, but is not limited to, nylon or polyester. In some embodiments, the blood filter can include pore sizes of ranging from 20-40 micrometers, 25-40 micrometers, 30-40 micrometers or any size or range therein in increments of 1 micron, to remove microaggregates within the aspirated blood. In some embodiments, the blood filter can have two or more stages of filtration, a first stage having a larger pore size to remove larger particles, and a second stage having a smaller pore size to remove smaller particles.

Blood Loss Monitoring Feature

Blood loss can also be monitored and controlled using a system designed to help control blood loss from a subject during a procedure. In some embodiments, a system taught herein can include a dynamic response monitor, wherein the valve closes dynamically at a maximum tolerable level of a blood loss from the subject measured using a real-time monitoring. In some embodiments, the system can include a blood aspiration canister with a float valve, wherein the maximum tolerable level of the blood loss from the subject is measured using the height of the float valve. Any device taught herein can have this blood monitoring feature.

In some embodiments, the frequency, dwell, and/or amount of vacuum can be adjusted through the blood loss monitoring feature. In some embodiments, the frequency, dwell, and or amount of vacuum can be dynamically adjusted, either manually, or automatically, for example, using the computer processor and memory systems taught herein. In some embodiments, the dynamic adjustment

occurs to reduce blood loss from the subject. In some embodiments, the dynamic adjustment occurs to stop blood loss from the subject by stopping the vacuum. In some embodiments, the dynamic adjustment occurs at a target tolerable level of blood loss from the subject. In some embodiments, the dynamic adjustment occurs to restore the frequency, dwell, and/or amount of vacuum in response to an acceptable amount of filtered blood being re-administered to the subject.

FIG. 20 illustrates a vascular cutter system that includes a component for monitoring blood loss from a subject during a vascular cutting procedure, according to some embodiments. The system 2000 can include a blood filter 1905 for filtering vascular tissue from the blood and a container 2010 for collecting the filtered blood, addition to the vascular cutter device 1800, and aspiration tube 1550. The container includes an inlet port 2015 a float 2020 that slidably translates 2030 up and down a measuring rod 2050 that provides a measurement of the amount of filtered blood in the container.

A vacuum source is used, in some embodiments, to aspirate cut vascular tissue from a blood vessel of the subject, and the vacuum source can be located outside the vascular cutter system, outside the vascular cutter device, inside the vascular cutter device, or inside the vascular cutter handle. An example of a vacuum source located outside the vascular cutter system can be a vacuum source that is shared by more than one location, for example, an inline vacuum system in a hospital. An example of a vacuum source located outside the vascular cutter device can be a vacuum source that is located at the site of the procedure but fixed at the location and attached to the vascular cutter device at the time of the procedure. An example of a vacuum source that is located inside the vascular cutter device can be a small vacuum source that is located in a handle of a vascular cutter device.

FIG. 21 illustrates a vascular cutter system that includes a vacuum source inside the handle, according to some embodiments. As shown in FIG. 21, the vascular cutter system 2100 has a handle 1700 has the vacuum source 2150 in the handle 1700 and, optionally, a blood filter 1905 for removing cut tissue 2189 such as a thrombus to create a filtered blood 2199. The system 2100 also includes an aspiration tube 1550 for transporting filtered blood 2199 to a blood aspiration canister 2010 with a float valve 2020. The aspiration tube 1550 can have an aspiration valve 2160, a one-way valve 2165 located between the blood filter 1905 and the blood aspiration canister. The vacuum 2155 is applied to the system through a splitter 2165, such as a Y-fitting or T-fitting, located between the blood filter and the aspiration valve that carries the vacuum from the vacuum source, through a vacuum channel 2175 in the handle 1700, through the splitter 2165, and through the blood filter 1905 to aspirate cut tissue 2189 from a subject. As blood is collected, filtered blood flows through the filter and past the one-way aspiration valve into the blood aspiration canister.

In some embodiments, the systems taught herein further comprise a blood return catheter for delivering the filtered blood back to the subject. In some embodiments, the systems taught herein comprises a dynamic response monitor, wherein the vacuum valve closes dynamically at a maximum tolerable level of a blood loss measured using a real-time monitoring, and the vacuum valve opens dynamically when the level of blood loss returns to a level that is less than a tolerable level of blood loss as blood is returned to the subject through the blood return catheter. In some embodiments, the systems taught herein comprise a dynamic

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response monitor with a blood aspiration canister having a float valve, wherein the maximum tolerable level of the blood loss is measured using a first height of the float valve, and the minimum tolerable level of blood loss is measured using a second height of the float valve.

Methods of Use

Methods of each and any of the devices and systems are also provided, and any blood vessel, venous or arterial, can be treated using a form of the devices and systems taught herein. In some embodiments, a method of removing a vascular tissue with a vascular cutter device is provided, the method including

creating a point of entry in a vascular lumen of the subject;
inserting the cutter head of a vascular cutter device into the vascular lumen;
delivering the vascular cutter device to the target site, the target site having the tissue for removal;
cutting the tissue from the vascular lumen with the cutter head of the vascular cutter device;
and,
removing the cutter head of the vascular cutter device from the vascular lumen of the subject.

In some embodiments, methods of performing a thrombectomy using a thrombectomy device is provided. The method can include:

creating a point of entry in a vascular lumen of the subject;
inserting the cutter head of a thrombectomy device into the vascular lumen;
delivering the cutter head to the target site in the delivery sheath, the target site having the tissue for removal;
cutting the tissue from the vascular lumen with the cutter head of the thrombectomy device;
and,
removing the cutter head of the thrombectomy device from the vascular lumen of the subject.

In some embodiments, a method of removing a vascular tissue with a vascular cutter device having a deflection distance of the cutter head is provided, the method including

creating a point of entry in a vascular lumen of the subject;
inserting a cutter head of the vascular cutter device into the vascular lumen, the vascular cutter device having a deflection distance;
delivering the cutter head to the target site, the target site having the tissue for removal;
cutting tissue from the vascular lumen with the cutter head of the vascular cutter device;
and,
removing the cutter head of the vascular cutter device from the vascular lumen of the subject.

In some embodiments, a method of removing a vascular tissue with a thrombectomy device having a deflection distance of the cutter head is provided, the method including

creating a point of entry in a vascular lumen of the subject;
inserting a cutter head of the thrombectomy device into the vascular lumen, the cutter head of the thrombectomy device having a deflection distance;
delivering the cutter head to the target site, the target site having the tissue for removal;
cutting tissue from the vascular lumen with the cutter head of the thrombectomy device;
and,
removing the cutter head of the vascular cutter device from the vascular lumen of the subject.

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removing the cutter head of the thrombectomy device from the vascular lumen of the subject.

In some embodiments, the method further includes discharging a cut tissue from the vascular lumen with a vacuum. In some embodiments, the vacuum can be a pulsating vacuum.

In some embodiments, a method of removing a vascular tissue with a vascular cutter device having a deflection distance of the cutter head is provided, the method including creating a point of entry in a vascular lumen of the subject;

inserting a cutter head of the vascular cutter device into the vascular lumen, the vascular cutter device having a deflection distance;
delivering the cutter head to the target site, the target site having the tissue for removal;
cutting tissue from the vascular lumen with the cutter head of the vascular cutter device;
discharging the tissue from the vascular lumen with the pulsating vacuum;
and,
removing the cutter head of the vascular cutter device from the vascular lumen of the subject.

In some embodiments, a method of removing a vascular tissue with a thrombectomy device having a deflection distance of the cutter head is provided, the method including creating a point of entry in a vascular lumen of the subject;

inserting a cutter head of the thrombectomy device into the vascular lumen, the cutter head of the thrombectomy device having a deflection distance;
delivering the cutter head to the target site, the target site having the tissue for removal;
cutting tissue from the vascular lumen with the cutter head of the thrombectomy device;
discharging the tissue from the vascular lumen with the pulsating vacuum;
and,
removing the cutter head of the vascular cutter device from the vascular lumen of the subject.

In some embodiments, the methods include using a delivery sheath for a safe delivery of the cutter to the target site for the tissue removal. The proviso, however, is that in some embodiments, the methods do not include using a delivery sheath, either by preference or by design.

In some embodiments, the method further includes inserting a guidewire in the point of entry and delivering the guidewire to a target tissue location, such as the location of a thrombus, and guiding the cutter head of the vascular cutter device to the target tissue location on the guidewire. The proviso, however, is that in some embodiments, the methods do not include using a guidewire, either by preference or by design.

In some embodiments, the methods include filtering the aspirated blood and, in some embodiments, the methods include filtering the aspirated blood and returning the filtered blood back to the subject. In some embodiments, the methods include filtering the aspirated blood, collecting the filtered blood in an aspiration container, and returning the filtered blood back to the subject. In some embodiments, the methods include filtering the aspirated blood, collecting the filtered blood in an aspiration container, measuring the amount of blood collected in the aspiration container, and returning the filtered blood back to the subject when a select amount of blood is collected.

The devices, systems, and methods taught herein can be used in the treatment of any vascular condition. In some

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embodiments, the treatment includes the treatment of a deep vein thrombosis. In some embodiments, the treatment includes the treatment of a pulmonary embolism. In some embodiments, the treatment includes the treatment of a peripheral vascular disease. In some embodiments, the treatment includes the treatment of a saphenous vein graft.

We claim:

1. A thrombectomy system, comprising
 - a flexible rotating shaft with a proximal end, a distal end, and a lumen;
 - a curved, tubular cutter operably connected to the distal end of the flexible rotating shaft and configured to remove a tissue from a target site in a subject, the tubular cutter including a head with an axis and a mouth having a z-axis, a neck connected to the head and having an axis of rotation, and an angle, θ_z , ranging from 90° to 135° , between the z-axis of the mouth and the axis of rotation, the mouth configured with
 - a leading edge having a highest height, h-leading;
 - a trailing edge having a lowest height, h-trailing;
 - a vertical bias ratio of h-leading/h-trailing ranging from 1.05 to 2.0;
 - a distance, B_L , between the axis of rotation and the leading edge;
 - a distance, B_T , between the axis of rotation and the trailing edge; and,
 - a lateral bias ratio of B_L/B_T ranging from 1.05 to 2.0;
 - and,
 - a pulsating vacuum control assembly having
 - a valve configured to open and close at one or more desired frequencies to create a pulsating vacuum that fluctuates the vacuum pressure applied to the cutter during operation of the system; and,
 - a port configured for operable communication with a vacuum source to facilitate transport of the tissue out of the subject.
2. The system of claim 1, wherein the one or more desired frequencies have a frequency ranging from 1 Hz to 5 Hz.
3. The system of claim 1, wherein the vacuum pressure fluctuates between two or more pressures including a high pressure and a low pressure, the high pressure ranging from 100 mmHg to 800 mmHg, and the low pressure ranging from 0 mmHg to 100 mmHg.
4. The system of claim 1, wherein the cutter is deflected from the axis of rotation to create a deflection distance between the leading edge of the cutter and the axis of rotation.
5. The system of claim 1, wherein the axis of the head is at a vertical angle, ϕ , with the axis of rotation, wherein ϕ ranges from 0° to 90° to create a deflection distance between the leading edge of the cutter and the axis of rotation.
6. The system of claim 1, wherein the axis of the head is at a lateral angle, θ_0 , with the axis of rotation, wherein θ_0 ranges from 0° to 90° to create a deflection distance between the leading edge of the cutter and the axis of rotation.
7. A method of performing a thrombectomy using the system of claim 1, the method comprising:
 - creating a point of entry in a vascular lumen of the subject;
 - inserting the curved, tubular cutter into the vascular lumen;
 - delivering the cutter to the target site having the tissue for removal;
 - cutting the tissue from the vascular lumen with the cutter;
 - discharging the tissue from the vascular lumen with the pulsating vacuum;

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and,
removing the cutter from the vascular lumen of the subject.

8. The system of claim 1, the system further comprising a handle that includes
 - a motor operably connected to a drive assembly that includes the flexible rotating shaft and the cutter, the motor configured to provide a rotational movement to the flexible, rotating shaft for rotating the cutter; and,
 - the pulsating vacuum control assembly.
9. The system of claim 8 further including a combination actuator for actuating the motor and the pulsating vacuum control assembly simultaneously to create the pulsating vacuum simultaneously with actuation of the cutter to an on-position and closes the pulsating vacuum upon actuation of the cutter to an off-position.
10. The system of claim 8, wherein
 - the flexible rotating shaft has a wall with an aspiration port;
 - the pulsating vacuum control assembly includes
 - a sleeve bearing having a proximal end, a distal end, and a wall with an aspiration outlet, wherein the flexible rotating shaft is rotatably translatable within the sleeve bearing; and
 - an aspiration manifold having a proximal end and a distal end and a lumen, the aspiration manifold configured to house the sleeve bearing and operably connect with the vacuum source to facilitate the transport of the tissue out of the subject when the aspiration port of the flexible rotating shaft aligns with the aspiration outlet of the sleeve bearing.
11. The system of claim 10 further including a combination actuator for actuating the motor and the pulsating vacuum control assembly simultaneously to create the pulsating vacuum simultaneously with actuation of the cutter to an on-position and closes the pulsating vacuum upon actuation of the cutter to an off-position.
12. The system of claim 1, wherein the flexible rotating shaft is deflected at an angle, θ_m , with the axis of rotation to form a curved region in the flexible rotating shaft, wherein θ_m ranges from 0 - 90 degrees to create a deflection distance between the leading edge of the cutter and the axis of rotation.
13. The system of claim 12, further comprising a straightening sheath.
14. The system of claim 12, the system further comprising a handle that includes
 - a motor operably connected to a drive assembly that includes the flexible rotating shaft and the cutter, the motor configured to provide a rotational movement to the flexible, rotating shaft for rotating the cutter; and,
 - the pulsating vacuum control assembly.
15. The system of claim 14, wherein
 - the flexible rotating shaft has a wall with an aspiration port;
 - the pulsating vacuum control assembly includes
 - a sleeve bearing having a proximal end, a distal end, and a wall with an aspiration outlet, wherein the flexible rotating shaft is rotatably translatable within the sleeve bearing; and
 - an aspiration manifold having a proximal end and a distal end and a lumen, the aspiration manifold configured to house the sleeve bearing and operably connect with the vacuum source to facilitate the transport of the tissue out of the subject when the aspiration port of the flexible rotating shaft aligns with the aspiration outlet of the sleeve bearing.

16. A method of performing a thrombectomy in the subject using the system of claim 15, wherein the system of claim 15 further includes a combination actuator for actuating the motor and the pulsating vacuum control assembly simultaneously to create the pulsating vacuum simultaneously with actuation of the cutter to an on-position and closes the pulsating vacuum upon actuation of the cutter to an off-position; and the method comprises:

- creating a point of entry in a vascular lumen of the subject;
- inserting the curved, tubular cutter into the vascular lumen;
- delivering the cutter to the target site having the tissue for removal;
- actuating the combination actuator;
- cutting the tissue from the vascular lumen with the cutter;
- discharging the tissue from the vascular lumen with the pulsating vacuum; and,
- removing the cutter from the vascular lumen of the subject.

17. The method of claim 16, wherein the thrombectomy includes the treatment of a deep vein thrombosis.

18. The method of claim 16, wherein the thrombectomy includes the treatment of a pulmonary embolism.

19. The method of claim 16, wherein the thrombectomy includes the treatment of a peripheral vascular disease.

20. The method of claim 16, wherein the thrombectomy includes the treatment of a saphenous vein graft.

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